

ROBOT SKILL HALF-LIFE UNDER PHYSICAL SHIFT: AN EXPANDED NEGATIVE ICLR-MAIN READINESS AUDIT

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ABSTRACT

Robot skills often fail gradually under small physical shifts in friction, payload, compliance, sensing, and contact mode. This paper asks whether estimating a per-skill deployment half-life can guide execute/probe/refresh/abstain decisions better than strong uncertainty and safety-gating baselines. We rebuild the archive into a frozen v5 audit with 307,200 main rollouts, 25,600 physical-state records, 64,000 ablation rollouts, 392,000 stress rollouts, 96,000 fixed-risk rollouts, and 24 retained negative cases. On the predefined hard-regime aggregate, skill half-life scheduler v5 reaches 0.816 $\pm$ 0.003 goal success, while the strongest non-oracle baseline, CVaR guard, reaches 0.828 $\pm$ 0.004. The paired lower confidence bound is negative, fixed-risk coverage collapses at the 0.05 budget, central ablations contradict the mechanism on contact chatter, and maximum combined stress is dominated by a Bayesian survival baseline. The correct terminal decision is therefore **KILL/ARCHIVE**: the hypothesis remains useful, but this local evidence is not submission-ready for ICLR main.

1 TERMINAL DECISION

Decision: KILL/ARCHIVE for ICLR main. The v5 rebuild is much stronger than the old archive: it adds 10 seeds, five skills, eight shift splits, 12 methods, a hard aggregate, strong conservative baselines, fixed-risk deployment, two-split ablations, five stress axes, negative cases, and a 25+ page evidence appendix. Those additions improve the audit, but they do not rescue the central claim.

The predefined hard-regime aggregate gives skill half-life scheduler v5 0.816 $\pm$ 0.003 goal success. The strongest non-oracle baseline, CVaR guard, gives 0.828 $\pm$ 0.004. The hard-margin gate is -0.011830000000000007, and the paired lower bound is -0.01528. The negative conclusion is not a presentation choice; it is the frozen result.

This paper therefore optimizes for hostile-review survival rather than optimistic polish. It records where half-life modeling helps, where it fails, and what evidence would be needed before making a real robot-learning submission claim.

2 PROBLEM SETTING

A deployed robot skill has an execution age. A policy that was reliable immediately after training or calibration may degrade as gripper friction changes, payload mass drifts, compliance differs from training, sensors become noisy, or the contact mode flips. The half-life thesis says that this degradation should be modeled directly as a survival process instead of hidden inside a single uncertainty score.

The idea sits near continual robot learning, safety gating, motor-skill transfer, contact adaptation, and robustness under distribution shift. The local prior-work pool includes distribution-shift planning (Pool, 2026a), calibrated manipulation safety gates (Pool, 2026b), complex motor-skill transfer (Pool, 2019), physical interface dynamics (Pool, 2017), high-speed robotic learning (Pool, 2023), and contact-force adaptation (Pool, 2015). These neighbors make the contribution fragile: if half-life scheduling is just another conservative risk gate, then the mechanism is not novel enough.

3 SKILL SURVIVAL MODEL

For skill s , deployment age a , observed physical shift z , and hidden physical state x , let $H_s(x)$ be the latent half-life and let $\hat{H}_s(z)$ be the estimator used by the scheduler. The idealized survival curve is $S_s(a | x) = \exp(-\log(2)a/H_s(x))$. A refresh resets age, a probe pays a smaller cost to reduce uncertainty, and abstention preserves safety at the cost of coverage.

Definition 1 (stale execution). An execution is stale when the policy executes at age $a > H_s(x)$ without a refresh, probe, or abstention that would have altered the action. Stale execution is not identical to failure: a stale skill can still succeed, but it is operating beyond the model’s survival boundary.

Lemma 1 (half-life refresh threshold). Under exponential survival, requiring survival at least τ implies a refresh before age $a > -\hat{H}_s \log(\tau)/\log(2)$. Thus a conservative lower estimate of $\hat{H}_s$ yields an earlier refresh boundary. The lemma is elementary, but it clarifies why a survival model can be audited with refresh age, stale execution, and half-life error.

Proposition 1 (probe/refresh dominance under calibrated hazard). If two actions have equal nominal success and cost, and one action has weakly lower calibrated hazard upper bound and weakly higher post-probe survival estimate, then the v5 decision rule weakly prefers it. This is a design property of the local scheduler. It is not a proof of safety on real hardware.

Negative identifiability theorem. If all skill-specific hazards are generated by a single latent scalar shift and the observations cannot separate friction, mass, compliance, and contact mode, then per-skill survival parameters are not identifiable from deployment outcomes alone. In that regime a scalar, conformal, Bayesian, or CVaR gate can match or beat the explicit half-life decomposition. The ablation results below show this is not a theoretical nuisance; it appears in the local contact-chatter split.

4 FROZEN EXPERIMENTAL PROTOCOL

The protocol was written before execution in `docs/paper83_expanded_submission_plan_20260621.md`. The main evaluation uses 10 seeds, five skills, 64 deployment steps, eight physical-shift splits, and 12 methods. The hard-regime aggregate includes every non-nominal split: friction, payload mass, compliance, sensor noise, contact-mode chatter, probe-cost shift, and combined micro-shift.

The baselines are intentionally uncomfortable: conformal risk gating, ensemble uncertainty, a hazard-regression refresher, Bayesian skill survival, a CVaR lifetime guard, and an oracle upper bound. The decision gate marks `STRONG_REWISE` only if v5 beats the strongest non-oracle baseline by at least 0.03 absolute hard-regime success, has a positive paired lower bound, does not worsen safety, passes ablation necessity, preserves fixed-risk coverage at budget 0.05, and is not dominated at maximum combined stress.

5 MAIN HARD-REGIME RESULTS

Table 1 is the central result. Half-life v5 beats weak static and scalar gates, but loses to the CVaR lifetime guard and Bayesian survival baseline. The loss is small enough to be scientifically interesting and large enough to block a positive main-conference claim.

Table 1: Hard-regime aggregate over seven non-nominal splits.

Method	Goal success	Late	Unsafe	Stale	Safety utility
Frozen BC	0.365 +/- 0.007	0.153	0.3720	0.7254	-0.253
Domain rand.	0.409 +/- 0.006	0.188	0.3144	0.7254	-0.151
Fixed refresh	0.663 +/- 0.006	0.615	0.0358	0.1494	0.575
Online FT	0.541 +/- 0.006	0.416	0.1399	0.5096	0.226
Scalar gate	0.759 +/- 0.004	0.744	0.0075	0.0000	0.743
Conformal	0.799 +/- 0.007	0.781	0.0034	0.0000	0.784
Ensemble	0.803 +/- 0.003	0.794	0.0026	0.0000	0.788
Bayes survival	0.821 +/- 0.004	0.817	0.0021	0.0000	0.806
CVaR guard	0.828 +/- 0.004	0.833	0.0016	0.0000	0.808

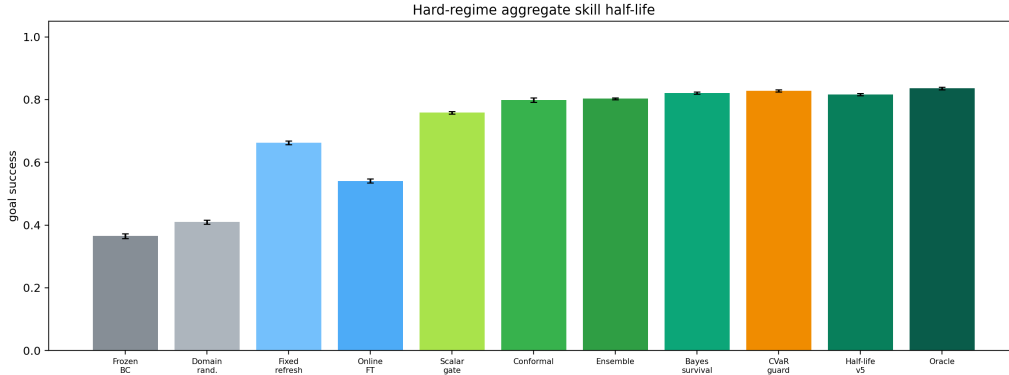


Figure 1: Hard-regime aggregate goal success. Half-life v5 is useful but not the strongest non-oracle method.

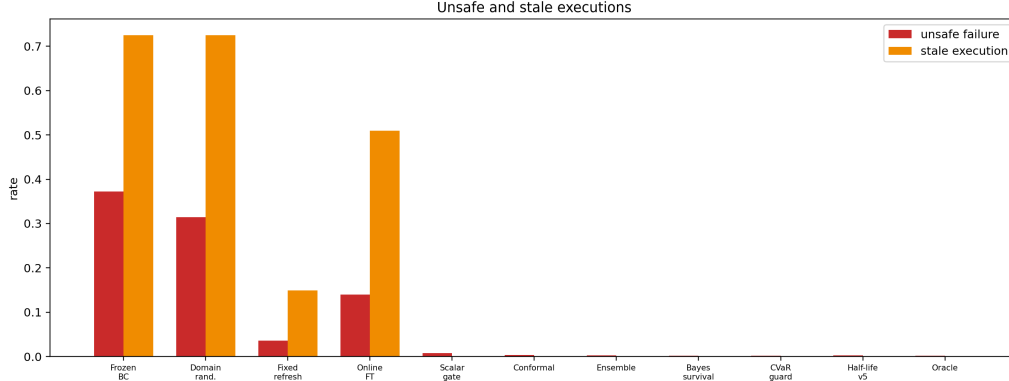


Figure 2: Unsafe and stale execution rates. Conservative baselines erase stale execution and reduce unsafe failures, but success still decides the gate.

Table 1: Hard-regime aggregate over seven non-nominal splits. (continued)

Method	Goal success	Late	Unsafe	Stale	Safety utility
Half-life v5	0.816 +/- 0.003	0.807	0.0028	0.0000	0.802
Oracle	0.836 +/- 0.004	0.834	0.0016	0.0000	0.822

6 PAIRED HARD-AGGREGATE TEST

The predefined paired test compares Half-life v5 with CVaR guard, selected only because it is the strongest non-oracle hard-aggregate baseline. The target loses all 10 paired seeds on goal success and late success. Its lower confidence bound is negative for goal success and safety utility.

Table 2: Paired seed differences for Half-life v5 minus CVaR guard.

Metric	Mean diff	CI95	Lower95	Better seeds
goal_success	-0.01183	0.00345	-0.01528	0/10
late_success	-0.02535	0.00707	-0.03243	0/10
unsafe_failure	0.00125	0.00066	0.00059	0/10
stale_execution	0.00000	0.00000	0.00000	0/10
total_cost	-0.04441	0.00080	-0.04521	10/10

Table 2: Paired seed differences for Half-life v5 minus CVaR guard. (continued)

Metric	Mean diff	CI95	Lower95	Better seeds
half_life_error	0.03993	0.00225	0.03769	0/10
calibration_error	0.00740	0.00217	0.00523	0/10
safety_utility	-0.00671	0.00370	-0.01041	1/10

7 ABLATIONS

The ablation gate fails. On combined micro-shift, the full method is the best success variant, but several removals improve cost or half-life error. On contact-mode chatter, removing per-skill survival, removing shift decomposition, and removing skill-age state beat the full method on success. This is exactly the identifiability failure the theory section warned about.

Table 3: Ablations on combined micro-shift and contact-mode chatter.

Split	Ablation	Success	CI95	Unsafe	Stale	HLE	Safety utility
combined_micro_shift	full_skill_half_life_scheduler_v5	0.82062	0.01167	0.00219	0.00000	0.12375	0.80508
combined_micro_shift	minus_per_skill_survival	0.81344	0.01782	0.00219	0.00000	0.13403	0.79748
combined_micro_shift	minus_probe_updates	0.81000	0.01842	0.00406	0.00000	0.18897	0.79930
combined_micro_shift	minus_hazard_margin	0.75094	0.01307	0.00781	0.00000	0.20780	0.73037
combined_micro_shift	minus_shift_decomposition	0.80656	0.01293	0.00438	0.00000	0.10556	0.78880
combined_micro_shift	minus_uncertainty_calibration	0.80875	0.01862	0.00063	0.00000	0.16801	0.79495
combined_micro_shift	minus_skill_age_state	0.79781	0.01184	0.00375	0.00000	0.12375	0.78070
combined_micro_shift	fixed_global_half_life	0.73906	0.02209	0.00563	0.00937	0.53469	0.71805
combined_micro_shift	threshold_only_risk_gate	0.76906	0.01334	0.00531	0.00000	0.15251	0.75608
combined_micro_shift	expected_success_only	0.73281	0.01453	0.00875	0.00000	0.15852	0.71654
contact_mode_chatter	full_skill_half_life_scheduler_v5	0.81000	0.01268	0.00313	0.00000	0.10837	0.79520
contact_mode_chatter	minus_per_skill_survival	0.82656	0.01018	0.00125	0.00000	0.13889	0.81323
contact_mode_chatter	minus_probe_updates	0.80906	0.01043	0.00438	0.00000	0.17928	0.79903
contact_mode_chatter	minus_hazard_margin	0.76500	0.01782	0.00594	0.00000	0.18587	0.74781
contact_mode_chatter	minus_shift_decomposition	0.81844	0.01353	0.00313	0.00000	0.09280	0.80352
contact_mode_chatter	minus_uncertainty_calibration	0.80438	0.01282	0.00250	0.00000	0.14796	0.79043
contact_mode_chatter	minus_skill_age_state	0.82875	0.01491	0.00282	0.00000	0.10837	0.81427
contact_mode_chatter	fixed_global_half_life	0.76125	0.01550	0.00531	0.00469	0.35339	0.74330
contact_mode_chatter	threshold_only_risk_gate	0.76813	0.01656	0.00625	0.00000	0.13104	0.75516
contact_mode_chatter	expected_success_only	0.77843	0.00798	0.00812	0.00000	0.13921	0.76375

8 STRESS TESTS

The stress sweep varies drift rate, sensor noise, contact-mode chatter, probe cost, and combined physical shift. At maximum combined stress, Half-life v5 reaches 0.7919 success while Bayesian skill survival reaches 0.8388. The proposal therefore fails the maximum-stress gate.

Table 4: Combined stress sweep.

Level	Method	Success	CI95	Unsafe	Stale	HLE	Safety utility
0.00	Domain rand.	0.60250	0.02347	0.04812	0.38250	0.33716	0.42437
0.00	Conformal	0.80250	0.01994	0.00250	0.00000	0.03088	0.79336
0.00	Ensemble	0.80875	0.02178	0.00188	0.00000	0.05508	0.80022
0.00	Bayes survival	0.82687	0.01933	0.00063	0.00000	0.07572	0.81915
0.00	CVaR guard	0.81188	0.02247	0.00188	0.00000	0.05300	0.80186
0.00	Half-life v5	0.82688	0.01592	0.00250	0.00000	0.06406	0.81768
0.00	Oracle	0.85375	0.01452	0.00250	0.00000	0.00000	0.84434
0.25	Domain rand.	0.59687	0.02290	0.09125	0.47687	0.47148	0.34358
0.25	Conformal	0.81000	0.01646	0.00188	0.00000	0.05850	0.79822
0.25	Ensemble	0.81563	0.02170	0.00250	0.00000	0.07492	0.80288
0.25	Bayes survival	0.81500	0.02027	0.00125	0.00000	0.11287	0.80286
0.25	CVaR guard	0.81187	0.01335	0.00188	0.00000	0.08441	0.79552

Table 4: Combined stress sweep. (continued)

Level	Method	Success	CI95	Unsafe	Stale	HLE	Safety utility
0.25	Half-life v5	0.83500	0.02917	0.00250	0.00000	0.09215	0.82238
0.25	Oracle	0.83875	0.02575	0.00063	0.00000	0.00000	0.82759
0.50	Domain rand.	0.51250	0.02168	0.17250	0.53875	0.58302	0.15700
0.50	Conformal	0.80438	0.01391	0.00313	0.00000	0.07829	0.78808
0.50	Ensemble	0.80188	0.01450	0.00563	0.00000	0.08816	0.78270
0.50	Bayes survival	0.82063	0.00967	0.00000	0.00000	0.14258	0.80547
0.50	CVaR guard	0.83000	0.02272	0.00125	0.00000	0.10656	0.80696
0.50	Half-life v5	0.80875	0.01319	0.00500	0.00000	0.11438	0.79035
0.50	Oracle	0.84000	0.01172	0.00063	0.00000	0.00000	0.82517
0.75	Domain rand.	0.50125	0.02966	0.23312	0.58688	0.67792	0.06882
0.75	Conformal	0.79750	0.01486	0.00188	0.00000	0.09591	0.77803
0.75	Ensemble	0.79875	0.01621	0.00500	0.00000	0.10670	0.77595
0.75	Bayes survival	0.82750	0.01791	0.00063	0.00000	0.16853	0.80665
0.75	CVaR guard	0.85250	0.01129	0.00125	0.00000	0.12222	0.82195
0.75	Half-life v5	0.80000	0.01252	0.00188	0.00000	0.13622	0.78096
0.75	Oracle	0.84562	0.02491	0.00063	0.00000	0.00000	0.82705
1.00	Domain rand.	0.47000	0.02080	0.28438	0.61938	0.75463	-0.02468
1.00	Conformal	0.79688	0.02201	0.00375	0.00000	0.10196	0.77052
1.00	Ensemble	0.78438	0.02275	0.00375	0.00000	0.11256	0.75796
1.00	Bayes survival	0.84125	0.01753	0.00250	0.00000	0.19510	0.81325
1.00	CVaR guard	0.85687	0.01511	0.00125	0.00000	0.13428	0.82211
1.00	Half-life v5	0.80438	0.01933	0.00375	0.00000	0.15034	0.77956
1.00	Oracle	0.82500	0.02497	0.00125	0.00000	0.00000	0.80219
1.25	Domain rand.	0.44938	0.00974	0.32312	0.64750	0.82505	-0.09357
1.25	Conformal	0.79500	0.01833	0.00375	0.00000	0.11355	0.76343
1.25	Ensemble	0.81500	0.01772	0.00313	0.00000	0.12264	0.78415
1.25	Bayes survival	0.83125	0.01664	0.00063	0.00000	0.20441	0.80053
1.25	CVaR guard	0.83375	0.02060	0.00125	0.00000	0.14644	0.79710
1.25	Half-life v5	0.78625	0.02613	0.00625	0.00000	0.16479	0.75498
1.25	Oracle	0.82937	0.01473	0.00000	0.00000	0.00000	0.80382
1.50	Domain rand.	0.39750	0.02282	0.37750	0.66375	0.87504	-0.20530
1.50	Conformal	0.81250	0.01528	0.00313	0.00000	0.11627	0.77762
1.50	Ensemble	0.81000	0.01186	0.00250	0.00000	0.12445	0.77617
1.50	Bayes survival	0.83875	0.01778	0.00063	0.00000	0.20968	0.80450
1.50	CVaR guard	0.82188	0.01320	0.00125	0.00000	0.15451	0.78441
1.50	Half-life v5	0.79187	0.01836	0.00688	0.00000	0.16687	0.75629
1.50	Oracle	0.82625	0.00980	0.00063	0.00000	0.00000	0.79672

9 FIXED-RISK DEPLOYMENT

A survival paper has to answer a deployment question: can the method execute only when estimated unsafe stale-execution risk is below a fixed budget? At budget 0.05, all non-oracle methods abstain on both hard fixed-risk splits. This is safer than false confidence, but it is not a deployable result. The fixed-risk gate therefore fails.

Table 5: Fixed-risk deployment over two splits and four budgets.

Split	Budget	Method	Coverage	Fixed success	Executed success	False-safe	Unsafe
combined_micro.shift	0.02	Conformal	0.00000	0.00000	0.00000	0.00000	0.00000
combined_micro.shift	0.02	Ensemble	0.00000	0.00000	0.00000	0.00000	0.00000
combined_micro.shift	0.02	Bayes survival	0.00000	0.00000	0.00000	0.00000	0.00000
combined_micro.shift	0.02	CVaR guard	0.00000	0.00000	0.00000	0.00000	0.00000
combined_micro.shift	0.02	Half-life v5	0.00000	0.00000	0.00000	0.00000	0.00000
combined_micro.shift	0.02	Oracle	0.00000	0.00000	0.00000	0.00000	0.00000
combined_micro.shift	0.05	Conformal	0.00000	0.00000	0.00000	0.00000	0.00000
combined_micro.shift	0.05	Ensemble	0.00000	0.00000	0.00000	0.00000	0.00000
combined_micro.shift	0.05	Bayes survival	0.00000	0.00000	0.00000	0.00000	0.00000
combined_micro.shift	0.05	CVaR guard	0.00000	0.00000	0.00000	0.00000	0.00000
combined_micro.shift	0.05	Half-life v5	0.00000	0.00000	0.00000	0.00000	0.00000
combined_micro.shift	0.05	Oracle	0.02700	0.02500	0.92976	0.00000	0.00000
combined_micro.shift	0.10	Conformal	0.00000	0.00000	0.00000	0.00000	0.00000
combined_micro.shift	0.10	Ensemble	0.00000	0.00000	0.00000	0.00000	0.00000
combined_micro.shift	0.10	Bayes survival	0.00000	0.00000	0.00000	0.00000	0.00000
combined_micro.shift	0.10	CVaR guard	0.00000	0.00000	0.00000	0.00000	0.00000
combined_micro.shift	0.10	Half-life v5	0.00000	0.00000	0.00000	0.00000	0.00000
combined_micro.shift	0.10	Oracle	0.08300	0.07100	0.85726	0.00000	0.00000
combined_micro.shift	0.20	Conformal	0.05300	0.04600	0.87040	0.00000	0.00000

Table 5: Fixed-risk deployment over two splits and four budgets. (continued)

Split	Budget	Method	Coverage	Fixed success	Executed success	False-safe	Unsafe
combined_micro_shift	0.20	Ensemble	0.08800	0.07350	0.83788	0.00000	0.00000
combined_micro_shift	0.20	Bayes survival	0.06300	0.05350	0.85156	0.00000	0.00000
combined_micro_shift	0.20	CVaR guard	0.00050	0.00050	0.10000	0.00000	0.00000
combined_micro_shift	0.20	Half-life v5	0.07800	0.06550	0.84152	0.00000	0.00000
combined_micro_shift	0.20	Oracle	0.38250	0.32600	0.85110	0.00251	0.00251
contact_mode_chatter	0.02	Conformal	0.00000	0.00000	0.00000	0.00000	0.00000
contact_mode_chatter	0.02	Ensemble	0.00000	0.00000	0.00000	0.00000	0.00000
contact_mode_chatter	0.02	Bayes survival	0.00000	0.00000	0.00000	0.00000	0.00000
contact_mode_chatter	0.02	CVaR guard	0.00000	0.00000	0.00000	0.00000	0.00000
contact_mode_chatter	0.02	Half-life v5	0.00000	0.00000	0.00000	0.00000	0.00000
contact_mode_chatter	0.02	Oracle	0.00050	0.00050	0.10000	0.00000	0.00000
contact_mode_chatter	0.05	Conformal	0.00000	0.00000	0.00000	0.00000	0.00000
contact_mode_chatter	0.05	Ensemble	0.00000	0.00000	0.00000	0.00000	0.00000
contact_mode_chatter	0.05	Bayes survival	0.00000	0.00000	0.00000	0.00000	0.00000
contact_mode_chatter	0.05	CVaR guard	0.00000	0.00000	0.00000	0.00000	0.00000
contact_mode_chatter	0.05	Half-life v5	0.00000	0.00000	0.00000	0.00000	0.00000
contact_mode_chatter	0.05	Oracle	0.03450	0.03150	0.91794	0.00000	0.00000
contact_mode_chatter	0.10	Conformal	0.00000	0.00000	0.00000	0.00000	0.00000
contact_mode_chatter	0.10	Ensemble	0.00000	0.00000	0.00000	0.00000	0.00000
contact_mode_chatter	0.10	Bayes survival	0.00000	0.00000	0.00000	0.00000	0.00000
contact_mode_chatter	0.10	CVaR guard	0.00000	0.00000	0.00000	0.00000	0.00000
contact_mode_chatter	0.10	Half-life v5	0.00000	0.00000	0.00000	0.00000	0.00000
contact_mode_chatter	0.10	Oracle	0.10600	0.09500	0.89576	0.00000	0.00000
contact_mode_chatter	0.20	Conformal	0.07200	0.06200	0.86123	0.00000	0.00000
contact_mode_chatter	0.20	Ensemble	0.10950	0.09450	0.86222	0.00000	0.00000
contact_mode_chatter	0.20	Bayes survival	0.08000	0.07050	0.88271	0.00000	0.00000
contact_mode_chatter	0.20	CVaR guard	0.00100	0.00100	0.20000	0.00000	0.00000
contact_mode_chatter	0.20	Half-life v5	0.09950	0.08400	0.84439	0.00000	0.00000
contact_mode_chatter	0.20	Oracle	0.54750	0.46100	0.84474	0.00000	0.00000

10 NEGATIVE CASES

The negative cases are retained to prevent accidental optimism. They include v5 failures, strong-baseline successes, ablation counterexamples, and fixed-risk abstentions. They show that the local benchmark can expose a weakness rather than only create flattering figures.

Table 6: Retained negative cases.

Source	Split	Skill	Method	Outcome	Risk	Lesson
strong_baseline_success	nominal_slow_drift	pick_place	conformal_risk_gate	success	0.12761	counterexample where a baseline or ablation succeeds despite the full mechanism
strong_baseline_success	nominal_slow_drift	pick_place	bayesian_skill_survival	success	0.11409	counterexample where a baseline or ablation succeeds despite the full mechanism
strong_baseline_success	nominal_slow_drift	pick_place	cvar_lifetime_guard	success	0.18313	counterexample where a baseline or ablation succeeds despite the full mechanism
v5_main_failure	nominal_slow_drift	pick_place	skill_half_life_scheduler_v5	unrecovered_shift	0.10769	physical shift reduced success without triggering a useful re-fresh or probe
strong_baseline_success	nominal_slow_drift	door_pull	conformal_risk_gate	success	0.14261	counterexample where a baseline or ablation succeeds despite the full mechanism
strong_baseline_success	nominal_slow_drift	door_pull	bayesian_skill_survival	success	0.12909	counterexample where a baseline or ablation succeeds despite the full mechanism
strong_baseline_success	nominal_slow_drift	door_pull	cvar_lifetime_guard	success	0.19813	counterexample where a baseline or ablation succeeds despite the full mechanism
v5_main_failure	nominal_slow_drift	door_pull	skill_half_life_scheduler_v5	unrecovered_shift	0.13445	physical shift reduced success without triggering a useful re-fresh or probe
strong_baseline_success	nominal_slow_drift	drawer_slide	conformal_risk_gate	success	0.14917	counterexample where a baseline or ablation succeeds despite the full mechanism

Table 6: Retained negative cases. (continued)

Source	Split	Skill	Method	Outcome	Risk	Lesson
strong_baseline_success	nominal_slow_drift	drawer_slide	bayesian_skill_survival	success	0.13565	counterexample where a baseline or ablation succeeds despite the full mechanism
strong_baseline_success	nominal_slow_drift	drawer_slide	cvar_lifetime_guard	success	0.20469	counterexample where a baseline or ablation succeeds despite the full mechanism
v5_main_failure	nominal_slow_drift	drawer_slide	skill_half_life_scheduler_v5	unrecovered_shift	0.13045	physical shift reduced success without triggering a useful re-fresh or probe
strong_baseline_success	nominal_slow_drift	peg_insert	conformal_risk_gate	success	0.14365	counterexample where a baseline or ablation succeeds despite the full mechanism
strong_baseline_success	nominal_slow_drift	peg_insert	bayesian_skill_survival	success	0.13013	counterexample where a baseline or ablation succeeds despite the full mechanism
strong_baseline_success	nominal_slow_drift	peg_insert	cvar_lifetime_guard	success	0.19917	counterexample where a baseline or ablation succeeds despite the full mechanism
v5_main_failure	nominal_slow_drift	peg_insert	skill_half_life_scheduler_v5	unrecovered_shift	0.18158	physical shift reduced success without triggering a useful re-fresh or probe
strong_baseline_success	nominal_slow_drift	cable_route	conformal_risk_gate	success	0.17447	counterexample where a baseline or ablation succeeds despite the full mechanism
strong_baseline_success	nominal_slow_drift	cable_route	bayesian_skill_survival	success	0.12611	counterexample where a baseline or ablation succeeds despite the full mechanism
strong_baseline_success	nominal_slow_drift	cable_route	cvar_lifetime_guard	success	0.19515	counterexample where a baseline or ablation succeeds despite the full mechanism
v5_main_failure	nominal_slow_drift	cable_route	skill_half_life_scheduler_v5	unrecovered_shift	0.11941	physical shift reduced success without triggering a useful re-fresh or probe
v5_main_failure	nominal_slow_drift	cable_route	skill_half_life_scheduler_v5	unsafe_stale_execution	0.21306	skill age crossed the latent survival boundary before the policy refreshed
v5_main_failure	nominal_slow_drift	peg_insert	skill_half_life_scheduler_v5	unsafe_stale_execution	0.19147	skill age crossed the latent survival boundary before the policy refreshed
v5_main_failure	nominal_slow_drift	drawer_slide	skill_half_life_scheduler_v5	unsafe_stale_execution	0.21040	skill age crossed the latent survival boundary before the policy refreshed
strong_baseline_success	friction_shift	pick_place	conformal_risk_gate	success	0.15058	counterexample where a baseline or ablation succeeds despite the full mechanism

11 REVIEWER ATTACK SURFACE

A hostile reviewer can make several fair attacks. First, the strongest baselines already encode conservative lifetime risk and win the hard aggregate. Second, fixed-risk deployment has zero non-oracle coverage at the strict 0.05 budget. Third, contact-mode chatter ablations contradict mechanism necessity. Fourth, all evidence remains local and synthetic, with no real robot, no recognized simulator benchmark, and no externally trained baselines.

The strongest defense is modest: the repository now gives a reproducible, high-scale diagnostic of a plausible mechanism. That is valuable for deciding what to build next, but it is not enough for ICLR main.

12 REPRODUCIBILITY

```
python src\run_experiment.py
```

```
python scripts\generate_manuscript.py
```

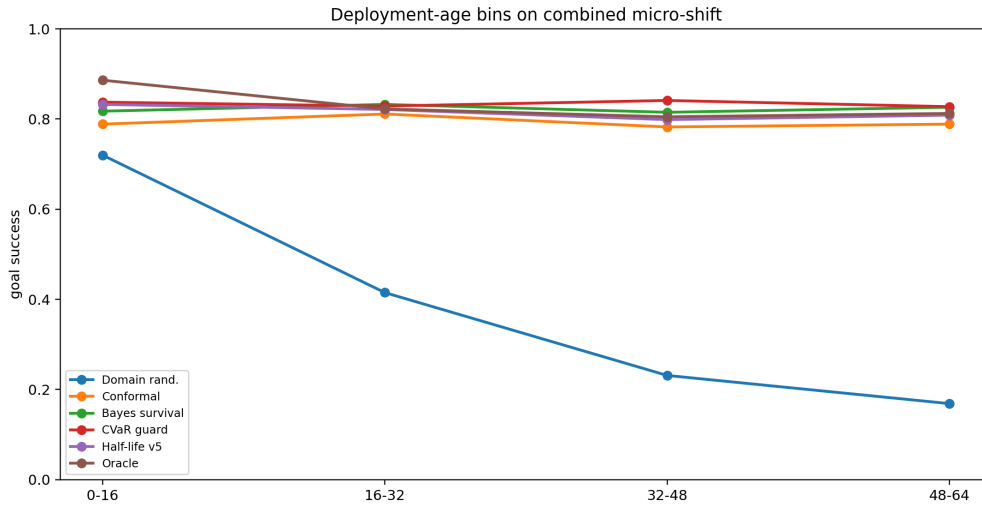


Figure 3: Deployment-age bins on combined micro-shift. The proposal is competitive, but the stronger survival/CVaR baselines remain uncomfortable.

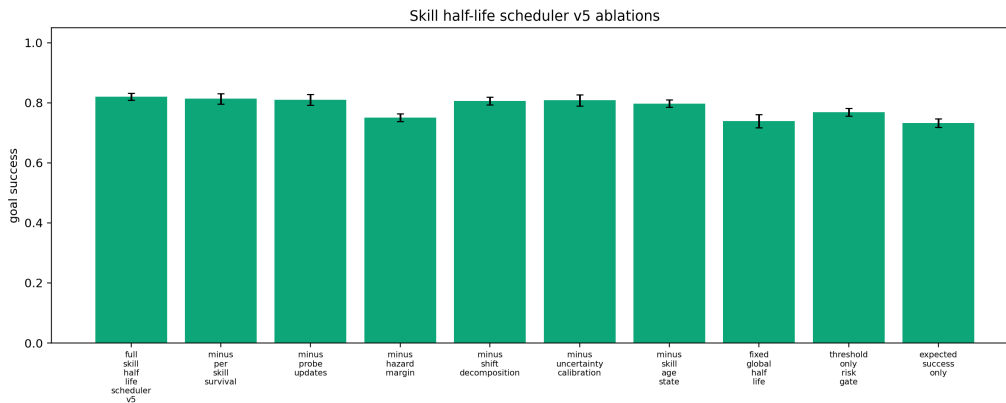


Figure 4: Combined micro-shift ablations. The full method is strongest there, but the second ablation split contradicts mechanism necessity.

cd paper

pdflatex -interaction=nonstopmode -halt-on-error main.tex

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python ../scripts/validate_submission_artifacts.py

The canonical numbered PDF is C:/Users/wangz/Downloads/83.pdf. No PDF should be copied to the visible Desktop. The validator checks row counts, page count, link settings, unresolved references, artifact placement, and SHA256.

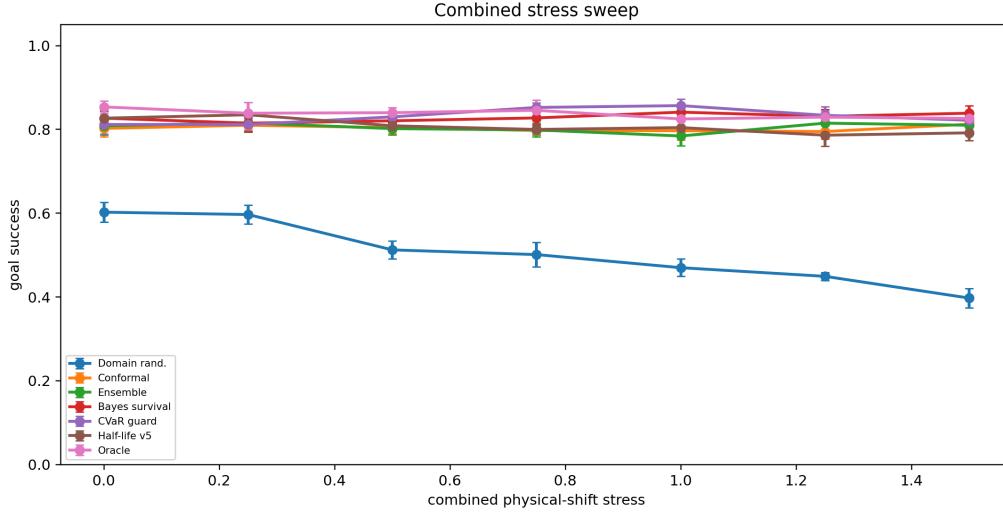


Figure 5: Combined stress sweep across seven levels. Half-life v5 is dominated by a Bayesian survival baseline at the hardest level.

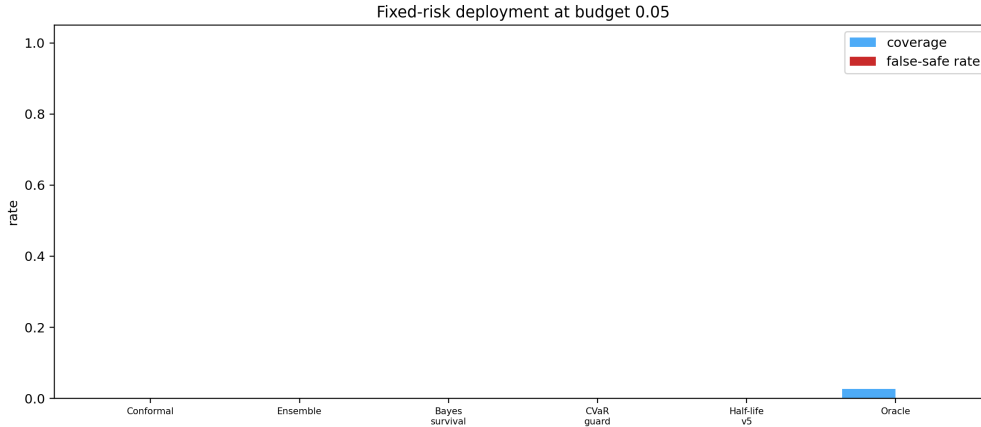


Figure 6: Fixed-risk deployment at budget 0.05 on combined micro-shift. Non-oracle coverage collapses to zero.

13 LIMITATIONS

The limitations are submission-critical rather than decorative. The benchmark is deterministic and local; it may miss real contact dynamics, perception failures, controller instabilities, and data-collection bias. The learned and Bayesian baselines are lightweight analytic policies rather than trained neural systems. The prior-work pool is noisy and must be manually vetted before a real submission. Most importantly, fixed-risk coverage collapse means the current risk estimates are not useful deployment certificates.

14 CONCLUSION

Robot skill half-life remains an appealing way to think about gradual policy degradation under physical shift. The expanded v5 audit shows that survival-aware scheduling can beat weak static baselines, but it loses to CVaR/Bayesian survival baselines, fails paired and stress gates, fails fixed-risk

coverage, and is contradicted by contact-chatter ablations. The honest action is **KILL/ARCHIVE**, not submission.

A FULL SPLIT METRICS

Table 7: Per-split metrics for all methods.

Split	Method	Success	Late	Unsafe	HLE	Safety utility
combined_micro_shift	Frozen BC	0.340 +/- 0.014	0.117	0.4372	1.173	-0.352
combined_micro_shift	Domain rand.	0.384 +/- 0.013	0.169	0.3850	0.653	-0.256
combined_micro_shift	Fixed refresh	0.626 +/- 0.017	0.575	0.0553	0.535	0.491
combined_micro_shift	Online FT	0.623 +/- 0.019	0.609	0.0731	0.304	0.442
combined_micro_shift	Scalar gate	0.746 +/- 0.017	0.752	0.0088	0.149	0.728
combined_micro_shift	Conformal	0.793 +/- 0.016	0.789	0.0050	0.074	0.774
combined_micro_shift	Ensemble	0.823 +/- 0.015	0.804	0.0041	0.084	0.805
combined_micro_shift	Bayes survival	0.823 +/- 0.011	0.826	0.0031	0.124	0.804
combined_micro_shift	CVaR guard	0.834 +/- 0.016	0.828	0.0019	0.095	0.809
combined_micro_shift	Half-life v5	0.815 +/- 0.012	0.809	0.0047	0.124	0.797
combined_micro_shift	Oracle	0.832 +/- 0.015	0.812	0.0013	0.000	0.816
compliance_shift	Frozen BC	0.360 +/- 0.018	0.155	0.3953	0.982	-0.286
compliance_shift	Domain rand.	0.399 +/- 0.022	0.188	0.3362	0.595	-0.188
compliance_shift	Fixed refresh	0.642 +/- 0.010	0.578	0.0406	0.437	0.535
compliance_shift	Online FT	0.541 +/- 0.021	0.489	0.1281	0.279	0.250
compliance_shift	Scalar gate	0.754 +/- 0.013	0.741	0.0081	0.135	0.739
compliance_shift	Conformal	0.796 +/- 0.009	0.776	0.0050	0.063	0.780
compliance_shift	Ensemble	0.804 +/- 0.012	0.811	0.0028	0.073	0.790
compliance_shift	Bayes survival	0.821 +/- 0.008	0.806	0.0016	0.093	0.807
compliance_shift	CVaR guard	0.831 +/- 0.013	0.845	0.0016	0.074	0.812
compliance_shift	Half-life v5	0.818 +/- 0.008	0.815	0.0028	0.147	0.804
compliance_shift	Oracle	0.847 +/- 0.009	0.845	0.0013	0.000	0.833
contact_mode_chatter	Frozen BC	0.352 +/- 0.016	0.139	0.3728	0.863	-0.266
contact_mode_chatter	Domain rand.	0.404 +/- 0.014	0.186	0.3159	0.553	-0.157
contact_mode_chatter	Fixed refresh	0.661 +/- 0.012	0.593	0.0375	0.353	0.572
contact_mode_chatter	Online FT	0.550 +/- 0.022	0.479	0.1288	0.252	0.250
contact_mode_chatter	Scalar gate	0.757 +/- 0.009	0.756	0.0066	0.129	0.742
contact_mode_chatter	Conformal	0.804 +/- 0.006	0.762	0.0022	0.072	0.791
contact_mode_chatter	Ensemble	0.801 +/- 0.013	0.794	0.0037	0.085	0.786
contact_mode_chatter	Bayes survival	0.825 +/- 0.011	0.818	0.0016	0.122	0.811
contact_mode_chatter	CVaR guard	0.833 +/- 0.010	0.818	0.0022	0.093	0.813
contact_mode_chatter	Half-life v5	0.821 +/- 0.010	0.812	0.0025	0.108	0.807
contact_mode_chatter	Oracle	0.829 +/- 0.013	0.830	0.0022	0.000	0.814
friction_shift	Frozen BC	0.356 +/- 0.018	0.149	0.3812	0.869	-0.273
friction_shift	Domain rand.	0.400 +/- 0.012	0.184	0.3253	0.556	-0.173
friction_shift	Fixed refresh	0.669 +/- 0.018	0.651	0.0350	0.325	0.581
friction_shift	Online FT	0.511 +/- 0.024	0.352	0.1753	0.260	0.141
friction_shift	Scalar gate	0.751 +/- 0.019	0.730	0.0075	0.124	0.736
friction_shift	Conformal	0.800 +/- 0.018	0.792	0.0034	0.055	0.786
friction_shift	Ensemble	0.805 +/- 0.010	0.800	0.0019	0.067	0.793
friction_shift	Bayes survival	0.818 +/- 0.017	0.834	0.0016	0.084	0.806
friction_shift	CVaR guard	0.823 +/- 0.018	0.831	0.0013	0.064	0.806
friction_shift	Half-life v5	0.818 +/- 0.012	0.824	0.0028	0.172	0.805
friction_shift	Oracle	0.835 +/- 0.011	0.840	0.0028	0.000	0.821
nominal_slow_drift	Frozen BC	0.437 +/- 0.022	0.234	0.2175	0.337	-0.005
nominal_slow_drift	Domain rand.	0.492 +/- 0.013	0.305	0.1675	0.322	0.101
nominal_slow_drift	Fixed refresh	0.718 +/- 0.010	0.690	0.0147	0.138	0.698
nominal_slow_drift	Online FT	0.569 +/- 0.013	0.381	0.0894	0.130	0.302
nominal_slow_drift	Scalar gate	0.759 +/- 0.010	0.734	0.0063	0.050	0.747
nominal_slow_drift	Conformal	0.799 +/- 0.013	0.782	0.0025	0.037	0.789
nominal_slow_drift	Ensemble	0.816 +/- 0.012	0.821	0.0037	0.059	0.804
nominal_slow_drift	Bayes survival	0.828 +/- 0.008	0.816	0.0022	0.091	0.817
nominal_slow_drift	CVaR guard	0.819 +/- 0.012	0.816	0.0022	0.063	0.807
nominal_slow_drift	Half-life v5	0.826 +/- 0.007	0.821	0.0013	0.063	0.816
nominal_slow_drift	Oracle	0.857 +/- 0.014	0.846	0.0022	0.000	0.847
payload_mass_shift	Frozen BC	0.383 +/- 0.020	0.179	0.3400	0.721	-0.201
payload_mass_shift	Domain rand.	0.425 +/- 0.011	0.180	0.2906	0.500	-0.109
payload_mass_shift	Fixed refresh	0.677 +/- 0.019	0.618	0.0272	0.247	0.611
payload_mass_shift	Online FT	0.500 +/- 0.014	0.258	0.1791	0.229	0.114
payload_mass_shift	Scalar gate	0.745 +/- 0.011	0.689	0.0097	0.104	0.728
payload_mass_shift	Conformal	0.792 +/- 0.011	0.766	0.0028	0.048	0.780
payload_mass_shift	Ensemble	0.789 +/- 0.014	0.769	0.0022	0.063	0.777
payload_mass_shift	Bayes survival	0.825 +/- 0.017	0.819	0.0016	0.090	0.812
payload_mass_shift	CVaR guard	0.808 +/- 0.012	0.828	0.0025	0.067	0.792
payload_mass_shift	Half-life v5	0.811 +/- 0.018	0.802	0.0031	0.155	0.798
payload_mass_shift	Oracle	0.841 +/- 0.013	0.845	0.0016	0.000	0.829
probe_cost_shift	Frozen BC	0.367 +/- 0.010	0.156	0.3659	0.775	-0.243
probe_cost_shift	Domain rand.	0.426 +/- 0.023	0.206	0.2906	0.519	-0.108
probe_cost_shift	Fixed refresh	0.673 +/- 0.017	0.630	0.0275	0.294	0.601

Table 7: Per-split metrics for all methods. (continued)

Split	Method	Success	Late	Unsafe	HLE	Safety utility
probe_cost_shift	Online FT	0.523 +/- 0.016	0.366	0.1547	0.237	0.175
probe_cost_shift	Scalar gate	0.776 +/- 0.014	0.765	0.0069	0.113	0.757
probe_cost_shift	Conformal	0.801 +/- 0.009	0.786	0.0031	0.064	0.781
probe_cost_shift	Ensemble	0.796 +/- 0.012	0.782	0.0016	0.077	0.777
probe_cost_shift	Bayes survival	0.812 +/- 0.011	0.791	0.0031	0.114	0.790
probe_cost_shift	CVaR guard	0.830 +/- 0.011	0.830	0.0006	0.086	0.805
probe_cost_shift	Half-life v5	0.806 +/- 0.009	0.794	0.0019	0.100	0.786
probe_cost_shift	Oracle	0.834 +/- 0.011	0.823	0.0009	0.000	0.815
sensor_noise_shift	Frozen BC	0.397 +/- 0.013	0.175	0.3116	0.647	-0.154
sensor_noise_shift	Domain rand.	0.428 +/- 0.014	0.206	0.2569	0.438	-0.068
sensor_noise_shift	Fixed refresh	0.693 +/- 0.016	0.659	0.0272	0.227	0.632
sensor_noise_shift	Online FT	0.538 +/- 0.011	0.360	0.1400	0.185	0.213
sensor_noise_shift	Scalar gate	0.781 +/- 0.016	0.772	0.0050	0.099	0.768
sensor_noise_shift	Conformal	0.808 +/- 0.011	0.792	0.0025	0.103	0.796
sensor_noise_shift	Ensemble	0.804 +/- 0.008	0.799	0.0022	0.113	0.791
sensor_noise_shift	Bayes survival	0.825 +/- 0.013	0.823	0.0022	0.206	0.811
sensor_noise_shift	CVaR guard	0.839 +/- 0.014	0.851	0.0009	0.151	0.821
sensor_noise_shift	Half-life v5	0.826 +/- 0.013	0.796	0.0019	0.105	0.814
sensor_noise_shift	Oracle	0.835 +/- 0.014	0.844	0.0013	0.000	0.823

B FULL HARD-AGGREGATE PAIRWISE TABLE

Table 8: All hard-regime aggregate paired seed differences for Half-life v5.

Reference	Metric	Mean diff	CI95	Lower95	Better seeds
Bayes survival	goal_success	-0.00505	0.00502	-0.01007	3/10
Bayes survival	late_success	-0.00911	0.01010	-0.01920	2/10
Bayes survival	unsafe_failure	0.00071	0.00109	-0.00037	3/10
Bayes survival	stale_execution	0.00000	0.00000	0.00000	0/10
Bayes survival	total_cost	-0.00925	0.00104	-0.01029	10/10
Bayes survival	half_life_error	0.01085	0.00204	0.00881	0/10
Bayes survival	calibration_error	-0.00127	0.00303	-0.00430	4/10
Bayes survival	safety_utility	-0.00438	0.00540	-0.00978	3/10
Conformal	goal_success	0.01714	0.00612	0.01102	10/10
Conformal	late_success	0.02679	0.01566	0.01113	9/10
Conformal	unsafe_failure	-0.00063	0.00139	-0.00201	6/10
Conformal	stale_execution	0.00000	0.00000	0.00000	0/10
Conformal	total_cost	0.00009	0.00128	-0.00119	5/10
Conformal	half_life_error	0.06166	0.00192	0.05973	0/10
Conformal	calibration_error	-0.02147	0.00365	-0.02512	10/10
Conformal	safety_utility	0.01768	0.00625	0.01143	10/10
CVaR guard	goal_success	-0.01183	0.00345	-0.01528	0/10
CVaR guard	late_success	-0.02535	0.00707	-0.03243	0/10
CVaR guard	unsafe_failure	0.00125	0.00066	0.00059	0/10
CVaR guard	stale_execution	0.00000	0.00000	0.00000	0/10
CVaR guard	total_cost	-0.04441	0.00080	-0.04521	10/10
CVaR guard	half_life_error	0.03993	0.00225	0.03769	0/10
CVaR guard	calibration_error	0.00740	0.00217	0.00523	0/10
CVaR guard	safety_utility	-0.00671	0.00370	-0.01041	1/10
Domain rand.	goal_success	0.40692	0.00817	0.39875	10/10
Domain rand.	late_success	0.61911	0.00964	0.60947	10/10
Domain rand.	unsafe_failure	-0.31156	0.00352	-0.31509	10/10
Domain rand.	stale_execution	-0.72536	0.00489	-0.73025	10/10
Domain rand.	total_cost	-0.17916	0.00292	-0.18208	10/10
Domain rand.	half_life_error	-0.41491	0.00207	-0.41698	10/10
Domain rand.	calibration_error	-0.11012	0.00295	-0.11307	10/10
Domain rand.	safety_utility	0.95280	0.01101	0.94179	10/10
Ensemble	goal_success	0.01326	0.00486	0.00840	10/10
Ensemble	late_success	0.01339	0.01197	0.00142	6/10
Ensemble	unsafe_failure	0.00018	0.00087	-0.00069	4/10
Ensemble	stale_execution	0.00000	0.00000	0.00000	0/10
Ensemble	total_cost	-0.00106	0.00080	-0.00186	8/10
Ensemble	half_life_error	0.04972	0.00137	0.04834	0/10
Ensemble	calibration_error	-0.01647	0.00282	-0.01928	10/10
Ensemble	safety_utility	0.01325	0.00491	0.00834	10/10
Fixed refresh	goal_success	0.15339	0.00567	0.14772	10/10
Fixed refresh	late_success	0.19286	0.00815	0.18471	10/10
Fixed refresh	unsafe_failure	-0.03295	0.00317	-0.03611	10/10
Fixed refresh	stale_execution	-0.14942	0.00260	-0.15201	10/10

Table 8: All hard-regime aggregate paired seed differences for Half-life v5. (continued)

Reference	Metric	Mean diff	CI95	Lower95	Better seeds
Fixed refresh	total_cost	0.04548	0.00273	0.04274	0/10
Fixed refresh	half_life.error	-0.21552	0.00660	-0.22212	10/10
Fixed refresh	calibration.error	-0.10208	0.00220	-0.10427	10/10
Fixed refresh	safety_utility	0.22682	0.00687	0.21995	10/10
Frozen BC	goal_success	0.45156	0.00737	0.44419	10/10
Frozen BC	late_success	0.65464	0.01066	0.64399	10/10
Frozen BC	unsafe_failure	-0.36920	0.00476	-0.37395	10/10
Frozen BC	stale_execution	-0.72536	0.00489	-0.73025	10/10
Frozen BC	total_cost	-0.22815	0.00399	-0.23214	10/10
Frozen BC	half_life.error	-0.73150	0.00328	-0.73478	10/10
Frozen BC	calibration.error	-0.12473	0.00314	-0.12787	10/10
Frozen BC	safety_utility	1.05502	0.01068	1.04434	10/10
Hazard reg.	goal_success	0.03094	0.00605	0.02489	10/10
Hazard reg.	late_success	0.03053	0.01607	0.01447	9/10
Hazard reg.	unsafe_failure	-0.00246	0.00132	-0.00377	9/10
Hazard reg.	stale_execution	0.00000	0.00000	0.00000	0/10
Hazard reg.	total_cost	0.04858	0.00119	0.04739	0/10
Hazard reg.	half_life.error	-0.05588	0.00254	-0.05842	10/10
Hazard reg.	calibration.error	-0.02254	0.00328	-0.02582	10/10
Hazard reg.	safety_utility	0.02630	0.00652	0.01978	10/10
Online FT	goal_success	0.27549	0.00630	0.26919	10/10
Online FT	late_success	0.39143	0.02336	0.36807	10/10
Online FT	unsafe_failure	-0.13705	0.00494	-0.14199	10/10
Online FT	stale_execution	-0.50964	0.01277	-0.52242	10/10
Online FT	total_cost	-0.04135	0.00425	-0.04560	10/10
Online FT	half_life.error	-0.11973	0.00156	-0.12129	10/10
Online FT	calibration.error	-0.12545	0.00271	-0.12816	10/10
Online FT	safety_utility	0.57517	0.01246	0.56271	10/10
Oracle	goal_success	-0.01955	0.00626	-0.02581	0/10
Oracle	late_success	-0.02661	0.00867	-0.03528	0/10
Oracle	unsafe_failure	0.00120	0.00080	0.00040	1/10
Oracle	stale_execution	0.00000	0.00000	0.00000	0/10
Oracle	total_cost	-0.00478	0.00069	-0.00547	10/10
Oracle	half_life.error	0.13000	0.00147	0.12853	0/10
Oracle	calibration.error	0.02042	0.00349	0.01693	0/10
Oracle	safety_utility	-0.01995	0.00638	-0.02633	0/10
Scalar gate	goal_success	0.05781	0.00648	0.05133	10/10
Scalar gate	late_success	0.06375	0.00678	0.05697	10/10
Scalar gate	unsafe_failure	-0.00468	0.00120	-0.00588	10/10
Scalar gate	stale_execution	0.00000	0.00000	0.00000	0/10
Scalar gate	total_cost	0.02194	0.00122	0.02073	0/10
Scalar gate	half_life.error	0.00811	0.00139	0.00672	0/10
Scalar gate	calibration.error	-0.05145	0.00334	-0.05479	10/10
Scalar gate	safety_utility	0.05886	0.00606	0.05281	10/10

C FULL STRESS SWEEP ACROSS ALL AXES

Table 9: All stress axes, levels, and methods.

Axis	Level	Method	Success	CI95	Unsafe	Stale	Safety utility
drift_rate	0.00	Domain rand.	0.53687	0.02299	0.17250	0.56938	0.17096
drift_rate	0.00	Conformal	0.78812	0.02110	0.00188	0.00000	0.77359
drift_rate	0.00	Ensemble	0.78812	0.02133	0.00313	0.00000	0.77195
drift_rate	0.00	Bayes survival	0.81375	0.00770	0.00313	0.00000	0.79600
drift_rate	0.00	CVaR guard	0.82625	0.01896	0.00125	0.00000	0.80325
drift_rate	0.00	Half-life v5	0.80813	0.01484	0.00313	0.00000	0.79232
drift_rate	0.00	Oracle	0.83375	0.01541	0.00313	0.00000	0.81686
drift_rate	0.25	Domain rand.	0.53000	0.02301	0.19875	0.57188	0.13701
drift_rate	0.25	Conformal	0.78375	0.01440	0.00500	0.00000	0.76582
drift_rate	0.25	Ensemble	0.82000	0.02594	0.00313	0.00000	0.80375
drift_rate	0.25	Bayes survival	0.81500	0.01214	0.00250	0.00000	0.79767
drift_rate	0.25	CVaR guard	0.82562	0.03096	0.00063	0.00000	0.80242
drift_rate	0.25	Half-life v5	0.79312	0.02299	0.00313	0.00000	0.77720
drift_rate	0.25	Oracle	0.83625	0.01526	0.00125	0.00000	0.82102
drift_rate	0.50	Domain rand.	0.50313	0.02027	0.20375	0.57500	0.10408
drift_rate	0.50	Conformal	0.79688	0.01706	0.00125	0.00000	0.78261
drift_rate	0.50	Ensemble	0.80188	0.01403	0.00188	0.00000	0.78667
drift_rate	0.50	Bayes survival	0.82812	0.01676	0.00125	0.00000	0.81185
drift_rate	0.50	CVaR guard	0.82500	0.01703	0.00000	0.00000	0.80209

Table 9: All stress axes, levels, and methods. (continued)

Axis	Level	Method	Success	CI95	Unsafe	Stale	Safety utility
drift_rate	0.50	Half-life v5	0.80688	0.02426	0.00250	0.00000	0.79141
drift_rate	0.50	Oracle	0.83063	0.01678	0.00125	0.00000	0.81529
drift_rate	0.75	Domain rand.	0.52000	0.02213	0.20312	0.57750	0.12073
drift_rate	0.75	Conformal	0.80688	0.01928	0.00250	0.00000	0.79113
drift_rate	0.75	Ensemble	0.80563	0.02349	0.00250	0.00000	0.78965
drift_rate	0.75	Bayes survival	0.81438	0.01654	0.00250	0.00000	0.79668
drift_rate	0.75	CVaR guard	0.81937	0.01617	0.00125	0.00000	0.79457
drift_rate	0.75	Half-life v5	0.81563	0.02019	0.00313	0.00000	0.79945
drift_rate	0.75	Oracle	0.83063	0.01628	0.00063	0.00000	0.81582
drift_rate	1.00	Domain rand.	0.50062	0.02446	0.21938	0.58500	0.08257
drift_rate	1.00	Conformal	0.79625	0.01934	0.00437	0.00000	0.77828
drift_rate	1.00	Ensemble	0.80187	0.01050	0.00375	0.00000	0.78442
drift_rate	1.00	Bayes survival	0.81563	0.02084	0.00313	0.00000	0.79701
drift_rate	1.00	CVaR guard	0.83813	0.02086	0.00188	0.00000	0.81215
drift_rate	1.00	Half-life v5	0.81125	0.02144	0.00250	0.00000	0.79559
drift_rate	1.00	Oracle	0.83625	0.01852	0.00063	0.00000	0.82125
drift_rate	1.25	Domain rand.	0.49562	0.01827	0.22437	0.59062	0.07066
drift_rate	1.25	Conformal	0.78750	0.03189	0.00813	0.00000	0.76585
drift_rate	1.25	Ensemble	0.80563	0.01848	0.00500	0.00000	0.78681
drift_rate	1.25	Bayes survival	0.81687	0.01528	0.00063	0.00000	0.80065
drift_rate	1.25	CVaR guard	0.83500	0.01763	0.00063	0.00000	0.80967
drift_rate	1.25	Half-life v5	0.79438	0.01688	0.00375	0.00000	0.77729
drift_rate	1.25	Oracle	0.83500	0.02789	0.00250	0.00000	0.81799
drift_rate	1.50	Domain rand.	0.50062	0.02156	0.22437	0.59813	0.07311
drift_rate	1.50	Conformal	0.78562	0.01664	0.00438	0.00000	0.76735
drift_rate	1.50	Ensemble	0.79063	0.00899	0.00375	0.00000	0.77279
drift_rate	1.50	Bayes survival	0.82937	0.01292	0.00125	0.00000	0.81231
drift_rate	1.50	CVaR guard	0.83625	0.01504	0.00188	0.00000	0.80935
drift_rate	1.50	Half-life v5	0.79312	0.01408	0.00250	0.00000	0.77710
drift_rate	1.50	Oracle	0.83187	0.02624	0.00063	0.00000	0.81657
sensor_noise	0.00	Domain rand.	0.50000	0.01539	0.19500	0.57125	0.11097
sensor_noise	0.00	Conformal	0.79375	0.01732	0.00625	0.00000	0.77507
sensor_noise	0.00	Ensemble	0.80563	0.01586	0.00375	0.00000	0.78917
sensor_noise	0.00	Bayes survival	0.81875	0.02295	0.00375	0.00000	0.80108
sensor_noise	0.00	CVaR guard	0.82125	0.01114	0.00000	0.00000	0.80251
sensor_noise	0.00	Half-life v5	0.81063	0.02596	0.00250	0.00000	0.79556
sensor_noise	0.00	Oracle	0.83625	0.01289	0.00125	0.00000	0.82116
sensor_noise	0.25	Domain rand.	0.47313	0.02332	0.21500	0.56875	0.06496
sensor_noise	0.25	Conformal	0.78812	0.03218	0.00125	0.00000	0.77413
sensor_noise	0.25	Ensemble	0.80563	0.02674	0.00188	0.00000	0.79080
sensor_noise	0.25	Bayes survival	0.82188	0.01819	0.00313	0.00000	0.80417
sensor_noise	0.25	CVaR guard	0.83813	0.01928	0.00188	0.00000	0.81496
sensor_noise	0.25	Half-life v5	0.81563	0.02458	0.00000	0.00000	0.80287
sensor_noise	0.25	Oracle	0.83125	0.01403	0.00063	0.00000	0.81669
sensor_noise	0.50	Domain rand.	0.52125	0.02996	0.19813	0.57188	0.12889
sensor_noise	0.50	Conformal	0.79000	0.02147	0.00250	0.00000	0.77417
sensor_noise	0.50	Ensemble	0.79250	0.01263	0.00125	0.00000	0.77776
sensor_noise	0.50	Bayes survival	0.82625	0.01974	0.00438	0.00000	0.80642
sensor_noise	0.50	CVaR guard	0.83875	0.01931	0.00063	0.00000	0.81384
sensor_noise	0.50	Half-life v5	0.81688	0.02001	0.00187	0.00000	0.80194
sensor_noise	0.50	Oracle	0.82375	0.02898	0.00188	0.00000	0.80771
sensor_noise	0.75	Domain rand.	0.50438	0.01415	0.19875	0.57062	0.11181
sensor_noise	0.75	Conformal	0.79937	0.01987	0.00437	0.00000	0.78100
sensor_noise	0.75	Ensemble	0.80312	0.01441	0.00188	0.00000	0.78731
sensor_noise	0.75	Bayes survival	0.84312	0.01271	0.00063	0.00000	0.82618
sensor_noise	0.75	CVaR guard	0.85563	0.00957	0.00188	0.00000	0.82640
sensor_noise	0.75	Half-life v5	0.81125	0.01778	0.00125	0.00000	0.79665
sensor_noise	0.75	Oracle	0.82438	0.01617	0.00188	0.00000	0.80839
sensor_noise	1.00	Domain rand.	0.50313	0.02376	0.20875	0.57312	0.09972
sensor_noise	1.00	Conformal	0.79062	0.02283	0.00250	0.00000	0.77317
sensor_noise	1.00	Ensemble	0.82000	0.01652	0.00250	0.00000	0.80277
sensor_noise	1.00	Bayes survival	0.82687	0.01582	0.00125	0.00000	0.80812
sensor_noise	1.00	CVaR guard	0.85563	0.01360	0.00063	0.00000	0.82489
sensor_noise	1.00	Half-life v5	0.81687	0.01924	0.00250	0.00000	0.80041
sensor_noise	1.00	Oracle	0.83750	0.01461	0.00125	0.00000	0.82189
sensor_noise	1.25	Domain rand.	0.51812	0.02896	0.20000	0.57812	0.12176
sensor_noise	1.25	Conformal	0.80000	0.01984	0.00313	0.00000	0.78091
sensor_noise	1.25	Ensemble	0.81875	0.02281	0.00125	0.00000	0.80191
sensor_noise	1.25	Bayes survival	0.83938	0.01714	0.00000	0.00000	0.82048
sensor_noise	1.25	CVaR guard	0.85437	0.01762	0.00063	0.00000	0.82109
sensor_noise	1.25	Half-life v5	0.81000	0.01819	0.00313	0.00000	0.79265
sensor_noise	1.25	Oracle	0.85062	0.01793	0.00125	0.00000	0.83501
sensor_noise	1.50	Domain rand.	0.50188	0.02090	0.21937	0.57625	0.08680
sensor_noise	1.50	Conformal	0.81812	0.01962	0.00125	0.00000	0.79960
sensor_noise	1.50	Ensemble	0.81063	0.01817	0.00313	0.00000	0.79078
sensor_noise	1.50	Bayes survival	0.84187	0.01305	0.00063	0.00000	0.82088
sensor_noise	1.50	CVaR guard	0.86812	0.01511	0.00063	0.00000	0.83327

Table 9: All stress axes, levels, and methods. (continued)

Axis	Level	Method	Success	CI95	Unsafe	Stale	Safety utility
sensor_noise	1.50	Half-life v5	0.83000	0.01672	0.00188	0.00000	0.81373
sensor_noise	1.50	Oracle	0.83813	0.01857	0.00125	0.00000	0.82237
mode_chatter	0.00	Domain rand.	0.51312	0.03069	0.18562	0.56375	0.13601
mode_chatter	0.00	Conformal	0.80813	0.01827	0.00250	0.00000	0.79287
mode_chatter	0.00	Ensemble	0.78375	0.01519	0.00437	0.00000	0.76637
mode_chatter	0.00	Bayes survival	0.82375	0.01493	0.00438	0.00000	0.80472
mode_chatter	0.00	CVaR guard	0.84188	0.02001	0.00063	0.00000	0.81907
mode_chatter	0.00	Half-life v5	0.80437	0.01592	0.00250	0.00000	0.78927
mode_chatter	0.00	Oracle	0.84750	0.02060	0.00125	0.00000	0.83261
mode_chatter	0.25	Domain rand.	0.51813	0.02180	0.20875	0.57375	0.11451
mode_chatter	0.25	Conformal	0.79125	0.02208	0.00313	0.00000	0.77536
mode_chatter	0.25	Ensemble	0.80062	0.02335	0.00375	0.00000	0.78373
mode_chatter	0.25	Bayes survival	0.82000	0.02024	0.00063	0.00000	0.80460
mode_chatter	0.25	CVaR guard	0.83437	0.02084	0.00063	0.00000	0.81103
mode_chatter	0.25	Half-life v5	0.80563	0.01875	0.00250	0.00000	0.79031
mode_chatter	0.25	Oracle	0.83813	0.02053	0.00063	0.00000	0.82359
mode_chatter	0.50	Domain rand.	0.49125	0.01656	0.22438	0.58437	0.06841
mode_chatter	0.50	Conformal	0.79187	0.02381	0.00313	0.00000	0.77545
mode_chatter	0.50	Ensemble	0.79000	0.01900	0.00375	0.00000	0.77278
mode_chatter	0.50	Bayes survival	0.81313	0.02217	0.00375	0.00000	0.79413
mode_chatter	0.50	CVaR guard	0.83938	0.02653	0.00125	0.00000	0.81467
mode_chatter	0.50	Half-life v5	0.81938	0.02133	0.00500	0.00000	0.80118
mode_chatter	0.50	Oracle	0.85125	0.01493	0.00125	0.00000	0.83570
mode_chatter	0.75	Domain rand.	0.49813	0.02259	0.22812	0.58938	0.06984
mode_chatter	0.75	Conformal	0.78125	0.01889	0.00750	0.00000	0.76033
mode_chatter	0.75	Ensemble	0.81125	0.02024	0.00500	0.00000	0.79258
mode_chatter	0.75	Bayes survival	0.81250	0.02251	0.00375	0.00000	0.79324
mode_chatter	0.75	CVaR guard	0.83313	0.02640	0.00000	0.00000	0.80896
mode_chatter	0.75	Half-life v5	0.79562	0.01317	0.00437	0.00000	0.77794
mode_chatter	0.75	Oracle	0.83187	0.01658	0.00063	0.00000	0.81667
mode_chatter	1.00	Domain rand.	0.48938	0.01292	0.24438	0.59125	0.04422
mode_chatter	1.00	Conformal	0.78000	0.01682	0.00313	0.00000	0.76307
mode_chatter	1.00	Ensemble	0.79250	0.02007	0.00437	0.00000	0.77406
mode_chatter	1.00	Bayes survival	0.80937	0.01666	0.00313	0.00000	0.79029
mode_chatter	1.00	CVaR guard	0.80625	0.02339	0.00125	0.00000	0.78040
mode_chatter	1.00	Half-life v5	0.80250	0.01099	0.00250	0.00000	0.78633
mode_chatter	1.00	Oracle	0.83313	0.01694	0.00063	0.00000	0.81770
mode_chatter	1.25	Domain rand.	0.46625	0.01486	0.25187	0.60000	0.01063
mode_chatter	1.25	Conformal	0.79187	0.02026	0.00313	0.00000	0.77462
mode_chatter	1.25	Ensemble	0.78875	0.01314	0.00688	0.00000	0.76755
mode_chatter	1.25	Bayes survival	0.81625	0.01846	0.00125	0.00000	0.79885
mode_chatter	1.25	CVaR guard	0.83000	0.01662	0.00125	0.00000	0.80367
mode_chatter	1.25	Half-life v5	0.81063	0.01253	0.00250	0.00000	0.79421
mode_chatter	1.25	Oracle	0.83000	0.01352	0.00188	0.00000	0.81303
mode_chatter	1.50	Domain rand.	0.47375	0.02040	0.26625	0.60375	0.00249
mode_chatter	1.50	Conformal	0.78875	0.02220	0.00437	0.00000	0.76983
mode_chatter	1.50	Ensemble	0.77812	0.02092	0.00625	0.00000	0.75714
mode_chatter	1.50	Bayes survival	0.80812	0.01112	0.00313	0.00000	0.78828
mode_chatter	1.50	CVaR guard	0.81875	0.01291	0.00187	0.00000	0.79103
mode_chatter	1.50	Half-life v5	0.80187	0.01959	0.00188	0.00000	0.78587
mode_chatter	1.50	Oracle	0.82500	0.02339	0.00125	0.00000	0.80841
probe_cost	0.00	Domain rand.	0.50062	0.03186	0.20875	0.57125	0.09786
probe_cost	0.00	Conformal	0.79063	0.01541	0.00625	0.00000	0.77578
probe_cost	0.00	Ensemble	0.80813	0.02138	0.00562	0.00000	0.79386
probe_cost	0.00	Bayes survival	0.82188	0.01115	0.00125	0.00000	0.81059
probe_cost	0.00	CVaR guard	0.82625	0.02476	0.00125	0.00000	0.80689
probe_cost	0.00	Half-life v5	0.81438	0.02647	0.00562	0.00000	0.80049
probe_cost	0.00	Oracle	0.84125	0.02051	0.00063	0.00000	0.83116
probe_cost	0.25	Domain rand.	0.50187	0.02576	0.20938	0.57000	0.09891
probe_cost	0.25	Conformal	0.80500	0.02689	0.00250	0.00000	0.79065
probe_cost	0.25	Ensemble	0.80250	0.01369	0.00437	0.00000	0.78603
probe_cost	0.25	Bayes survival	0.81687	0.01212	0.00250	0.00000	0.80053
probe_cost	0.25	CVaR guard	0.83938	0.01924	0.00125	0.00000	0.81622
probe_cost	0.25	Half-life v5	0.80437	0.02199	0.00375	0.00000	0.78889
probe_cost	0.25	Oracle	0.83000	0.01400	0.00125	0.00000	0.81591
probe_cost	0.50	Domain rand.	0.49250	0.02205	0.19750	0.56875	0.10182
probe_cost	0.50	Conformal	0.79125	0.02603	0.00500	0.00000	0.77103
probe_cost	0.50	Ensemble	0.81187	0.01793	0.00250	0.00000	0.79372
probe_cost	0.50	Bayes survival	0.82938	0.02288	0.00313	0.00000	0.80862
probe_cost	0.50	CVaR guard	0.83813	0.01775	0.00063	0.00000	0.81191
probe_cost	0.50	Half-life v5	0.80750	0.01447	0.00688	0.00000	0.78550
probe_cost	0.50	Oracle	0.81812	0.01040	0.00125	0.00000	0.80055
probe_cost	0.75	Domain rand.	0.52375	0.02949	0.20125	0.57063	0.12869
probe_cost	0.75	Conformal	0.79563	0.02697	0.00375	0.00000	0.77329
probe_cost	0.75	Ensemble	0.81438	0.01933	0.00500	0.00000	0.79045
probe_cost	0.75	Bayes survival	0.81125	0.01672	0.00688	0.00000	0.78297
probe_cost	0.75	CVaR guard	0.84125	0.01405	0.00188	0.00000	0.81021

Table 9: All stress axes, levels, and methods. (continued)

Axis	Level	Method	Success	CI95	Unsafe	Stale	Safety utility
probe_cost	0.75	Half-life v5	0.81312	0.01848	0.00188	0.00000	0.79253
probe_cost	0.75	Oracle	0.83125	0.02082	0.00188	0.00000	0.80975
probe_cost	1.00	Domain rand.	0.51687	0.02192	0.20312	0.57125	0.11973
probe_cost	1.00	Conformal	0.80000	0.01975	0.00250	0.00000	0.77556
probe_cost	1.00	Ensemble	0.80000	0.02665	0.00375	0.00000	0.77390
probe_cost	1.00	Bayes survival	0.82312	0.01198	0.00000	0.00000	0.79812
probe_cost	1.00	CVaR guard	0.84688	0.01952	0.00125	0.00000	0.81302
probe_cost	1.00	Half-life v5	0.79875	0.01701	0.00250	0.00000	0.77410
probe_cost	1.00	Oracle	0.83500	0.01381	0.00188	0.00000	0.81017
probe_cost	1.25	Domain rand.	0.50562	0.01902	0.21688	0.57125	0.09474
probe_cost	1.25	Conformal	0.81500	0.01615	0.00313	0.00000	0.78624
probe_cost	1.25	Ensemble	0.80937	0.01917	0.00313	0.00000	0.77986
probe_cost	1.25	Bayes survival	0.83188	0.01544	0.00313	0.00000	0.79959
probe_cost	1.25	CVaR guard	0.83437	0.01666	0.00188	0.00000	0.79685
probe_cost	1.25	Half-life v5	0.81312	0.01954	0.00313	0.00000	0.78393
probe_cost	1.25	Oracle	0.84000	0.01475	0.00125	0.00000	0.81257
probe_cost	1.50	Domain rand.	0.50625	0.02627	0.18625	0.57188	0.12575
probe_cost	1.50	Conformal	0.77563	0.02156	0.00500	0.00000	0.74222
probe_cost	1.50	Ensemble	0.78937	0.01379	0.00500	0.00000	0.75506
probe_cost	1.50	Bayes survival	0.83250	0.01701	0.00437	0.00000	0.79503
probe_cost	1.50	CVaR guard	0.83688	0.01893	0.00125	0.00000	0.79677
probe_cost	1.50	Half-life v5	0.80875	0.02728	0.00375	0.00000	0.77588
probe_cost	1.50	Oracle	0.84063	0.02027	0.00125	0.00000	0.80971
combined	0.00	Domain rand.	0.60250	0.02347	0.04812	0.38250	0.42437
combined	0.00	Conformal	0.80250	0.01994	0.00250	0.00000	0.79336
combined	0.00	Ensemble	0.80875	0.02178	0.00188	0.00000	0.80022
combined	0.00	Bayes survival	0.82687	0.01933	0.00063	0.00000	0.81915
combined	0.00	CVaR guard	0.81188	0.02247	0.00188	0.00000	0.80186
combined	0.00	Half-life v5	0.82688	0.01592	0.00250	0.00000	0.81768
combined	0.00	Oracle	0.85375	0.01452	0.00250	0.00000	0.84434
combined	0.25	Domain rand.	0.59687	0.02290	0.09125	0.47687	0.34358
combined	0.25	Conformal	0.81000	0.01646	0.00188	0.00000	0.79822
combined	0.25	Ensemble	0.81563	0.02170	0.00250	0.00000	0.80288
combined	0.25	Bayes survival	0.81500	0.02027	0.00125	0.00000	0.80286
combined	0.25	CVaR guard	0.81187	0.01335	0.00188	0.00000	0.79552
combined	0.25	Half-life v5	0.83500	0.02917	0.00250	0.00000	0.82238
combined	0.25	Oracle	0.83875	0.02575	0.00063	0.00000	0.82759
combined	0.50	Domain rand.	0.51250	0.02168	0.17250	0.53875	0.15700
combined	0.50	Conformal	0.80438	0.01391	0.00313	0.00000	0.78808
combined	0.50	Ensemble	0.80188	0.01450	0.00563	0.00000	0.78270
combined	0.50	Bayes survival	0.82063	0.00967	0.00000	0.00000	0.80547
combined	0.50	CVaR guard	0.83000	0.02272	0.00125	0.00000	0.80696
combined	0.50	Half-life v5	0.80875	0.01319	0.00500	0.00000	0.79035
combined	0.50	Oracle	0.84000	0.01172	0.00063	0.00000	0.82517
combined	0.75	Domain rand.	0.50125	0.02966	0.23312	0.58688	0.06882
combined	0.75	Conformal	0.79750	0.01486	0.00188	0.00000	0.77803
combined	0.75	Ensemble	0.79875	0.01621	0.00500	0.00000	0.77595
combined	0.75	Bayes survival	0.82750	0.01791	0.00063	0.00000	0.80665
combined	0.75	CVaR guard	0.85250	0.01129	0.00125	0.00000	0.82195
combined	0.75	Half-life v5	0.80000	0.01252	0.00188	0.00000	0.78096
combined	0.75	Oracle	0.84562	0.02491	0.00063	0.00000	0.82705
combined	1.00	Domain rand.	0.47000	0.02080	0.28438	0.61938	-0.02468
combined	1.00	Conformal	0.79688	0.02201	0.00375	0.00000	0.77052
combined	1.00	Ensemble	0.78438	0.02275	0.00375	0.00000	0.75796
combined	1.00	Bayes survival	0.84125	0.01753	0.00250	0.00000	0.81325
combined	1.00	CVaR guard	0.85687	0.01511	0.00125	0.00000	0.82211
combined	1.00	Half-life v5	0.80438	0.01933	0.00375	0.00000	0.77956
combined	1.00	Oracle	0.82500	0.02497	0.00125	0.00000	0.80219
combined	1.25	Domain rand.	0.44938	0.00974	0.32312	0.64750	-0.09357
combined	1.25	Conformal	0.79500	0.01833	0.00375	0.00000	0.76343
combined	1.25	Ensemble	0.81500	0.01772	0.00313	0.00000	0.78415
combined	1.25	Bayes survival	0.83125	0.01664	0.00063	0.00000	0.80053
combined	1.25	CVaR guard	0.83375	0.02060	0.00125	0.00000	0.79710
combined	1.25	Half-life v5	0.78625	0.02613	0.00625	0.00000	0.75498
combined	1.25	Oracle	0.82937	0.01473	0.00000	0.00000	0.80382
combined	1.50	Domain rand.	0.39750	0.02282	0.37750	0.66375	-0.20530
combined	1.50	Conformal	0.81250	0.01528	0.00313	0.00000	0.77762
combined	1.50	Ensemble	0.81000	0.01186	0.00250	0.00000	0.77617
combined	1.50	Bayes survival	0.83875	0.01778	0.00063	0.00000	0.80450
combined	1.50	CVaR guard	0.82188	0.01320	0.00125	0.00000	0.78441
combined	1.50	Half-life v5	0.79187	0.01836	0.00688	0.00000	0.75629
combined	1.50	Oracle	0.82625	0.00980	0.00063	0.00000	0.79672

D STRESS SEED-LEVEL METRICS

Table 10: Seed-level stress metrics.

Axis	Level	Method	Seed	Success	Unsafe	Stale	Safety utility
drift_rate	0.00	Domain rand.	0	0.53750	0.20000	0.55625	0.14857
drift_rate	0.00	Domain rand.	1	0.56250	0.19375	0.55625	0.17982
drift_rate	0.00	Domain rand.	2	0.50625	0.13125	0.56875	0.18176
drift_rate	0.00	Domain rand.	3	0.51250	0.18125	0.57500	0.13593
drift_rate	0.00	Domain rand.	4	0.55000	0.16875	0.58750	0.18167
drift_rate	0.00	Domain rand.	5	0.46875	0.20625	0.58750	0.06296
drift_rate	0.00	Domain rand.	6	0.56875	0.13125	0.58125	0.24001
drift_rate	0.00	Domain rand.	7	0.57500	0.16250	0.56875	0.21929
drift_rate	0.00	Domain rand.	8	0.58125	0.15625	0.55625	0.23603
drift_rate	0.00	Domain rand.	9	0.50625	0.19375	0.55625	0.12357
drift_rate	0.00	Conformal	0	0.84375	0.00000	0.00000	0.83104
drift_rate	0.00	Conformal	1	0.80625	0.00000	0.00000	0.79354
drift_rate	0.00	Conformal	2	0.77500	0.00000	0.00000	0.76227
drift_rate	0.00	Conformal	3	0.80625	0.00000	0.00000	0.79383
drift_rate	0.00	Conformal	4	0.76250	0.00000	0.00000	0.75002
drift_rate	0.00	Conformal	5	0.73125	0.00000	0.00000	0.71879
drift_rate	0.00	Conformal	6	0.83125	0.00000	0.00000	0.81848
drift_rate	0.00	Conformal	7	0.78750	0.00625	0.00000	0.76815
drift_rate	0.00	Conformal	8	0.76875	0.00625	0.00000	0.74959
drift_rate	0.00	Conformal	9	0.76875	0.00625	0.00000	0.75019
drift_rate	0.00	Ensemble	0	0.83750	0.00625	0.00000	0.81821
drift_rate	0.00	Ensemble	1	0.76875	0.00625	0.00000	0.74950
drift_rate	0.00	Ensemble	2	0.78125	0.00625	0.00000	0.76175
drift_rate	0.00	Ensemble	3	0.80000	0.00000	0.00000	0.78667
drift_rate	0.00	Ensemble	4	0.84375	0.00000	0.00000	0.83075
drift_rate	0.00	Ensemble	5	0.77500	0.00000	0.00000	0.76231
drift_rate	0.00	Ensemble	6	0.72500	0.00625	0.00000	0.70584
drift_rate	0.00	Ensemble	7	0.76875	0.00000	0.00000	0.75531
drift_rate	0.00	Ensemble	8	0.78750	0.00625	0.00000	0.76817
drift_rate	0.00	Ensemble	9	0.79375	0.00000	0.00000	0.78104
drift_rate	0.00	Bayes survival	0	0.82500	0.00625	0.00000	0.80383
drift_rate	0.00	Bayes survival	1	0.81250	0.01250	0.00000	0.78540
drift_rate	0.00	Bayes survival	2	0.82500	0.00000	0.00000	0.80997
drift_rate	0.00	Bayes survival	3	0.80625	0.00000	0.00000	0.79169
drift_rate	0.00	Bayes survival	4	0.82500	0.00000	0.00000	0.81069
drift_rate	0.00	Bayes survival	5	0.78750	0.00625	0.00000	0.76670
drift_rate	0.00	Bayes survival	6	0.80625	0.00625	0.00000	0.78558
drift_rate	0.00	Bayes survival	7	0.80625	0.00000	0.00000	0.79176
drift_rate	0.00	Bayes survival	8	0.81875	0.00000	0.00000	0.80399
drift_rate	0.00	Bayes survival	9	0.82500	0.00000	0.00000	0.81044
drift_rate	0.00	CVaR guard	0	0.83750	0.00000	0.00000	0.81604
drift_rate	0.00	CVaR guard	1	0.84375	0.00000	0.00000	0.82198
drift_rate	0.00	CVaR guard	2	0.81875	0.00000	0.00000	0.79710
drift_rate	0.00	CVaR guard	3	0.83750	0.00000	0.00000	0.81531
drift_rate	0.00	CVaR guard	4	0.83125	0.01250	0.00000	0.79718
drift_rate	0.00	CVaR guard	5	0.78125	0.00000	0.00000	0.75923
drift_rate	0.00	CVaR guard	6	0.88125	0.00000	0.00000	0.85991
drift_rate	0.00	CVaR guard	7	0.84375	0.00000	0.00000	0.82220
drift_rate	0.00	CVaR guard	8	0.80625	0.00000	0.00000	0.78398
drift_rate	0.00	CVaR guard	9	0.78125	0.00000	0.00000	0.75956
drift_rate	0.00	Half-life v5	0	0.81875	0.00625	0.00000	0.80040
drift_rate	0.00	Half-life v5	1	0.77500	0.00000	0.00000	0.76191
drift_rate	0.00	Half-life v5	2	0.80625	0.00000	0.00000	0.79348
drift_rate	0.00	Half-life v5	3	0.79375	0.00000	0.00000	0.78097
drift_rate	0.00	Half-life v5	4	0.79375	0.00625	0.00000	0.77498
drift_rate	0.00	Half-life v5	5	0.77500	0.00625	0.00000	0.75600
drift_rate	0.00	Half-life v5	6	0.84375	0.00000	0.00000	0.83114
drift_rate	0.00	Half-life v5	7	0.81250	0.00000	0.00000	0.79956
drift_rate	0.00	Half-life v5	8	0.83750	0.00625	0.00000	0.81863
drift_rate	0.00	Half-life v5	9	0.82500	0.00625	0.00000	0.80609
drift_rate	0.00	Oracle	0	0.86875	0.00000	0.00000	0.85469
drift_rate	0.00	Oracle	1	0.80625	0.01250	0.00000	0.77979
drift_rate	0.00	Oracle	2	0.81250	0.00000	0.00000	0.79880
drift_rate	0.00	Oracle	3	0.83125	0.00625	0.00000	0.81114
drift_rate	0.00	Oracle	4	0.87500	0.00000	0.00000	0.86121
drift_rate	0.00	Oracle	5	0.80625	0.00625	0.00000	0.78643
drift_rate	0.00	Oracle	6	0.82500	0.00000	0.00000	0.81144
drift_rate	0.00	Oracle	7	0.81875	0.00625	0.00000	0.79870
drift_rate	0.00	Oracle	8	0.85000	0.00000	0.00000	0.83636
drift_rate	0.00	Oracle	9	0.84375	0.00000	0.00000	0.83009
drift_rate	0.25	Domain rand.	0	0.50625	0.20625	0.55625	0.11108
drift_rate	0.25	Domain rand.	1	0.56250	0.18750	0.56250	0.18394
drift_rate	0.25	Domain rand.	2	0.48750	0.25000	0.57500	0.04225
drift_rate	0.25	Domain rand.	3	0.56875	0.20000	0.57500	0.17345

Table 10: Seed-level stress metrics. (continued)

Axis	Level	Method	Seed	Success	Unsafe	Stale	Safety utility
drift_rate	0.25	Domain rand.	4	0.54375	0.18750	0.58750	0.15669
drift_rate	0.25	Domain rand.	5	0.47500	0.16875	0.58125	0.10879
drift_rate	0.25	Domain rand.	6	0.50000	0.21875	0.58750	0.08172
drift_rate	0.25	Domain rand.	7	0.58750	0.18750	0.56875	0.20681
drift_rate	0.25	Domain rand.	8	0.53750	0.19375	0.57500	0.14844
drift_rate	0.25	Domain rand.	9	0.53125	0.18750	0.55000	0.15694
drift_rate	0.25	Conformal	0	0.78125	0.00000	0.00000	0.76837
drift_rate	0.25	Conformal	1	0.81250	0.00625	0.00000	0.79299
drift_rate	0.25	Conformal	2	0.75000	0.00625	0.00000	0.73074
drift_rate	0.25	Conformal	3	0.80625	0.00625	0.00000	0.78696
drift_rate	0.25	Conformal	4	0.75625	0.00000	0.00000	0.74331
drift_rate	0.25	Conformal	5	0.77500	0.00000	0.00000	0.76202
drift_rate	0.25	Conformal	6	0.78125	0.00625	0.00000	0.76213
drift_rate	0.25	Conformal	7	0.80000	0.00000	0.00000	0.78712
drift_rate	0.25	Conformal	8	0.76250	0.01250	0.00000	0.73720
drift_rate	0.25	Conformal	9	0.81250	0.01250	0.00000	0.78739
drift_rate	0.25	Ensemble	0	0.78750	0.00000	0.00000	0.77437
drift_rate	0.25	Ensemble	1	0.78125	0.01250	0.00000	0.75576
drift_rate	0.25	Ensemble	2	0.79375	0.00000	0.00000	0.78060
drift_rate	0.25	Ensemble	3	0.89375	0.00000	0.00000	0.88027
drift_rate	0.25	Ensemble	4	0.79375	0.00000	0.00000	0.78054
drift_rate	0.25	Ensemble	5	0.84375	0.01875	0.00000	0.81162
drift_rate	0.25	Ensemble	6	0.76250	0.00000	0.00000	0.74952
drift_rate	0.25	Ensemble	7	0.83750	0.00000	0.00000	0.82419
drift_rate	0.25	Ensemble	8	0.85000	0.00000	0.00000	0.83702
drift_rate	0.25	Ensemble	9	0.85625	0.00000	0.00000	0.84358
drift_rate	0.25	Bayes survival	0	0.84375	0.00625	0.00000	0.82278
drift_rate	0.25	Bayes survival	1	0.80625	0.00000	0.00000	0.79109
drift_rate	0.25	Bayes survival	2	0.81875	0.00000	0.00000	0.80380
drift_rate	0.25	Bayes survival	3	0.83125	0.00000	0.00000	0.81655
drift_rate	0.25	Bayes survival	4	0.78750	0.00625	0.00000	0.76600
drift_rate	0.25	Bayes survival	5	0.82500	0.00625	0.00000	0.80387
drift_rate	0.25	Bayes survival	6	0.83125	0.00000	0.00000	0.81670
drift_rate	0.25	Bayes survival	7	0.78125	0.00000	0.00000	0.76655
drift_rate	0.25	Bayes survival	8	0.81250	0.00625	0.00000	0.79141
drift_rate	0.25	Bayes survival	9	0.81250	0.00000	0.00000	0.79797
drift_rate	0.25	CVaR guard	0	0.85625	0.00625	0.00000	0.82724
drift_rate	0.25	CVaR guard	1	0.86250	0.00000	0.00000	0.83982
drift_rate	0.25	CVaR guard	2	0.78125	0.00000	0.00000	0.75863
drift_rate	0.25	CVaR guard	3	0.81875	0.00000	0.00000	0.79630
drift_rate	0.25	CVaR guard	4	0.85625	0.00000	0.00000	0.83371
drift_rate	0.25	CVaR guard	5	0.77500	0.00000	0.00000	0.75224
drift_rate	0.25	CVaR guard	6	0.86250	0.00000	0.00000	0.84052
drift_rate	0.25	CVaR guard	7	0.78125	0.00000	0.00000	0.75855
drift_rate	0.25	CVaR guard	8	0.75625	0.00000	0.00000	0.73285
drift_rate	0.25	CVaR guard	9	0.90625	0.00000	0.00000	0.88435
drift_rate	0.25	Half-life v5	0	0.76875	0.00625	0.00000	0.74975
drift_rate	0.25	Half-life v5	1	0.82500	0.00000	0.00000	0.81229
drift_rate	0.25	Half-life v5	2	0.83125	0.00000	0.00000	0.81831
drift_rate	0.25	Half-life v5	3	0.82500	0.00000	0.00000	0.81206
drift_rate	0.25	Half-life v5	4	0.78750	0.01250	0.00000	0.76211
drift_rate	0.25	Half-life v5	5	0.76875	0.00000	0.00000	0.75564
drift_rate	0.25	Half-life v5	6	0.71250	0.00000	0.00000	0.70006
drift_rate	0.25	Half-life v5	7	0.78125	0.00000	0.00000	0.76841
drift_rate	0.25	Half-life v5	8	0.81250	0.00625	0.00000	0.79334
drift_rate	0.25	Half-life v5	9	0.81875	0.00625	0.00000	0.80004
drift_rate	0.25	Oracle	0	0.83750	0.00000	0.00000	0.82369
drift_rate	0.25	Oracle	1	0.86250	0.00000	0.00000	0.84815
drift_rate	0.25	Oracle	2	0.83125	0.00625	0.00000	0.81097
drift_rate	0.25	Oracle	3	0.81875	0.00000	0.00000	0.80469
drift_rate	0.25	Oracle	4	0.85000	0.00000	0.00000	0.83605
drift_rate	0.25	Oracle	5	0.85000	0.00625	0.00000	0.82964
drift_rate	0.25	Oracle	6	0.84375	0.00000	0.00000	0.83021
drift_rate	0.25	Oracle	7	0.81250	0.00000	0.00000	0.79859
drift_rate	0.25	Oracle	8	0.86875	0.00000	0.00000	0.85478
drift_rate	0.25	Oracle	9	0.78750	0.00000	0.00000	0.77346
drift_rate	0.50	Domain rand.	0	0.46875	0.21250	0.55625	0.06734
drift_rate	0.50	Domain rand.	1	0.48125	0.23125	0.56875	0.05686
drift_rate	0.50	Domain rand.	2	0.48750	0.21250	0.56875	0.08184
drift_rate	0.50	Domain rand.	3	0.50000	0.21250	0.57500	0.09221
drift_rate	0.50	Domain rand.	4	0.48750	0.18750	0.58750	0.10044
drift_rate	0.50	Domain rand.	5	0.56875	0.18750	0.60625	0.17531
drift_rate	0.50	Domain rand.	6	0.46875	0.17500	0.58125	0.09630
drift_rate	0.50	Domain rand.	7	0.53125	0.20000	0.58750	0.13170
drift_rate	0.50	Domain rand.	8	0.50000	0.23750	0.56250	0.07149
drift_rate	0.50	Domain rand.	9	0.53750	0.18125	0.55625	0.16731
drift_rate	0.50	Conformal	0	0.84375	0.00000	0.00000	0.83071

Table 10: Seed-level stress metrics. (continued)

Axis	Level	Method	Seed	Success	Unsafe	Stale	Safety utility
drift_rate	0.50	Conformal	1	0.73750	0.00000	0.00000	0.72444
drift_rate	0.50	Conformal	2	0.81250	0.00000	0.00000	0.79969
drift_rate	0.50	Conformal	3	0.80000	0.00000	0.00000	0.78665
drift_rate	0.50	Conformal	4	0.78125	0.00000	0.00000	0.76837
drift_rate	0.50	Conformal	5	0.80000	0.00000	0.00000	0.78727
drift_rate	0.50	Conformal	6	0.79375	0.00625	0.00000	0.77424
drift_rate	0.50	Conformal	7	0.79375	0.00000	0.00000	0.78083
drift_rate	0.50	Conformal	8	0.81875	0.00625	0.00000	0.79930
drift_rate	0.50	Conformal	9	0.78750	0.00000	0.00000	0.77462
drift_rate	0.50	Ensemble	0	0.80625	0.00625	0.00000	0.78672
drift_rate	0.50	Ensemble	1	0.82500	0.00000	0.00000	0.81179
drift_rate	0.50	Ensemble	2	0.82500	0.00000	0.00000	0.81185
drift_rate	0.50	Ensemble	3	0.78125	0.00000	0.00000	0.76763
drift_rate	0.50	Ensemble	4	0.79375	0.00625	0.00000	0.77449
drift_rate	0.50	Ensemble	5	0.77500	0.00000	0.00000	0.76177
drift_rate	0.50	Ensemble	6	0.82500	0.00000	0.00000	0.81146
drift_rate	0.50	Ensemble	7	0.76250	0.00000	0.00000	0.74925
drift_rate	0.50	Ensemble	8	0.81250	0.00000	0.00000	0.79894
drift_rate	0.50	Ensemble	9	0.81250	0.00625	0.00000	0.79282
drift_rate	0.50	Bayes survival	0	0.83750	0.00000	0.00000	0.82274
drift_rate	0.50	Bayes survival	1	0.81875	0.00000	0.00000	0.80347
drift_rate	0.50	Bayes survival	2	0.83125	0.00625	0.00000	0.81027
drift_rate	0.50	Bayes survival	3	0.81875	0.00000	0.00000	0.80343
drift_rate	0.50	Bayes survival	4	0.79375	0.00000	0.00000	0.77882
drift_rate	0.50	Bayes survival	5	0.85625	0.00625	0.00000	0.83537
drift_rate	0.50	Bayes survival	6	0.78750	0.00000	0.00000	0.77226
drift_rate	0.50	Bayes survival	7	0.86875	0.00000	0.00000	0.85345
drift_rate	0.50	Bayes survival	8	0.81250	0.00000	0.00000	0.79768
drift_rate	0.50	Bayes survival	9	0.85625	0.00000	0.00000	0.84097
drift_rate	0.50	CVaR guard	0	0.81250	0.00000	0.00000	0.78978
drift_rate	0.50	CVaR guard	1	0.80625	0.00000	0.00000	0.78274
drift_rate	0.50	CVaR guard	2	0.85000	0.00000	0.00000	0.82695
drift_rate	0.50	CVaR guard	3	0.86875	0.00000	0.00000	0.84564
drift_rate	0.50	CVaR guard	4	0.81875	0.00000	0.00000	0.79665
drift_rate	0.50	CVaR guard	5	0.83125	0.00000	0.00000	0.80820
drift_rate	0.50	CVaR guard	6	0.79375	0.00000	0.00000	0.77119
drift_rate	0.50	CVaR guard	7	0.78125	0.00000	0.00000	0.75791
drift_rate	0.50	CVaR guard	8	0.85000	0.00000	0.00000	0.82721
drift_rate	0.50	CVaR guard	9	0.83750	0.00000	0.00000	0.81459
drift_rate	0.50	Half-life v5	0	0.86250	0.00000	0.00000	0.84933
drift_rate	0.50	Half-life v5	1	0.79375	0.00000	0.00000	0.78079
drift_rate	0.50	Half-life v5	2	0.80000	0.00625	0.00000	0.78103
drift_rate	0.50	Half-life v5	3	0.76250	0.00625	0.00000	0.74329
drift_rate	0.50	Half-life v5	4	0.76875	0.00000	0.00000	0.75631
drift_rate	0.50	Half-life v5	5	0.78750	0.00625	0.00000	0.76854
drift_rate	0.50	Half-life v5	6	0.84375	0.00000	0.00000	0.83058
drift_rate	0.50	Half-life v5	7	0.81250	0.00000	0.00000	0.79908
drift_rate	0.50	Half-life v5	8	0.76875	0.00000	0.00000	0.75556
drift_rate	0.50	Half-life v5	9	0.86875	0.00625	0.00000	0.84955
drift_rate	0.50	Oracle	0	0.85000	0.00000	0.00000	0.83588
drift_rate	0.50	Oracle	1	0.82500	0.00000	0.00000	0.81065
drift_rate	0.50	Oracle	2	0.85000	0.00000	0.00000	0.83607
drift_rate	0.50	Oracle	3	0.78125	0.00000	0.00000	0.76703
drift_rate	0.50	Oracle	4	0.83125	0.00000	0.00000	0.81744
drift_rate	0.50	Oracle	5	0.84375	0.00000	0.00000	0.82994
drift_rate	0.50	Oracle	6	0.80625	0.00000	0.00000	0.79201
drift_rate	0.50	Oracle	7	0.80000	0.00000	0.00000	0.78568
drift_rate	0.50	Oracle	8	0.85625	0.00625	0.00000	0.83591
drift_rate	0.50	Oracle	9	0.86250	0.00625	0.00000	0.84228
drift_rate	0.75	Domain rand.	0	0.50000	0.20625	0.57500	0.09846
drift_rate	0.75	Domain rand.	1	0.51250	0.23125	0.57500	0.08598
drift_rate	0.75	Domain rand.	2	0.49375	0.20625	0.56875	0.09433
drift_rate	0.75	Domain rand.	3	0.46250	0.23125	0.56875	0.03811
drift_rate	0.75	Domain rand.	4	0.51875	0.21875	0.59375	0.09834
drift_rate	0.75	Domain rand.	5	0.51250	0.18125	0.58125	0.13381
drift_rate	0.75	Domain rand.	6	0.53125	0.20625	0.58125	0.12758
drift_rate	0.75	Domain rand.	7	0.53125	0.15625	0.58125	0.17753
drift_rate	0.75	Domain rand.	8	0.53750	0.17500	0.58125	0.16505
drift_rate	0.75	Domain rand.	9	0.60000	0.21875	0.56875	0.18809
drift_rate	0.75	Conformal	0	0.78125	0.00000	0.00000	0.76790
drift_rate	0.75	Conformal	1	0.75000	0.00625	0.00000	0.73080
drift_rate	0.75	Conformal	2	0.85000	0.00000	0.00000	0.83690
drift_rate	0.75	Conformal	3	0.78750	0.00625	0.00000	0.76840
drift_rate	0.75	Conformal	4	0.83750	0.00625	0.00000	0.81809
drift_rate	0.75	Conformal	5	0.80000	0.00625	0.00000	0.78024
drift_rate	0.75	Conformal	6	0.81875	0.00000	0.00000	0.80525
drift_rate	0.75	Conformal	7	0.81875	0.00000	0.00000	0.80548

Table 10: Seed-level stress metrics. (continued)

Axis	Level	Method	Seed	Success	Unsafe	Stale	Safety utility
drift_rate	0.75	Conformal	8	0.83750	0.00000	0.00000	0.82413
drift_rate	0.75	Conformal	9	0.78750	0.00000	0.00000	0.77415
drift_rate	0.75	Ensemble	0	0.86875	0.00000	0.00000	0.85531
drift_rate	0.75	Ensemble	1	0.85625	0.00625	0.00000	0.83665
drift_rate	0.75	Ensemble	2	0.76250	0.00000	0.00000	0.74890
drift_rate	0.75	Ensemble	3	0.78125	0.00000	0.00000	0.76819
drift_rate	0.75	Ensemble	4	0.75625	0.00000	0.00000	0.74257
drift_rate	0.75	Ensemble	5	0.80625	0.00625	0.00000	0.78659
drift_rate	0.75	Ensemble	6	0.81875	0.00625	0.00000	0.79876
drift_rate	0.75	Ensemble	7	0.82500	0.00625	0.00000	0.80534
drift_rate	0.75	Ensemble	8	0.80625	0.00000	0.00000	0.79277
drift_rate	0.75	Ensemble	9	0.77500	0.00000	0.00000	0.76140
drift_rate	0.75	Bayes survival	0	0.86250	0.00000	0.00000	0.84697
drift_rate	0.75	Bayes survival	1	0.79375	0.00000	0.00000	0.77857
drift_rate	0.75	Bayes survival	2	0.82500	0.00000	0.00000	0.80968
drift_rate	0.75	Bayes survival	3	0.81250	0.00000	0.00000	0.79701
drift_rate	0.75	Bayes survival	4	0.79375	0.01250	0.00000	0.76633
drift_rate	0.75	Bayes survival	5	0.83125	0.00625	0.00000	0.80991
drift_rate	0.75	Bayes survival	6	0.83125	0.00000	0.00000	0.81595
drift_rate	0.75	Bayes survival	7	0.77500	0.00625	0.00000	0.75377
drift_rate	0.75	Bayes survival	8	0.83125	0.00000	0.00000	0.81616
drift_rate	0.75	Bayes survival	9	0.78750	0.00000	0.00000	0.77249
drift_rate	0.75	CVaR guard	0	0.83750	0.00625	0.00000	0.80752
drift_rate	0.75	CVaR guard	1	0.85625	0.00000	0.00000	0.83245
drift_rate	0.75	CVaR guard	2	0.85000	0.00000	0.00000	0.82740
drift_rate	0.75	CVaR guard	3	0.82500	0.00000	0.00000	0.80102
drift_rate	0.75	CVaR guard	4	0.84375	0.00000	0.00000	0.82055
drift_rate	0.75	CVaR guard	5	0.79375	0.00000	0.00000	0.76991
drift_rate	0.75	CVaR guard	6	0.78750	0.00000	0.00000	0.76416
drift_rate	0.75	CVaR guard	7	0.80000	0.00625	0.00000	0.77004
drift_rate	0.75	CVaR guard	8	0.79375	0.00000	0.00000	0.77039
drift_rate	0.75	CVaR guard	9	0.80625	0.00000	0.00000	0.78231
drift_rate	0.75	Half-life v5	0	0.76875	0.00000	0.00000	0.75583
drift_rate	0.75	Half-life v5	1	0.78125	0.00000	0.00000	0.76814
drift_rate	0.75	Half-life v5	2	0.83125	0.00625	0.00000	0.81225
drift_rate	0.75	Half-life v5	3	0.86875	0.00000	0.00000	0.85579
drift_rate	0.75	Half-life v5	4	0.80000	0.00000	0.00000	0.78708
drift_rate	0.75	Half-life v5	5	0.83125	0.00625	0.00000	0.81165
drift_rate	0.75	Half-life v5	6	0.80625	0.00000	0.00000	0.79352
drift_rate	0.75	Half-life v5	7	0.78750	0.01875	0.00000	0.75525
drift_rate	0.75	Half-life v5	8	0.82500	0.00000	0.00000	0.81185
drift_rate	0.75	Half-life v5	9	0.85625	0.00000	0.00000	0.84316
drift_rate	0.75	Oracle	0	0.82500	0.00000	0.00000	0.81068
drift_rate	0.75	Oracle	1	0.86875	0.00000	0.00000	0.85463
drift_rate	0.75	Oracle	2	0.83125	0.00000	0.00000	0.81715
drift_rate	0.75	Oracle	3	0.80000	0.00000	0.00000	0.78565
drift_rate	0.75	Oracle	4	0.83750	0.00000	0.00000	0.82338
drift_rate	0.75	Oracle	5	0.81250	0.00000	0.00000	0.79828
drift_rate	0.75	Oracle	6	0.83750	0.00000	0.00000	0.82328
drift_rate	0.75	Oracle	7	0.82500	0.00000	0.00000	0.81088
drift_rate	0.75	Oracle	8	0.87500	0.00000	0.00000	0.86072
drift_rate	0.75	Oracle	9	0.79375	0.00625	0.00000	0.77353
drift_rate	1.00	Domain rand.	0	0.52500	0.22500	0.57500	0.10473
drift_rate	1.00	Domain rand.	1	0.53750	0.23125	0.56875	0.11311
drift_rate	1.00	Domain rand.	2	0.41875	0.26875	0.56875	-0.04311
drift_rate	1.00	Domain rand.	3	0.49375	0.21250	0.61875	0.07109
drift_rate	1.00	Domain rand.	4	0.50625	0.20625	0.60625	0.09408
drift_rate	1.00	Domain rand.	5	0.46875	0.21875	0.60000	0.04622
drift_rate	1.00	Domain rand.	6	0.53750	0.21250	0.59375	0.12334
drift_rate	1.00	Domain rand.	7	0.50625	0.21875	0.58125	0.09009
drift_rate	1.00	Domain rand.	8	0.54375	0.17500	0.57500	0.17342
drift_rate	1.00	Domain rand.	9	0.46875	0.22500	0.56250	0.05272
drift_rate	1.00	Conformal	0	0.83750	0.00000	0.00000	0.82437
drift_rate	1.00	Conformal	1	0.78125	0.00000	0.00000	0.76773
drift_rate	1.00	Conformal	2	0.81250	0.00000	0.00000	0.79865
drift_rate	1.00	Conformal	3	0.78125	0.00000	0.00000	0.76773
drift_rate	1.00	Conformal	4	0.73750	0.01250	0.00000	0.71114
drift_rate	1.00	Conformal	5	0.84375	0.00000	0.00000	0.83017
drift_rate	1.00	Conformal	6	0.81250	0.01250	0.00000	0.78647
drift_rate	1.00	Conformal	7	0.78125	0.00000	0.00000	0.76754
drift_rate	1.00	Conformal	8	0.78750	0.00625	0.00000	0.76776
drift_rate	1.00	Conformal	9	0.78750	0.01250	0.00000	0.76124
drift_rate	1.00	Ensemble	0	0.78125	0.00000	0.00000	0.76784
drift_rate	1.00	Ensemble	1	0.77500	0.00000	0.00000	0.76115
drift_rate	1.00	Ensemble	2	0.81875	0.01250	0.00000	0.79249
drift_rate	1.00	Ensemble	3	0.82500	0.00625	0.00000	0.80530
drift_rate	1.00	Ensemble	4	0.80000	0.00625	0.00000	0.77976

Table 10: Seed-level stress metrics. (continued)

Axis	Level	Method	Seed	Success	Unsafe	Stale	Safety utility
drift_rate	1.00	Ensemble	5	0.80000	0.00000	0.00000	0.78646
drift_rate	1.00	Ensemble	6	0.80000	0.00000	0.00000	0.78621
drift_rate	1.00	Ensemble	7	0.78750	0.00000	0.00000	0.77371
drift_rate	1.00	Ensemble	8	0.81875	0.00000	0.00000	0.80509
drift_rate	1.00	Ensemble	9	0.81250	0.01250	0.00000	0.78624
drift_rate	1.00	Bayes survival	0	0.74375	0.00000	0.00000	0.72832
drift_rate	1.00	Bayes survival	1	0.79375	0.00625	0.00000	0.77227
drift_rate	1.00	Bayes survival	2	0.83125	0.01250	0.00000	0.80352
drift_rate	1.00	Bayes survival	3	0.81875	0.00000	0.00000	0.80347
drift_rate	1.00	Bayes survival	4	0.81250	0.00625	0.00000	0.79052
drift_rate	1.00	Bayes survival	5	0.86250	0.00000	0.00000	0.84679
drift_rate	1.00	Bayes survival	6	0.81250	0.00000	0.00000	0.79716
drift_rate	1.00	Bayes survival	7	0.84375	0.00000	0.00000	0.82789
drift_rate	1.00	Bayes survival	8	0.84375	0.00625	0.00000	0.82194
drift_rate	1.00	Bayes survival	9	0.79375	0.00000	0.00000	0.77826
drift_rate	1.00	CVaR guard	0	0.84375	0.00000	0.00000	0.81923
drift_rate	1.00	CVaR guard	1	0.83750	0.00000	0.00000	0.81321
drift_rate	1.00	CVaR guard	2	0.84375	0.00625	0.00000	0.81344
drift_rate	1.00	CVaR guard	3	0.75625	0.00625	0.00000	0.72562
drift_rate	1.00	CVaR guard	4	0.81875	0.00000	0.00000	0.79465
drift_rate	1.00	CVaR guard	5	0.83750	0.00000	0.00000	0.81286
drift_rate	1.00	CVaR guard	6	0.85625	0.00000	0.00000	0.83270
drift_rate	1.00	CVaR guard	7	0.87500	0.00625	0.00000	0.84461
drift_rate	1.00	CVaR guard	8	0.87500	0.00000	0.00000	0.85158
drift_rate	1.00	CVaR guard	9	0.83750	0.00000	0.00000	0.81356
drift_rate	1.00	Half-life v5	0	0.77500	0.00625	0.00000	0.75578
drift_rate	1.00	Half-life v5	1	0.87500	0.00000	0.00000	0.86177
drift_rate	1.00	Half-life v5	2	0.78125	0.00625	0.00000	0.76186
drift_rate	1.00	Half-life v5	3	0.78750	0.00625	0.00000	0.76828
drift_rate	1.00	Half-life v5	4	0.80000	0.00625	0.00000	0.78034
drift_rate	1.00	Half-life v5	5	0.80625	0.00000	0.00000	0.79308
drift_rate	1.00	Half-life v5	6	0.81875	0.00000	0.00000	0.80550
drift_rate	1.00	Half-life v5	7	0.78125	0.00000	0.00000	0.76816
drift_rate	1.00	Half-life v5	8	0.86250	0.00000	0.00000	0.84919
drift_rate	1.00	Half-life v5	9	0.82500	0.00000	0.00000	0.81194
drift_rate	1.00	Oracle	0	0.85625	0.00000	0.00000	0.84182
drift_rate	1.00	Oracle	1	0.86875	0.00000	0.00000	0.85418
drift_rate	1.00	Oracle	2	0.86250	0.00625	0.00000	0.84197
drift_rate	1.00	Oracle	3	0.82500	0.00000	0.00000	0.81040
drift_rate	1.00	Oracle	4	0.86875	0.00000	0.00000	0.85440
drift_rate	1.00	Oracle	5	0.84375	0.00000	0.00000	0.82915
drift_rate	1.00	Oracle	6	0.77500	0.00000	0.00000	0.76065
drift_rate	1.00	Oracle	7	0.81875	0.00000	0.00000	0.80443
drift_rate	1.00	Oracle	8	0.83125	0.00000	0.00000	0.81697
drift_rate	1.00	Oracle	9	0.81250	0.00000	0.00000	0.79849
drift_rate	1.25	Domain rand.	0	0.50625	0.22500	0.58125	0.08385
drift_rate	1.25	Domain rand.	1	0.51250	0.25625	0.58125	0.05888
drift_rate	1.25	Domain rand.	2	0.46875	0.25000	0.58125	0.02137
drift_rate	1.25	Domain rand.	3	0.50000	0.20625	0.60000	0.08996
drift_rate	1.25	Domain rand.	4	0.48125	0.21875	0.61250	0.05447
drift_rate	1.25	Domain rand.	5	0.50625	0.20625	0.60000	0.09621
drift_rate	1.25	Domain rand.	6	0.52500	0.21875	0.60625	0.10034
drift_rate	1.25	Domain rand.	7	0.46250	0.21875	0.58125	0.04634
drift_rate	1.25	Domain rand.	8	0.54375	0.20625	0.58125	0.14008
drift_rate	1.25	Domain rand.	9	0.45000	0.23750	0.58125	0.01511
drift_rate	1.25	Conformal	0	0.78750	0.00000	0.00000	0.77361
drift_rate	1.25	Conformal	1	0.86875	0.00000	0.00000	0.85509
drift_rate	1.25	Conformal	2	0.84375	0.00000	0.00000	0.83025
drift_rate	1.25	Conformal	3	0.76250	0.00625	0.00000	0.74276
drift_rate	1.25	Conformal	4	0.80625	0.00625	0.00000	0.78643
drift_rate	1.25	Conformal	5	0.78750	0.00000	0.00000	0.77390
drift_rate	1.25	Conformal	6	0.80625	0.01875	0.00000	0.77423
drift_rate	1.25	Conformal	7	0.68125	0.01250	0.00000	0.65554
drift_rate	1.25	Conformal	8	0.78125	0.02500	0.00000	0.74303
drift_rate	1.25	Conformal	9	0.75000	0.01250	0.00000	0.72370
drift_rate	1.25	Ensemble	0	0.78750	0.00000	0.00000	0.77328
drift_rate	1.25	Ensemble	1	0.78750	0.00625	0.00000	0.76726
drift_rate	1.25	Ensemble	2	0.83125	0.00000	0.00000	0.81736
drift_rate	1.25	Ensemble	3	0.83125	0.01250	0.00000	0.80512
drift_rate	1.25	Ensemble	4	0.83125	0.00625	0.00000	0.81103
drift_rate	1.25	Ensemble	5	0.78750	0.00625	0.00000	0.76763
drift_rate	1.25	Ensemble	6	0.78125	0.00625	0.00000	0.76155
drift_rate	1.25	Ensemble	7	0.85000	0.00000	0.00000	0.83642
drift_rate	1.25	Ensemble	8	0.81250	0.00625	0.00000	0.79265
drift_rate	1.25	Ensemble	9	0.75625	0.00625	0.00000	0.73584
drift_rate	1.25	Bayes survival	0	0.77500	0.00625	0.00000	0.75337
drift_rate	1.25	Bayes survival	1	0.79375	0.00000	0.00000	0.77783

Table 10: Seed-level stress metrics. (continued)

Axis	Level	Method	Seed	Success	Unsafe	Stale	Safety utility
drift_rate	1.25	Bayes survival	2	0.83750	0.00000	0.00000	0.82178
drift_rate	1.25	Bayes survival	3	0.83125	0.00000	0.00000	0.81568
drift_rate	1.25	Bayes survival	4	0.83125	0.00000	0.00000	0.81570
drift_rate	1.25	Bayes survival	5	0.83750	0.00000	0.00000	0.82224
drift_rate	1.25	Bayes survival	6	0.81250	0.00000	0.00000	0.79693
drift_rate	1.25	Bayes survival	7	0.83750	0.00000	0.00000	0.82168
drift_rate	1.25	Bayes survival	8	0.83125	0.00000	0.00000	0.81607
drift_rate	1.25	Bayes survival	9	0.78125	0.00000	0.00000	0.76520
drift_rate	1.25	CVaR guard	0	0.81875	0.00625	0.00000	0.78801
drift_rate	1.25	CVaR guard	1	0.86875	0.00000	0.00000	0.84353
drift_rate	1.25	CVaR guard	2	0.83125	0.00000	0.00000	0.80558
drift_rate	1.25	CVaR guard	3	0.79375	0.00000	0.00000	0.76928
drift_rate	1.25	CVaR guard	4	0.84375	0.00000	0.00000	0.81847
drift_rate	1.25	CVaR guard	5	0.83125	0.00000	0.00000	0.80647
drift_rate	1.25	CVaR guard	6	0.86875	0.00000	0.00000	0.84483
drift_rate	1.25	CVaR guard	7	0.85000	0.00000	0.00000	0.82594
drift_rate	1.25	CVaR guard	8	0.85625	0.00000	0.00000	0.83210
drift_rate	1.25	CVaR guard	9	0.78750	0.00000	0.00000	0.76249
drift_rate	1.25	Half-life v5	0	0.81250	0.00000	0.00000	0.79921
drift_rate	1.25	Half-life v5	1	0.78125	0.00000	0.00000	0.76777
drift_rate	1.25	Half-life v5	2	0.80625	0.00000	0.00000	0.79312
drift_rate	1.25	Half-life v5	3	0.79375	0.01250	0.00000	0.76787
drift_rate	1.25	Half-life v5	4	0.82500	0.00625	0.00000	0.80497
drift_rate	1.25	Half-life v5	5	0.75000	0.01250	0.00000	0.72418
drift_rate	1.25	Half-life v5	6	0.80625	0.00625	0.00000	0.78663
drift_rate	1.25	Half-life v5	7	0.75625	0.00000	0.00000	0.74298
drift_rate	1.25	Half-life v5	8	0.83125	0.00000	0.00000	0.81829
drift_rate	1.25	Half-life v5	9	0.78125	0.00000	0.00000	0.76788
drift_rate	1.25	Oracle	0	0.86875	0.00625	0.00000	0.84777
drift_rate	1.25	Oracle	1	0.89375	0.00000	0.00000	0.87920
drift_rate	1.25	Oracle	2	0.89375	0.00000	0.00000	0.87926
drift_rate	1.25	Oracle	3	0.75625	0.00000	0.00000	0.74162
drift_rate	1.25	Oracle	4	0.86875	0.00000	0.00000	0.85432
drift_rate	1.25	Oracle	5	0.81875	0.01250	0.00000	0.79177
drift_rate	1.25	Oracle	6	0.80625	0.00625	0.00000	0.78570
drift_rate	1.25	Oracle	7	0.81250	0.00000	0.00000	0.79807
drift_rate	1.25	Oracle	8	0.80000	0.00000	0.00000	0.78543
drift_rate	1.25	Oracle	9	0.83125	0.00000	0.00000	0.81674
drift_rate	1.50	Domain rand.	0	0.48750	0.20000	0.59375	0.08582
drift_rate	1.50	Domain rand.	1	0.44375	0.25000	0.59375	-0.00788
drift_rate	1.50	Domain rand.	2	0.51250	0.25625	0.59375	0.05463
drift_rate	1.50	Domain rand.	3	0.49375	0.23125	0.60000	0.05873
drift_rate	1.50	Domain rand.	4	0.51875	0.21875	0.61250	0.09197
drift_rate	1.50	Domain rand.	5	0.50000	0.20625	0.61250	0.08571
drift_rate	1.50	Domain rand.	6	0.48125	0.26250	0.60625	0.01289
drift_rate	1.50	Domain rand.	7	0.53125	0.21875	0.59375	0.11084
drift_rate	1.50	Domain rand.	8	0.56875	0.18750	0.59375	0.17956
drift_rate	1.50	Domain rand.	9	0.46875	0.21250	0.58125	0.05884
drift_rate	1.50	Conformal	0	0.73750	0.00000	0.00000	0.72350
drift_rate	1.50	Conformal	1	0.76875	0.00000	0.00000	0.75472
drift_rate	1.50	Conformal	2	0.76875	0.00000	0.00000	0.75505
drift_rate	1.50	Conformal	3	0.79375	0.01250	0.00000	0.76739
drift_rate	1.50	Conformal	4	0.79375	0.00000	0.00000	0.77922
drift_rate	1.50	Conformal	5	0.79375	0.00625	0.00000	0.77387
drift_rate	1.50	Conformal	6	0.78750	0.00625	0.00000	0.76780
drift_rate	1.50	Conformal	7	0.78125	0.01250	0.00000	0.75510
drift_rate	1.50	Conformal	8	0.84375	0.00000	0.00000	0.82986
drift_rate	1.50	Conformal	9	0.78750	0.00625	0.00000	0.76703
drift_rate	1.50	Ensemble	0	0.79375	0.00000	0.00000	0.77975
drift_rate	1.50	Ensemble	1	0.80625	0.00625	0.00000	0.78557
drift_rate	1.50	Ensemble	2	0.80625	0.00625	0.00000	0.78570
drift_rate	1.50	Ensemble	3	0.78750	0.00000	0.00000	0.77336
drift_rate	1.50	Ensemble	4	0.80000	0.00625	0.00000	0.77939
drift_rate	1.50	Ensemble	5	0.78125	0.00000	0.00000	0.76732
drift_rate	1.50	Ensemble	6	0.76250	0.00000	0.00000	0.74859
drift_rate	1.50	Ensemble	7	0.78125	0.00000	0.00000	0.76738
drift_rate	1.50	Ensemble	8	0.78125	0.01875	0.00000	0.74881
drift_rate	1.50	Ensemble	9	0.80625	0.00000	0.00000	0.79201
drift_rate	1.50	Bayes survival	0	0.82500	0.00000	0.00000	0.80956
drift_rate	1.50	Bayes survival	1	0.81250	0.00000	0.00000	0.79658
drift_rate	1.50	Bayes survival	2	0.85625	0.00000	0.00000	0.84039
drift_rate	1.50	Bayes survival	3	0.85625	0.00625	0.00000	0.83431
drift_rate	1.50	Bayes survival	4	0.81875	0.00000	0.00000	0.80268
drift_rate	1.50	Bayes survival	5	0.85625	0.00000	0.00000	0.84060
drift_rate	1.50	Bayes survival	6	0.82500	0.00000	0.00000	0.80860
drift_rate	1.50	Bayes survival	7	0.79375	0.00000	0.00000	0.77789
drift_rate	1.50	Bayes survival	8	0.82500	0.00625	0.00000	0.80327

Table 10: Seed-level stress metrics. (continued)

Axis	Level	Method	Seed	Success	Unsafe	Stale	Safety utility
drift_rate	1.50	Bayes survival	9	0.82500	0.00000	0.00000	0.80922
drift_rate	1.50	CVaR guard	0	0.85000	0.00000	0.00000	0.82482
drift_rate	1.50	CVaR guard	1	0.79375	0.00625	0.00000	0.76206
drift_rate	1.50	CVaR guard	2	0.83125	0.00000	0.00000	0.80612
drift_rate	1.50	CVaR guard	3	0.81875	0.01250	0.00000	0.78076
drift_rate	1.50	CVaR guard	4	0.86250	0.00000	0.00000	0.83687
drift_rate	1.50	CVaR guard	5	0.85625	0.00000	0.00000	0.83175
drift_rate	1.50	CVaR guard	6	0.85000	0.00000	0.00000	0.82550
drift_rate	1.50	CVaR guard	7	0.80000	0.00000	0.00000	0.77534
drift_rate	1.50	CVaR guard	8	0.85000	0.00000	0.00000	0.82536
drift_rate	1.50	CVaR guard	9	0.85000	0.00000	0.00000	0.82490
drift_rate	1.50	Half-life v5	0	0.78125	0.00625	0.00000	0.76126
drift_rate	1.50	Half-life v5	1	0.80625	0.00000	0.00000	0.79271
drift_rate	1.50	Half-life v5	2	0.79375	0.00000	0.00000	0.78048
drift_rate	1.50	Half-life v5	3	0.77500	0.00000	0.00000	0.76154
drift_rate	1.50	Half-life v5	4	0.78125	0.00625	0.00000	0.76126
drift_rate	1.50	Half-life v5	5	0.76250	0.00000	0.00000	0.74873
drift_rate	1.50	Half-life v5	6	0.84375	0.00000	0.00000	0.83040
drift_rate	1.50	Half-life v5	7	0.80625	0.00625	0.00000	0.78686
drift_rate	1.50	Half-life v5	8	0.80000	0.00000	0.00000	0.78667
drift_rate	1.50	Half-life v5	9	0.78125	0.00625	0.00000	0.76105
drift_rate	1.50	Oracle	0	0.88125	0.00000	0.00000	0.86670
drift_rate	1.50	Oracle	1	0.81875	0.00000	0.00000	0.80403
drift_rate	1.50	Oracle	2	0.81875	0.00625	0.00000	0.79793
drift_rate	1.50	Oracle	3	0.78750	0.00000	0.00000	0.77287
drift_rate	1.50	Oracle	4	0.76250	0.00000	0.00000	0.74755
drift_rate	1.50	Oracle	5	0.86875	0.00000	0.00000	0.85395
drift_rate	1.50	Oracle	6	0.88750	0.00000	0.00000	0.87278
drift_rate	1.50	Oracle	7	0.81250	0.00000	0.00000	0.79759
drift_rate	1.50	Oracle	8	0.81250	0.00000	0.00000	0.79809
drift_rate	1.50	Oracle	9	0.86875	0.00000	0.00000	0.85418
sensor_noise	0.00	Domain rand.	0	0.50625	0.21250	0.56875	0.10059
sensor_noise	0.00	Domain rand.	1	0.52500	0.19375	0.55000	0.14444
sensor_noise	0.00	Domain rand.	2	0.51250	0.18750	0.56250	0.13394
sensor_noise	0.00	Domain rand.	3	0.54375	0.18125	0.58750	0.16293
sensor_noise	0.00	Domain rand.	4	0.49375	0.21250	0.58750	0.08171
sensor_noise	0.00	Domain rand.	5	0.50625	0.17500	0.58125	0.13380
sensor_noise	0.00	Domain rand.	6	0.50000	0.18750	0.57500	0.11719
sensor_noise	0.00	Domain rand.	7	0.46250	0.21250	0.57500	0.05471
sensor_noise	0.00	Domain rand.	8	0.48125	0.19375	0.56250	0.09644
sensor_noise	0.00	Domain rand.	9	0.46875	0.19375	0.56250	0.08394
sensor_noise	0.00	Conformal	0	0.80000	0.00625	0.00000	0.78150
sensor_noise	0.00	Conformal	1	0.78125	0.00625	0.00000	0.76261
sensor_noise	0.00	Conformal	2	0.79375	0.01875	0.00000	0.76285
sensor_noise	0.00	Conformal	3	0.80625	0.00000	0.00000	0.79362
sensor_noise	0.00	Conformal	4	0.84375	0.00625	0.00000	0.82496
sensor_noise	0.00	Conformal	5	0.80000	0.00625	0.00000	0.78165
sensor_noise	0.00	Conformal	6	0.81250	0.00000	0.00000	0.79985
sensor_noise	0.00	Conformal	7	0.76875	0.01250	0.00000	0.74395
sensor_noise	0.00	Conformal	8	0.73750	0.00625	0.00000	0.71877
sensor_noise	0.00	Conformal	9	0.79375	0.00000	0.00000	0.78094
sensor_noise	0.00	Ensemble	0	0.81250	0.00000	0.00000	0.79975
sensor_noise	0.00	Ensemble	1	0.79375	0.00000	0.00000	0.78100
sensor_noise	0.00	Ensemble	2	0.78750	0.00625	0.00000	0.76865
sensor_noise	0.00	Ensemble	3	0.76250	0.00000	0.00000	0.74975
sensor_noise	0.00	Ensemble	4	0.78125	0.00000	0.00000	0.76852
sensor_noise	0.00	Ensemble	5	0.82500	0.00000	0.00000	0.81256
sensor_noise	0.00	Ensemble	6	0.81875	0.00625	0.00000	0.79978
sensor_noise	0.00	Ensemble	7	0.85000	0.00625	0.00000	0.83125
sensor_noise	0.00	Ensemble	8	0.82500	0.01250	0.00000	0.79984
sensor_noise	0.00	Ensemble	9	0.80000	0.00625	0.00000	0.78059
sensor_noise	0.00	Bayes survival	0	0.83125	0.00625	0.00000	0.81084
sensor_noise	0.00	Bayes survival	1	0.83750	0.00625	0.00000	0.81709
sensor_noise	0.00	Bayes survival	2	0.86875	0.00625	0.00000	0.84843
sensor_noise	0.00	Bayes survival	3	0.78750	0.00000	0.00000	0.77350
sensor_noise	0.00	Bayes survival	4	0.80000	0.00625	0.00000	0.78007
sensor_noise	0.00	Bayes survival	5	0.74375	0.00000	0.00000	0.73038
sensor_noise	0.00	Bayes survival	6	0.84375	0.00625	0.00000	0.82343
sensor_noise	0.00	Bayes survival	7	0.81875	0.00000	0.00000	0.80492
sensor_noise	0.00	Bayes survival	8	0.80000	0.00000	0.00000	0.78617
sensor_noise	0.00	Bayes survival	9	0.85625	0.00625	0.00000	0.83593
sensor_noise	0.00	CVaR guard	0	0.81250	0.00000	0.00000	0.79354
sensor_noise	0.00	CVaR guard	1	0.81875	0.00000	0.00000	0.79933
sensor_noise	0.00	CVaR guard	2	0.83750	0.00000	0.00000	0.81918
sensor_noise	0.00	CVaR guard	3	0.83125	0.00000	0.00000	0.81223
sensor_noise	0.00	CVaR guard	4	0.82500	0.00000	0.00000	0.80680
sensor_noise	0.00	CVaR guard	5	0.85000	0.00000	0.00000	0.83191

Table 10: Seed-level stress metrics. (continued)

Axis	Level	Method	Seed	Success	Unsafe	Stale	Safety utility
sensor_noise	0.00	CVaR guard	6	0.82500	0.00000	0.00000	0.80610
sensor_noise	0.00	CVaR guard	7	0.82500	0.00000	0.00000	0.80606
sensor_noise	0.00	CVaR guard	8	0.78750	0.00000	0.00000	0.76852
sensor_noise	0.00	CVaR guard	9	0.80000	0.00000	0.00000	0.78139
sensor_noise	0.00	Half-life v5	0	0.78125	0.01250	0.00000	0.75621
sensor_noise	0.00	Half-life v5	1	0.75625	0.00000	0.00000	0.74372
sensor_noise	0.00	Half-life v5	2	0.84375	0.00000	0.00000	0.83120
sensor_noise	0.00	Half-life v5	3	0.75000	0.00000	0.00000	0.73731
sensor_noise	0.00	Half-life v5	4	0.83750	0.00000	0.00000	0.82543
sensor_noise	0.00	Half-life v5	5	0.83750	0.00000	0.00000	0.82497
sensor_noise	0.00	Half-life v5	6	0.84375	0.00000	0.00000	0.83131
sensor_noise	0.00	Half-life v5	7	0.76875	0.00625	0.00000	0.74965
sensor_noise	0.00	Half-life v5	8	0.82500	0.00625	0.00000	0.80606
sensor_noise	0.00	Half-life v5	9	0.86250	0.00000	0.00000	0.84975
sensor_noise	0.00	Oracle	0	0.83125	0.00000	0.00000	0.81736
sensor_noise	0.00	Oracle	1	0.86250	0.00000	0.00000	0.84855
sensor_noise	0.00	Oracle	2	0.84375	0.00000	0.00000	0.83000
sensor_noise	0.00	Oracle	3	0.84375	0.00000	0.00000	0.82994
sensor_noise	0.00	Oracle	4	0.84375	0.00000	0.00000	0.83011
sensor_noise	0.00	Oracle	5	0.79375	0.00000	0.00000	0.77971
sensor_noise	0.00	Oracle	6	0.80625	0.00000	0.00000	0.79236
sensor_noise	0.00	Oracle	7	0.85000	0.01250	0.00000	0.82370
sensor_noise	0.00	Oracle	8	0.84375	0.00000	0.00000	0.83000
sensor_noise	0.00	Oracle	9	0.84375	0.00000	0.00000	0.82986
sensor_noise	0.25	Domain rand.	0	0.48125	0.21250	0.55625	0.07984
sensor_noise	0.25	Domain rand.	1	0.40625	0.21875	0.55625	-0.00141
sensor_noise	0.25	Domain rand.	2	0.49375	0.23750	0.56875	0.06311
sensor_noise	0.25	Domain rand.	3	0.46875	0.22500	0.57500	0.04847
sensor_noise	0.25	Domain rand.	4	0.49375	0.17500	0.57500	0.12342
sensor_noise	0.25	Domain rand.	5	0.43125	0.22500	0.58125	0.00885
sensor_noise	0.25	Domain rand.	6	0.48125	0.20625	0.57500	0.07971
sensor_noise	0.25	Domain rand.	7	0.52500	0.20000	0.57500	0.12970
sensor_noise	0.25	Domain rand.	8	0.43750	0.23750	0.56875	0.00686
sensor_noise	0.25	Domain rand.	9	0.51250	0.21250	0.55625	0.11109
sensor_noise	0.25	Conformal	0	0.86250	0.00625	0.00000	0.84355
sensor_noise	0.25	Conformal	1	0.80625	0.00000	0.00000	0.79327
sensor_noise	0.25	Conformal	2	0.75625	0.00000	0.00000	0.74323
sensor_noise	0.25	Conformal	3	0.78125	0.00000	0.00000	0.76865
sensor_noise	0.25	Conformal	4	0.81875	0.00000	0.00000	0.80579
sensor_noise	0.25	Conformal	5	0.67500	0.00000	0.00000	0.66254
sensor_noise	0.25	Conformal	6	0.80000	0.00000	0.00000	0.78727
sensor_noise	0.25	Conformal	7	0.75000	0.00000	0.00000	0.73752
sensor_noise	0.25	Conformal	8	0.80000	0.00000	0.00000	0.78719
sensor_noise	0.25	Conformal	9	0.83125	0.00625	0.00000	0.81228
sensor_noise	0.25	Ensemble	0	0.81875	0.00000	0.00000	0.80585
sensor_noise	0.25	Ensemble	1	0.75625	0.00625	0.00000	0.73709
sensor_noise	0.25	Ensemble	2	0.75000	0.01250	0.00000	0.72462
sensor_noise	0.25	Ensemble	3	0.85625	0.00000	0.00000	0.84344
sensor_noise	0.25	Ensemble	4	0.76875	0.00000	0.00000	0.75560
sensor_noise	0.25	Ensemble	5	0.83125	0.00000	0.00000	0.81852
sensor_noise	0.25	Ensemble	6	0.86250	0.00000	0.00000	0.84917
sensor_noise	0.25	Ensemble	7	0.78750	0.00000	0.00000	0.77466
sensor_noise	0.25	Ensemble	8	0.77500	0.00000	0.00000	0.76185
sensor_noise	0.25	Ensemble	9	0.85000	0.00000	0.00000	0.83716
sensor_noise	0.25	Bayes survival	0	0.83750	0.00000	0.00000	0.82288
sensor_noise	0.25	Bayes survival	1	0.76250	0.00625	0.00000	0.74181
sensor_noise	0.25	Bayes survival	2	0.79375	0.00000	0.00000	0.77872
sensor_noise	0.25	Bayes survival	3	0.80625	0.01250	0.00000	0.77923
sensor_noise	0.25	Bayes survival	4	0.82500	0.00000	0.00000	0.81007
sensor_noise	0.25	Bayes survival	5	0.82500	0.00000	0.00000	0.81072
sensor_noise	0.25	Bayes survival	6	0.86250	0.00625	0.00000	0.84164
sensor_noise	0.25	Bayes survival	7	0.85625	0.00000	0.00000	0.84180
sensor_noise	0.25	Bayes survival	8	0.81875	0.00625	0.00000	0.79812
sensor_noise	0.25	Bayes survival	9	0.83125	0.00000	0.00000	0.81672
sensor_noise	0.25	CVaR guard	0	0.90625	0.00000	0.00000	0.88539
sensor_noise	0.25	CVaR guard	1	0.81875	0.00000	0.00000	0.79673
sensor_noise	0.25	CVaR guard	2	0.81875	0.00000	0.00000	0.79698
sensor_noise	0.25	CVaR guard	3	0.84375	0.00625	0.00000	0.81644
sensor_noise	0.25	CVaR guard	4	0.81875	0.00000	0.00000	0.79704
sensor_noise	0.25	CVaR guard	5	0.87500	0.00000	0.00000	0.85424
sensor_noise	0.25	CVaR guard	6	0.81875	0.00000	0.00000	0.79795
sensor_noise	0.25	CVaR guard	7	0.81250	0.00625	0.00000	0.78539
sensor_noise	0.25	CVaR guard	8	0.81875	0.00625	0.00000	0.79069
sensor_noise	0.25	CVaR guard	9	0.85000	0.00000	0.00000	0.82876
sensor_noise	0.25	Half-life v5	0	0.80625	0.00000	0.00000	0.79341
sensor_noise	0.25	Half-life v5	1	0.80625	0.00000	0.00000	0.79331
sensor_noise	0.25	Half-life v5	2	0.75625	0.00000	0.00000	0.74341

Table 10: Seed-level stress metrics. (continued)

Axis	Level	Method	Seed	Success	Unsafe	Stale	Safety utility
sensor_noise	0.25	Half-life v5	3	0.80625	0.00000	0.00000	0.79381
sensor_noise	0.25	Half-life v5	4	0.81250	0.00000	0.00000	0.79956
sensor_noise	0.25	Half-life v5	5	0.85000	0.00000	0.00000	0.83731
sensor_noise	0.25	Half-life v5	6	0.86875	0.00000	0.00000	0.85606
sensor_noise	0.25	Half-life v5	7	0.81250	0.00000	0.00000	0.79981
sensor_noise	0.25	Half-life v5	8	0.76250	0.00000	0.00000	0.74975
sensor_noise	0.25	Half-life v5	9	0.87500	0.00000	0.00000	0.86223
sensor_noise	0.25	Oracle	0	0.80625	0.00000	0.00000	0.79224
sensor_noise	0.25	Oracle	1	0.85000	0.00000	0.00000	0.83572
sensor_noise	0.25	Oracle	2	0.83750	0.00000	0.00000	0.82363
sensor_noise	0.25	Oracle	3	0.85000	0.00000	0.00000	0.83603
sensor_noise	0.25	Oracle	4	0.81250	0.00000	0.00000	0.79844
sensor_noise	0.25	Oracle	5	0.82500	0.00000	0.00000	0.81105
sensor_noise	0.25	Oracle	6	0.86875	0.00000	0.00000	0.85486
sensor_noise	0.25	Oracle	7	0.83750	0.00000	0.00000	0.82361
sensor_noise	0.25	Oracle	8	0.79375	0.00625	0.00000	0.77387
sensor_noise	0.25	Oracle	9	0.83125	0.00000	0.00000	0.81744
sensor_noise	0.50	Domain rand.	0	0.53125	0.20000	0.56250	0.14020
sensor_noise	0.50	Domain rand.	1	0.53125	0.19375	0.56875	0.14432
sensor_noise	0.50	Domain rand.	2	0.58125	0.17500	0.56250	0.21518
sensor_noise	0.50	Domain rand.	3	0.52500	0.20625	0.57500	0.12346
sensor_noise	0.50	Domain rand.	4	0.53125	0.21875	0.58125	0.11509
sensor_noise	0.50	Domain rand.	5	0.43125	0.23125	0.57500	0.00473
sensor_noise	0.50	Domain rand.	6	0.50625	0.20000	0.58750	0.10670
sensor_noise	0.50	Domain rand.	7	0.50000	0.18750	0.57500	0.11719
sensor_noise	0.50	Domain rand.	8	0.47500	0.20625	0.56875	0.07558
sensor_noise	0.50	Domain rand.	9	0.60000	0.16250	0.56250	0.24641
sensor_noise	0.50	Conformal	0	0.73125	0.00625	0.00000	0.71186
sensor_noise	0.50	Conformal	1	0.78125	0.00625	0.00000	0.76169
sensor_noise	0.50	Conformal	2	0.85000	0.00000	0.00000	0.83663
sensor_noise	0.50	Conformal	3	0.76875	0.00000	0.00000	0.75509
sensor_noise	0.50	Conformal	4	0.78125	0.00000	0.00000	0.76790
sensor_noise	0.50	Conformal	5	0.81250	0.00000	0.00000	0.79900
sensor_noise	0.50	Conformal	6	0.77500	0.01250	0.00000	0.74891
sensor_noise	0.50	Conformal	7	0.78125	0.00000	0.00000	0.76792
sensor_noise	0.50	Conformal	8	0.83750	0.00000	0.00000	0.82460
sensor_noise	0.50	Conformal	9	0.78125	0.00000	0.00000	0.76812
sensor_noise	0.50	Ensemble	0	0.80000	0.00000	0.00000	0.78669
sensor_noise	0.50	Ensemble	1	0.81875	0.00000	0.00000	0.80521
sensor_noise	0.50	Ensemble	2	0.80000	0.00625	0.00000	0.78055
sensor_noise	0.50	Ensemble	3	0.78750	0.00000	0.00000	0.77394
sensor_noise	0.50	Ensemble	4	0.81250	0.00000	0.00000	0.79879
sensor_noise	0.50	Ensemble	5	0.80000	0.00000	0.00000	0.78640
sensor_noise	0.50	Ensemble	6	0.76875	0.00000	0.00000	0.75498
sensor_noise	0.50	Ensemble	7	0.78750	0.00000	0.00000	0.77406
sensor_noise	0.50	Ensemble	8	0.75000	0.00625	0.00000	0.73053
sensor_noise	0.50	Ensemble	9	0.80000	0.00000	0.00000	0.78644
sensor_noise	0.50	Bayes survival	0	0.81875	0.00625	0.00000	0.79735
sensor_noise	0.50	Bayes survival	1	0.83750	0.00000	0.00000	0.82203
sensor_noise	0.50	Bayes survival	2	0.86875	0.00625	0.00000	0.84700
sensor_noise	0.50	Bayes survival	3	0.88750	0.00625	0.00000	0.86542
sensor_noise	0.50	Bayes survival	4	0.83125	0.00625	0.00000	0.80948
sensor_noise	0.50	Bayes survival	5	0.80000	0.00625	0.00000	0.77796
sensor_noise	0.50	Bayes survival	6	0.78125	0.00625	0.00000	0.75937
sensor_noise	0.50	Bayes survival	7	0.80625	0.00625	0.00000	0.78458
sensor_noise	0.50	Bayes survival	8	0.81250	0.00000	0.00000	0.79763
sensor_noise	0.50	Bayes survival	9	0.81875	0.00000	0.00000	0.80337
sensor_noise	0.50	CVaR guard	0	0.80000	0.00625	0.00000	0.77019
sensor_noise	0.50	CVaR guard	1	0.84375	0.00000	0.00000	0.81915
sensor_noise	0.50	CVaR guard	2	0.85000	0.00000	0.00000	0.82579
sensor_noise	0.50	CVaR guard	3	0.79375	0.00000	0.00000	0.76890
sensor_noise	0.50	CVaR guard	4	0.81875	0.00000	0.00000	0.79400
sensor_noise	0.50	CVaR guard	5	0.86250	0.00000	0.00000	0.83848
sensor_noise	0.50	CVaR guard	6	0.88750	0.00000	0.00000	0.86335
sensor_noise	0.50	CVaR guard	7	0.87500	0.00000	0.00000	0.85083
sensor_noise	0.50	CVaR guard	8	0.83750	0.00000	0.00000	0.81412
sensor_noise	0.50	CVaR guard	9	0.81875	0.00000	0.00000	0.79355
sensor_noise	0.50	Half-life v5	0	0.88750	0.00000	0.00000	0.87450
sensor_noise	0.50	Half-life v5	1	0.82500	0.00000	0.00000	0.81183
sensor_noise	0.50	Half-life v5	2	0.81250	0.00000	0.00000	0.79960
sensor_noise	0.50	Half-life v5	3	0.80625	0.01875	0.00000	0.77446
sensor_noise	0.50	Half-life v5	4	0.85000	0.00000	0.00000	0.83675
sensor_noise	0.50	Half-life v5	5	0.80625	0.00000	0.00000	0.79294
sensor_noise	0.50	Half-life v5	6	0.79375	0.00000	0.00000	0.78060
sensor_noise	0.50	Half-life v5	7	0.81250	0.00000	0.00000	0.79939
sensor_noise	0.50	Half-life v5	8	0.80625	0.00000	0.00000	0.79358
sensor_noise	0.50	Half-life v5	9	0.76875	0.00000	0.00000	0.75575

Table 10: Seed-level stress metrics. (continued)

Axis	Level	Method	Seed	Success	Unsafe	Stale	Safety utility
sensor_noise	0.50	Oracle	0	0.83750	0.00000	0.00000	0.82361
sensor_noise	0.50	Oracle	1	0.81250	0.00625	0.00000	0.79164
sensor_noise	0.50	Oracle	2	0.90625	0.00000	0.00000	0.89238
sensor_noise	0.50	Oracle	3	0.78750	0.00625	0.00000	0.76708
sensor_noise	0.50	Oracle	4	0.78750	0.00000	0.00000	0.77346
sensor_noise	0.50	Oracle	5	0.82500	0.00000	0.00000	0.81072
sensor_noise	0.50	Oracle	6	0.76875	0.00000	0.00000	0.75432
sensor_noise	0.50	Oracle	7	0.89375	0.00000	0.00000	0.87994
sensor_noise	0.50	Oracle	8	0.78125	0.00000	0.00000	0.76694
sensor_noise	0.50	Oracle	9	0.83750	0.00625	0.00000	0.81706
sensor_noise	0.75	Domain rand.	0	0.49375	0.20625	0.56250	0.09646
sensor_noise	0.75	Domain rand.	1	0.50000	0.18125	0.55625	0.12981
sensor_noise	0.75	Domain rand.	2	0.50000	0.23125	0.57500	0.07348
sensor_noise	0.75	Domain rand.	3	0.51250	0.20625	0.56250	0.11521
sensor_noise	0.75	Domain rand.	4	0.46875	0.19375	0.58125	0.07757
sensor_noise	0.75	Domain rand.	5	0.47500	0.20625	0.58750	0.06921
sensor_noise	0.75	Domain rand.	6	0.53125	0.20625	0.58125	0.12758
sensor_noise	0.75	Domain rand.	7	0.54375	0.20625	0.56875	0.14433
sensor_noise	0.75	Domain rand.	8	0.51250	0.17500	0.56250	0.14642
sensor_noise	0.75	Domain rand.	9	0.50625	0.17500	0.56875	0.13805
sensor_noise	0.75	Conformal	0	0.80625	0.00000	0.00000	0.79232
sensor_noise	0.75	Conformal	1	0.77500	0.00625	0.00000	0.75464
sensor_noise	0.75	Conformal	2	0.83125	0.00000	0.00000	0.81723
sensor_noise	0.75	Conformal	3	0.75000	0.00000	0.00000	0.73609
sensor_noise	0.75	Conformal	4	0.77500	0.00000	0.00000	0.76129
sensor_noise	0.75	Conformal	5	0.83125	0.00000	0.00000	0.81744
sensor_noise	0.75	Conformal	6	0.76875	0.02500	0.00000	0.72976
sensor_noise	0.75	Conformal	7	0.80625	0.00000	0.00000	0.79203
sensor_noise	0.75	Conformal	8	0.80000	0.00625	0.00000	0.77968
sensor_noise	0.75	Conformal	9	0.85000	0.00625	0.00000	0.82956
sensor_noise	0.75	Ensemble	0	0.79375	0.00625	0.00000	0.77349
sensor_noise	0.75	Ensemble	1	0.80000	0.00000	0.00000	0.78592
sensor_noise	0.75	Ensemble	2	0.81250	0.00000	0.00000	0.79863
sensor_noise	0.75	Ensemble	3	0.76875	0.00000	0.00000	0.75498
sensor_noise	0.75	Ensemble	4	0.83125	0.00000	0.00000	0.81765
sensor_noise	0.75	Ensemble	5	0.78750	0.00625	0.00000	0.76763
sensor_noise	0.75	Ensemble	6	0.80000	0.00000	0.00000	0.78604
sensor_noise	0.75	Ensemble	7	0.82500	0.00625	0.00000	0.80437
sensor_noise	0.75	Ensemble	8	0.77500	0.00000	0.00000	0.76100
sensor_noise	0.75	Ensemble	9	0.83750	0.00000	0.00000	0.82344
sensor_noise	0.75	Bayes survival	0	0.83125	0.00000	0.00000	0.81533
sensor_noise	0.75	Bayes survival	1	0.86250	0.00000	0.00000	0.84639
sensor_noise	0.75	Bayes survival	2	0.85000	0.00000	0.00000	0.83376
sensor_noise	0.75	Bayes survival	3	0.85625	0.00625	0.00000	0.83390
sensor_noise	0.75	Bayes survival	4	0.85625	0.00000	0.00000	0.84002
sensor_noise	0.75	Bayes survival	5	0.86250	0.00000	0.00000	0.84610
sensor_noise	0.75	Bayes survival	6	0.80625	0.00000	0.00000	0.79014
sensor_noise	0.75	Bayes survival	7	0.83750	0.00000	0.00000	0.82032
sensor_noise	0.75	Bayes survival	8	0.85625	0.00000	0.00000	0.83979
sensor_noise	0.75	Bayes survival	9	0.81250	0.00000	0.00000	0.79604
sensor_noise	0.75	CVaR guard	0	0.87500	0.00000	0.00000	0.84798
sensor_noise	0.75	CVaR guard	1	0.84375	0.00000	0.00000	0.81717
sensor_noise	0.75	CVaR guard	2	0.87500	0.00000	0.00000	0.84763
sensor_noise	0.75	CVaR guard	3	0.83750	0.00000	0.00000	0.80999
sensor_noise	0.75	CVaR guard	4	0.85000	0.00625	0.00000	0.81654
sensor_noise	0.75	CVaR guard	5	0.86875	0.00000	0.00000	0.84130
sensor_noise	0.75	CVaR guard	6	0.83125	0.00625	0.00000	0.79779
sensor_noise	0.75	CVaR guard	7	0.85000	0.00000	0.00000	0.82162
sensor_noise	0.75	CVaR guard	8	0.86250	0.00625	0.00000	0.82846
sensor_noise	0.75	CVaR guard	9	0.86250	0.00000	0.00000	0.83551
sensor_noise	0.75	Half-life v5	0	0.76250	0.00000	0.00000	0.74914
sensor_noise	0.75	Half-life v5	1	0.83125	0.00000	0.00000	0.81783
sensor_noise	0.75	Half-life v5	2	0.79375	0.00625	0.00000	0.77417
sensor_noise	0.75	Half-life v5	3	0.78125	0.00625	0.00000	0.76186
sensor_noise	0.75	Half-life v5	4	0.86250	0.00000	0.00000	0.84927
sensor_noise	0.75	Half-life v5	5	0.80000	0.00000	0.00000	0.78669
sensor_noise	0.75	Half-life v5	6	0.81875	0.00000	0.00000	0.80550
sensor_noise	0.75	Half-life v5	7	0.80625	0.00000	0.00000	0.79271
sensor_noise	0.75	Half-life v5	8	0.83125	0.00000	0.00000	0.81779
sensor_noise	0.75	Half-life v5	9	0.82500	0.00000	0.00000	0.81158
sensor_noise	0.75	Oracle	0	0.82500	0.00000	0.00000	0.81080
sensor_noise	0.75	Oracle	1	0.85625	0.00625	0.00000	0.83603
sensor_noise	0.75	Oracle	2	0.81875	0.00625	0.00000	0.79818
sensor_noise	0.75	Oracle	3	0.83750	0.00625	0.00000	0.81728
sensor_noise	0.75	Oracle	4	0.85625	0.00000	0.00000	0.84250
sensor_noise	0.75	Oracle	5	0.83750	0.00000	0.00000	0.82347
sensor_noise	0.75	Oracle	6	0.78125	0.00000	0.00000	0.76719

Table 10: Seed-level stress metrics. (continued)

Axis	Level	Method	Seed	Success	Unsafe	Stale	Safety utility
sensor_noise	0.75	Oracle	7	0.78750	0.00000	0.00000	0.77322
sensor_noise	0.75	Oracle	8	0.80625	0.00000	0.00000	0.79199
sensor_noise	0.75	Oracle	9	0.83750	0.00000	0.00000	0.82328
sensor_noise	1.00	Domain rand.	0	0.55000	0.21875	0.56875	0.13809
sensor_noise	1.00	Domain rand.	1	0.53125	0.16875	0.56875	0.16929
sensor_noise	1.00	Domain rand.	2	0.48125	0.24375	0.57500	0.04224
sensor_noise	1.00	Domain rand.	3	0.50000	0.24375	0.58125	0.05887
sensor_noise	1.00	Domain rand.	4	0.48750	0.19375	0.58750	0.09419
sensor_noise	1.00	Domain rand.	5	0.55625	0.17500	0.59375	0.17955
sensor_noise	1.00	Domain rand.	6	0.53750	0.15625	0.57500	0.18591
sensor_noise	1.00	Domain rand.	7	0.48125	0.22500	0.56250	0.06523
sensor_noise	1.00	Domain rand.	8	0.45000	0.22500	0.56250	0.03398
sensor_noise	1.00	Domain rand.	9	0.45625	0.23750	0.55625	0.02986
sensor_noise	1.00	Conformal	0	0.76250	0.00000	0.00000	0.74732
sensor_noise	1.00	Conformal	1	0.80000	0.00625	0.00000	0.77897
sensor_noise	1.00	Conformal	2	0.81250	0.00000	0.00000	0.79728
sensor_noise	1.00	Conformal	3	0.76250	0.00000	0.00000	0.74774
sensor_noise	1.00	Conformal	4	0.81250	0.00625	0.00000	0.79152
sensor_noise	1.00	Conformal	5	0.74375	0.00625	0.00000	0.72268
sensor_noise	1.00	Conformal	6	0.75625	0.00000	0.00000	0.74109
sensor_noise	1.00	Conformal	7	0.77500	0.00000	0.00000	0.75974
sensor_noise	1.00	Conformal	8	0.81875	0.00625	0.00000	0.79793
sensor_noise	1.00	Conformal	9	0.86250	0.00000	0.00000	0.84747
sensor_noise	1.00	Ensemble	0	0.86250	0.00000	0.00000	0.84724
sensor_noise	1.00	Ensemble	1	0.78750	0.00000	0.00000	0.77305
sensor_noise	1.00	Ensemble	2	0.81250	0.00000	0.00000	0.79768
sensor_noise	1.00	Ensemble	3	0.78750	0.01250	0.00000	0.76046
sensor_noise	1.00	Ensemble	4	0.81250	0.00000	0.00000	0.79780
sensor_noise	1.00	Ensemble	5	0.86250	0.00000	0.00000	0.84796
sensor_noise	1.00	Ensemble	6	0.81875	0.00625	0.00000	0.79775
sensor_noise	1.00	Ensemble	7	0.81250	0.00625	0.00000	0.79129
sensor_noise	1.00	Ensemble	8	0.83750	0.00000	0.00000	0.82299
sensor_noise	1.00	Ensemble	9	0.80625	0.00000	0.00000	0.79145
sensor_noise	1.00	Bayes survival	0	0.82500	0.00000	0.00000	0.80750
sensor_noise	1.00	Bayes survival	1	0.88125	0.00625	0.00000	0.85724
sensor_noise	1.00	Bayes survival	2	0.81250	0.00000	0.00000	0.79455
sensor_noise	1.00	Bayes survival	3	0.78750	0.00000	0.00000	0.76999
sensor_noise	1.00	Bayes survival	4	0.80625	0.00000	0.00000	0.78948
sensor_noise	1.00	Bayes survival	5	0.81250	0.00000	0.00000	0.79554
sensor_noise	1.00	Bayes survival	6	0.84375	0.00625	0.00000	0.81999
sensor_noise	1.00	Bayes survival	7	0.83750	0.00000	0.00000	0.81972
sensor_noise	1.00	Bayes survival	8	0.82500	0.00000	0.00000	0.80751
sensor_noise	1.00	Bayes survival	9	0.83750	0.00000	0.00000	0.81969
sensor_noise	1.00	CVaR guard	0	0.87500	0.00000	0.00000	0.84510
sensor_noise	1.00	CVaR guard	1	0.84375	0.00000	0.00000	0.81312
sensor_noise	1.00	CVaR guard	2	0.90000	0.00625	0.00000	0.86313
sensor_noise	1.00	CVaR guard	3	0.83750	0.00000	0.00000	0.80789
sensor_noise	1.00	CVaR guard	4	0.85000	0.00000	0.00000	0.82039
sensor_noise	1.00	CVaR guard	5	0.84375	0.00000	0.00000	0.81312
sensor_noise	1.00	CVaR guard	6	0.88125	0.00000	0.00000	0.85091
sensor_noise	1.00	CVaR guard	7	0.83750	0.00000	0.00000	0.80731
sensor_noise	1.00	CVaR guard	8	0.83750	0.00000	0.00000	0.80782
sensor_noise	1.00	CVaR guard	9	0.85000	0.00000	0.00000	0.82010
sensor_noise	1.00	Half-life v5	0	0.86875	0.00000	0.00000	0.85438
sensor_noise	1.00	Half-life v5	1	0.77500	0.00625	0.00000	0.75522
sensor_noise	1.00	Half-life v5	2	0.80000	0.00000	0.00000	0.78621
sensor_noise	1.00	Half-life v5	3	0.82500	0.00000	0.00000	0.81094
sensor_noise	1.00	Half-life v5	4	0.81250	0.00000	0.00000	0.79885
sensor_noise	1.00	Half-life v5	5	0.80000	0.00625	0.00000	0.77986
sensor_noise	1.00	Half-life v5	6	0.79375	0.00625	0.00000	0.77388
sensor_noise	1.00	Half-life v5	7	0.80000	0.00625	0.00000	0.77949
sensor_noise	1.00	Half-life v5	8	0.86875	0.00000	0.00000	0.85448
sensor_noise	1.00	Half-life v5	9	0.82500	0.00000	0.00000	0.81084
sensor_noise	1.00	Oracle	0	0.83125	0.00625	0.00000	0.81039
sensor_noise	1.00	Oracle	1	0.83750	0.00000	0.00000	0.82332
sensor_noise	1.00	Oracle	2	0.82500	0.00000	0.00000	0.81065
sensor_noise	1.00	Oracle	3	0.83125	0.00000	0.00000	0.81688
sensor_noise	1.00	Oracle	4	0.83750	0.00000	0.00000	0.82330
sensor_noise	1.00	Oracle	5	0.88750	0.00000	0.00000	0.87338
sensor_noise	1.00	Oracle	6	0.83125	0.00625	0.00000	0.81060
sensor_noise	1.00	Oracle	7	0.81250	0.00000	0.00000	0.79799
sensor_noise	1.00	Oracle	8	0.86875	0.00000	0.00000	0.85424
sensor_noise	1.00	Oracle	9	0.81250	0.00000	0.00000	0.79819
sensor_noise	1.25	Domain rand.	0	0.51875	0.15625	0.56875	0.16928
sensor_noise	1.25	Domain rand.	1	0.42500	0.20000	0.56250	0.03395
sensor_noise	1.25	Domain rand.	2	0.53750	0.21875	0.57500	0.12347
sensor_noise	1.25	Domain rand.	3	0.55625	0.18750	0.57500	0.17344

Table 10: Seed-level stress metrics. (continued)

Axis	Level	Method	Seed	Success	Unsafe	Stale	Safety utility
sensor_noise	1.25	Domain rand.	4	0.50625	0.18750	0.57500	0.12344
sensor_noise	1.25	Domain rand.	5	0.54375	0.21250	0.59375	0.12959
sensor_noise	1.25	Domain rand.	6	0.60000	0.20625	0.58750	0.19421
sensor_noise	1.25	Domain rand.	7	0.50625	0.18750	0.58750	0.11919
sensor_noise	1.25	Domain rand.	8	0.50625	0.23750	0.58125	0.07136
sensor_noise	1.25	Domain rand.	9	0.48125	0.20625	0.57500	0.07971
sensor_noise	1.25	Conformal	0	0.76250	0.00000	0.00000	0.74651
sensor_noise	1.25	Conformal	1	0.85000	0.00000	0.00000	0.83412
sensor_noise	1.25	Conformal	2	0.83750	0.00000	0.00000	0.82164
sensor_noise	1.25	Conformal	3	0.79375	0.00625	0.00000	0.77136
sensor_noise	1.25	Conformal	4	0.81875	0.00625	0.00000	0.79666
sensor_noise	1.25	Conformal	5	0.80000	0.00625	0.00000	0.77771
sensor_noise	1.25	Conformal	6	0.79375	0.00625	0.00000	0.77130
sensor_noise	1.25	Conformal	7	0.77500	0.00000	0.00000	0.75962
sensor_noise	1.25	Conformal	8	0.81875	0.00000	0.00000	0.80272
sensor_noise	1.25	Conformal	9	0.75000	0.00625	0.00000	0.72750
sensor_noise	1.25	Ensemble	0	0.80000	0.00000	0.00000	0.78460
sensor_noise	1.25	Ensemble	1	0.86250	0.00625	0.00000	0.84069
sensor_noise	1.25	Ensemble	2	0.84375	0.00000	0.00000	0.82812
sensor_noise	1.25	Ensemble	3	0.85000	0.00000	0.00000	0.83416
sensor_noise	1.25	Ensemble	4	0.85000	0.00625	0.00000	0.82839
sensor_noise	1.25	Ensemble	5	0.83750	0.00000	0.00000	0.82160
sensor_noise	1.25	Ensemble	6	0.78750	0.00000	0.00000	0.77162
sensor_noise	1.25	Ensemble	7	0.74375	0.00000	0.00000	0.72895
sensor_noise	1.25	Ensemble	8	0.81250	0.00000	0.00000	0.79714
sensor_noise	1.25	Ensemble	9	0.80000	0.00000	0.00000	0.78383
sensor_noise	1.25	Bayes survival	0	0.85625	0.00000	0.00000	0.83696
sensor_noise	1.25	Bayes survival	1	0.83125	0.00000	0.00000	0.81208
sensor_noise	1.25	Bayes survival	2	0.81250	0.00000	0.00000	0.79331
sensor_noise	1.25	Bayes survival	3	0.86250	0.00000	0.00000	0.84366
sensor_noise	1.25	Bayes survival	4	0.87500	0.00000	0.00000	0.85622
sensor_noise	1.25	Bayes survival	5	0.85625	0.00000	0.00000	0.83818
sensor_noise	1.25	Bayes survival	6	0.80625	0.00000	0.00000	0.78780
sensor_noise	1.25	Bayes survival	7	0.80000	0.00000	0.00000	0.78151
sensor_noise	1.25	Bayes survival	8	0.82500	0.00000	0.00000	0.80616
sensor_noise	1.25	Bayes survival	9	0.86875	0.00000	0.00000	0.84892
sensor_noise	1.25	CVaR guard	0	0.88750	0.00000	0.00000	0.85514
sensor_noise	1.25	CVaR guard	1	0.88750	0.00000	0.00000	0.85558
sensor_noise	1.25	CVaR guard	2	0.83125	0.00625	0.00000	0.79222
sensor_noise	1.25	CVaR guard	3	0.82500	0.00000	0.00000	0.79279
sensor_noise	1.25	CVaR guard	4	0.87500	0.00000	0.00000	0.84250
sensor_noise	1.25	CVaR guard	5	0.83750	0.00000	0.00000	0.80428
sensor_noise	1.25	CVaR guard	6	0.86875	0.00000	0.00000	0.83639
sensor_noise	1.25	CVaR guard	7	0.81250	0.00000	0.00000	0.77928
sensor_noise	1.25	CVaR guard	8	0.88125	0.00000	0.00000	0.84788
sensor_noise	1.25	CVaR guard	9	0.83750	0.00000	0.00000	0.80485
sensor_noise	1.25	Half-life v5	0	0.85000	0.00000	0.00000	0.83588
sensor_noise	1.25	Half-life v5	1	0.79375	0.00625	0.00000	0.77322
sensor_noise	1.25	Half-life v5	2	0.83750	0.00000	0.00000	0.82296
sensor_noise	1.25	Half-life v5	3	0.79375	0.00000	0.00000	0.77928
sensor_noise	1.25	Half-life v5	4	0.78750	0.01250	0.00000	0.76085
sensor_noise	1.25	Half-life v5	5	0.76250	0.00000	0.00000	0.74857
sensor_noise	1.25	Half-life v5	6	0.78750	0.00625	0.00000	0.76695
sensor_noise	1.25	Half-life v5	7	0.84375	0.00000	0.00000	0.83038
sensor_noise	1.25	Half-life v5	8	0.81250	0.00625	0.00000	0.79207
sensor_noise	1.25	Half-life v5	9	0.83125	0.00000	0.00000	0.81632
sensor_noise	1.25	Oracle	0	0.84375	0.00000	0.00000	0.82930
sensor_noise	1.25	Oracle	1	0.83750	0.00000	0.00000	0.82324
sensor_noise	1.25	Oracle	2	0.88125	0.00000	0.00000	0.86676
sensor_noise	1.25	Oracle	3	0.81250	0.00625	0.00000	0.79199
sensor_noise	1.25	Oracle	4	0.81875	0.00000	0.00000	0.80430
sensor_noise	1.25	Oracle	5	0.91250	0.00000	0.00000	0.89799
sensor_noise	1.25	Oracle	6	0.85000	0.00000	0.00000	0.83557
sensor_noise	1.25	Oracle	7	0.85000	0.00000	0.00000	0.83596
sensor_noise	1.25	Oracle	8	0.84375	0.00000	0.00000	0.82940
sensor_noise	1.25	Oracle	9	0.85625	0.00625	0.00000	0.83564
sensor_noise	1.50	Domain rand.	0	0.48125	0.24375	0.56875	0.04437
sensor_noise	1.50	Domain rand.	1	0.53750	0.22500	0.57500	0.11722
sensor_noise	1.50	Domain rand.	2	0.55000	0.22500	0.59375	0.12335
sensor_noise	1.50	Domain rand.	3	0.47500	0.17500	0.58125	0.10255
sensor_noise	1.50	Domain rand.	4	0.43750	0.25625	0.58125	-0.01612
sensor_noise	1.50	Domain rand.	5	0.52500	0.18125	0.58125	0.14631
sensor_noise	1.50	Domain rand.	6	0.50000	0.20625	0.57500	0.09846
sensor_noise	1.50	Domain rand.	7	0.51250	0.23125	0.56875	0.08811
sensor_noise	1.50	Domain rand.	8	0.48125	0.23125	0.56875	0.05686
sensor_noise	1.50	Domain rand.	9	0.51875	0.21875	0.56875	0.10684
sensor_noise	1.50	Conformal	0	0.78125	0.00000	0.00000	0.76367

Table 10: Seed-level stress metrics. (continued)

Axis	Level	Method	Seed	Success	Unsafe	Stale	Safety utility
sensor_noise	1.50	Conformal	1	0.81875	0.00000	0.00000	0.80192
sensor_noise	1.50	Conformal	2	0.82500	0.00000	0.00000	0.80734
sensor_noise	1.50	Conformal	3	0.78125	0.00625	0.00000	0.75809
sensor_noise	1.50	Conformal	4	0.78750	0.00625	0.00000	0.76383
sensor_noise	1.50	Conformal	5	0.84375	0.00000	0.00000	0.82667
sensor_noise	1.50	Conformal	6	0.79375	0.00000	0.00000	0.77694
sensor_noise	1.50	Conformal	7	0.82500	0.00000	0.00000	0.80689
sensor_noise	1.50	Conformal	8	0.86875	0.00000	0.00000	0.85144
sensor_noise	1.50	Conformal	9	0.85625	0.00000	0.00000	0.83921
sensor_noise	1.50	Ensemble	0	0.81250	0.00000	0.00000	0.79523
sensor_noise	1.50	Ensemble	1	0.85000	0.00625	0.00000	0.82769
sensor_noise	1.50	Ensemble	2	0.83750	0.00625	0.00000	0.81419
sensor_noise	1.50	Ensemble	3	0.80000	0.00000	0.00000	0.78371
sensor_noise	1.50	Ensemble	4	0.77500	0.00000	0.00000	0.75803
sensor_noise	1.50	Ensemble	5	0.85625	0.00625	0.00000	0.83371
sensor_noise	1.50	Ensemble	6	0.80625	0.00000	0.00000	0.78973
sensor_noise	1.50	Ensemble	7	0.76875	0.00000	0.00000	0.75145
sensor_noise	1.50	Ensemble	8	0.80000	0.01250	0.00000	0.77066
sensor_noise	1.50	Ensemble	9	0.80000	0.00000	0.00000	0.78342
sensor_noise	1.50	Bayes survival	0	0.83125	0.00000	0.00000	0.81057
sensor_noise	1.50	Bayes survival	1	0.81875	0.00000	0.00000	0.79888
sensor_noise	1.50	Bayes survival	2	0.82500	0.00000	0.00000	0.80447
sensor_noise	1.50	Bayes survival	3	0.87500	0.00000	0.00000	0.85486
sensor_noise	1.50	Bayes survival	4	0.85000	0.00625	0.00000	0.82310
sensor_noise	1.50	Bayes survival	5	0.85000	0.00000	0.00000	0.82992
sensor_noise	1.50	Bayes survival	6	0.86250	0.00000	0.00000	0.84224
sensor_noise	1.50	Bayes survival	7	0.85000	0.00000	0.00000	0.82893
sensor_noise	1.50	Bayes survival	8	0.80625	0.00000	0.00000	0.78609
sensor_noise	1.50	Bayes survival	9	0.85000	0.00000	0.00000	0.82976
sensor_noise	1.50	CVaR guard	0	0.87500	0.00000	0.00000	0.84062
sensor_noise	1.50	CVaR guard	1	0.84375	0.00000	0.00000	0.80966
sensor_noise	1.50	CVaR guard	2	0.86250	0.00000	0.00000	0.82841
sensor_noise	1.50	CVaR guard	3	0.84375	0.00000	0.00000	0.80995
sensor_noise	1.50	CVaR guard	4	0.86250	0.00000	0.00000	0.82870
sensor_noise	1.50	CVaR guard	5	0.83125	0.00000	0.00000	0.79589
sensor_noise	1.50	CVaR guard	6	0.91250	0.00000	0.00000	0.87841
sensor_noise	1.50	CVaR guard	7	0.88750	0.00000	0.00000	0.85312
sensor_noise	1.50	CVaR guard	8	0.88125	0.00625	0.00000	0.84077
sensor_noise	1.50	CVaR guard	9	0.88125	0.00000	0.00000	0.84716
sensor_noise	1.50	Half-life v5	0	0.82500	0.00625	0.00000	0.80410
sensor_noise	1.50	Half-life v5	1	0.85000	0.00000	0.00000	0.83584
sensor_noise	1.50	Half-life v5	2	0.80000	0.00000	0.00000	0.78515
sensor_noise	1.50	Half-life v5	3	0.80000	0.00625	0.00000	0.77966
sensor_noise	1.50	Half-life v5	4	0.81250	0.00000	0.00000	0.79786
sensor_noise	1.50	Half-life v5	5	0.86875	0.00000	0.00000	0.85419
sensor_noise	1.50	Half-life v5	6	0.81250	0.00000	0.00000	0.79842
sensor_noise	1.50	Half-life v5	7	0.81875	0.00625	0.00000	0.79818
sensor_noise	1.50	Half-life v5	8	0.83750	0.00000	0.00000	0.82303
sensor_noise	1.50	Half-life v5	9	0.87500	0.00000	0.00000	0.86082
sensor_noise	1.50	Oracle	0	0.83750	0.00000	0.00000	0.82307
sensor_noise	1.50	Oracle	1	0.81875	0.00000	0.00000	0.80455
sensor_noise	1.50	Oracle	2	0.88750	0.00000	0.00000	0.87276
sensor_noise	1.50	Oracle	3	0.80000	0.00000	0.00000	0.78557
sensor_noise	1.50	Oracle	4	0.83750	0.00625	0.00000	0.81685
sensor_noise	1.50	Oracle	5	0.83750	0.00000	0.00000	0.82307
sensor_noise	1.50	Oracle	6	0.80000	0.00000	0.00000	0.78524
sensor_noise	1.50	Oracle	7	0.82500	0.00000	0.00000	0.81043
sensor_noise	1.50	Oracle	8	0.88125	0.00000	0.00000	0.86697
sensor_noise	1.50	Oracle	9	0.85625	0.00625	0.00000	0.83522
mode_chatter	0.00	Domain rand.	0	0.59375	0.15625	0.55625	0.24853
mode_chatter	0.00	Domain rand.	1	0.40625	0.20625	0.55625	0.01108
mode_chatter	0.00	Domain rand.	2	0.53125	0.17500	0.56875	0.16305
mode_chatter	0.00	Domain rand.	3	0.51875	0.17500	0.56875	0.15055
mode_chatter	0.00	Domain rand.	4	0.50000	0.17500	0.56875	0.13180
mode_chatter	0.00	Domain rand.	5	0.51250	0.20625	0.58125	0.10883
mode_chatter	0.00	Domain rand.	6	0.48750	0.18125	0.56875	0.11306
mode_chatter	0.00	Domain rand.	7	0.55000	0.18125	0.56250	0.17768
mode_chatter	0.00	Domain rand.	8	0.54375	0.15000	0.55625	0.20477
mode_chatter	0.00	Domain rand.	9	0.48750	0.25000	0.55000	0.05075
mode_chatter	0.00	Conformal	0	0.82500	0.00000	0.00000	0.81235
mode_chatter	0.00	Conformal	1	0.80000	0.00625	0.00000	0.78124
mode_chatter	0.00	Conformal	2	0.82500	0.00000	0.00000	0.81200
mode_chatter	0.00	Conformal	3	0.80625	0.00000	0.00000	0.79315
mode_chatter	0.00	Conformal	4	0.74375	0.00625	0.00000	0.72486
mode_chatter	0.00	Conformal	5	0.80625	0.00625	0.00000	0.78744
mode_chatter	0.00	Conformal	6	0.80625	0.00000	0.00000	0.79352
mode_chatter	0.00	Conformal	7	0.80000	0.00000	0.00000	0.78735

Table 10: Seed-level stress metrics. (continued)

Axis	Level	Method	Seed	Success	Unsafe	Stale	Safety utility
mode_chatter	0.00	Conformal	8	0.80625	0.00625	0.00000	0.78717
mode_chatter	0.00	Conformal	9	0.86250	0.00000	0.00000	0.84962
mode_chatter	0.00	Ensemble	0	0.78750	0.00625	0.00000	0.76859
mode_chatter	0.00	Ensemble	1	0.76875	0.00625	0.00000	0.74961
mode_chatter	0.00	Ensemble	2	0.81250	0.00000	0.00000	0.79908
mode_chatter	0.00	Ensemble	3	0.79375	0.00000	0.00000	0.78052
mode_chatter	0.00	Ensemble	4	0.76250	0.00625	0.00000	0.74359
mode_chatter	0.00	Ensemble	5	0.79375	0.00625	0.00000	0.77467
mode_chatter	0.00	Ensemble	6	0.80625	0.00000	0.00000	0.79325
mode_chatter	0.00	Ensemble	7	0.75625	0.00625	0.00000	0.73694
mode_chatter	0.00	Ensemble	8	0.81250	0.00625	0.00000	0.79300
mode_chatter	0.00	Ensemble	9	0.74375	0.00625	0.00000	0.72442
mode_chatter	0.00	Bayes survival	0	0.85000	0.00000	0.00000	0.83534
mode_chatter	0.00	Bayes survival	1	0.83750	0.00000	0.00000	0.82257
mode_chatter	0.00	Bayes survival	2	0.78750	0.01250	0.00000	0.76031
mode_chatter	0.00	Bayes survival	3	0.85625	0.00625	0.00000	0.83551
mode_chatter	0.00	Bayes survival	4	0.81250	0.00625	0.00000	0.79178
mode_chatter	0.00	Bayes survival	5	0.82500	0.00000	0.00000	0.81059
mode_chatter	0.00	Bayes survival	6	0.85000	0.00000	0.00000	0.83520
mode_chatter	0.00	Bayes survival	7	0.79375	0.01250	0.00000	0.76679
mode_chatter	0.00	Bayes survival	8	0.81250	0.00000	0.00000	0.79786
mode_chatter	0.00	Bayes survival	9	0.81250	0.00625	0.00000	0.79122
mode_chatter	0.00	CVaR guard	0	0.83125	0.00000	0.00000	0.80913
mode_chatter	0.00	CVaR guard	1	0.83750	0.00000	0.00000	0.81503
mode_chatter	0.00	CVaR guard	2	0.87500	0.00000	0.00000	0.85203
mode_chatter	0.00	CVaR guard	3	0.81250	0.00000	0.00000	0.78996
mode_chatter	0.00	CVaR guard	4	0.84375	0.00000	0.00000	0.82225
mode_chatter	0.00	CVaR guard	5	0.87500	0.00625	0.00000	0.84723
mode_chatter	0.00	CVaR guard	6	0.78125	0.00000	0.00000	0.75935
mode_chatter	0.00	CVaR guard	7	0.88125	0.00000	0.00000	0.85921
mode_chatter	0.00	CVaR guard	8	0.86250	0.00000	0.00000	0.84031
mode_chatter	0.00	CVaR guard	9	0.81875	0.00000	0.00000	0.79625
mode_chatter	0.00	Half-life v5	0	0.78125	0.00000	0.00000	0.76887
mode_chatter	0.00	Half-life v5	1	0.76875	0.00625	0.00000	0.74981
mode_chatter	0.00	Half-life v5	2	0.84375	0.00625	0.00000	0.82481
mode_chatter	0.00	Half-life v5	3	0.81250	0.00625	0.00000	0.79340
mode_chatter	0.00	Half-life v5	4	0.81250	0.00000	0.00000	0.80020
mode_chatter	0.00	Half-life v5	5	0.80625	0.00000	0.00000	0.79364
mode_chatter	0.00	Half-life v5	6	0.81250	0.00625	0.00000	0.79354
mode_chatter	0.00	Half-life v5	7	0.82500	0.00000	0.00000	0.81237
mode_chatter	0.00	Half-life v5	8	0.76250	0.00000	0.00000	0.74979
mode_chatter	0.00	Half-life v5	9	0.81875	0.00000	0.00000	0.80622
mode_chatter	0.00	Oracle	0	0.86875	0.00000	0.00000	0.85486
mode_chatter	0.00	Oracle	1	0.87500	0.00625	0.00000	0.85503
mode_chatter	0.00	Oracle	2	0.78750	0.00625	0.00000	0.76784
mode_chatter	0.00	Oracle	3	0.86875	0.00000	0.00000	0.85517
mode_chatter	0.00	Oracle	4	0.82500	0.00000	0.00000	0.81142
mode_chatter	0.00	Oracle	5	0.83750	0.00000	0.00000	0.82384
mode_chatter	0.00	Oracle	6	0.87500	0.00000	0.00000	0.86134
mode_chatter	0.00	Oracle	7	0.80625	0.00000	0.00000	0.79267
mode_chatter	0.00	Oracle	8	0.88750	0.00000	0.00000	0.87394
mode_chatter	0.00	Oracle	9	0.84375	0.00000	0.00000	0.83003
mode_chatter	0.25	Domain rand.	0	0.57500	0.20000	0.56250	0.18395
mode_chatter	0.25	Domain rand.	1	0.48125	0.21875	0.58125	0.06509
mode_chatter	0.25	Domain rand.	2	0.48750	0.25625	0.57500	0.03601
mode_chatter	0.25	Domain rand.	3	0.53750	0.18750	0.57500	0.15469
mode_chatter	0.25	Domain rand.	4	0.47500	0.22500	0.58125	0.05260
mode_chatter	0.25	Domain rand.	5	0.51250	0.22500	0.57500	0.09222
mode_chatter	0.25	Domain rand.	6	0.53750	0.17500	0.58750	0.16292
mode_chatter	0.25	Domain rand.	7	0.48750	0.20625	0.56875	0.08808
mode_chatter	0.25	Domain rand.	8	0.52500	0.19375	0.57500	0.13594
mode_chatter	0.25	Domain rand.	9	0.56250	0.20000	0.55625	0.17357
mode_chatter	0.25	Conformal	0	0.75000	0.00000	0.00000	0.73721
mode_chatter	0.25	Conformal	1	0.75625	0.00000	0.00000	0.74354
mode_chatter	0.25	Conformal	2	0.81875	0.00000	0.00000	0.80598
mode_chatter	0.25	Conformal	3	0.80000	0.00625	0.00000	0.78113
mode_chatter	0.25	Conformal	4	0.75000	0.01250	0.00000	0.72474
mode_chatter	0.25	Conformal	5	0.80000	0.00000	0.00000	0.78765
mode_chatter	0.25	Conformal	6	0.76250	0.01250	0.00000	0.73709
mode_chatter	0.25	Conformal	7	0.85625	0.00000	0.00000	0.84364
mode_chatter	0.25	Conformal	8	0.80000	0.00000	0.00000	0.78669
mode_chatter	0.25	Conformal	9	0.81875	0.00000	0.00000	0.80596
mode_chatter	0.25	Ensemble	0	0.81250	0.00625	0.00000	0.79350
mode_chatter	0.25	Ensemble	1	0.80625	0.00625	0.00000	0.78694
mode_chatter	0.25	Ensemble	2	0.83125	0.00625	0.00000	0.81174
mode_chatter	0.25	Ensemble	3	0.85625	0.00625	0.00000	0.83688
mode_chatter	0.25	Ensemble	4	0.75625	0.00000	0.00000	0.74321

Table 10: Seed-level stress metrics. (continued)

Axis	Level	Method	Seed	Success	Unsafe	Stale	Safety utility
mode_chatter	0.25	Ensemble	5	0.77500	0.00000	0.00000	0.76196
mode_chatter	0.25	Ensemble	6	0.81250	0.00000	0.00000	0.79925
mode_chatter	0.25	Ensemble	7	0.75625	0.00625	0.00000	0.73686
mode_chatter	0.25	Ensemble	8	0.84375	0.00000	0.00000	0.83010
mode_chatter	0.25	Ensemble	9	0.75625	0.00625	0.00000	0.73688
mode_chatter	0.25	Bayes survival	0	0.78750	0.00625	0.00000	0.76662
mode_chatter	0.25	Bayes survival	1	0.86875	0.00000	0.00000	0.85419
mode_chatter	0.25	Bayes survival	2	0.76250	0.00000	0.00000	0.74761
mode_chatter	0.25	Bayes survival	3	0.80000	0.00000	0.00000	0.78536
mode_chatter	0.25	Bayes survival	4	0.85625	0.00000	0.00000	0.84128
mode_chatter	0.25	Bayes survival	5	0.83750	0.00000	0.00000	0.82313
mode_chatter	0.25	Bayes survival	6	0.84375	0.00000	0.00000	0.82857
mode_chatter	0.25	Bayes survival	7	0.81250	0.00000	0.00000	0.79794
mode_chatter	0.25	Bayes survival	8	0.80625	0.00000	0.00000	0.79112
mode_chatter	0.25	Bayes survival	9	0.82500	0.00000	0.00000	0.81022
mode_chatter	0.25	CVaR guard	0	0.83125	0.00000	0.00000	0.80832
mode_chatter	0.25	CVaR guard	1	0.84375	0.00000	0.00000	0.82039
mode_chatter	0.25	CVaR guard	2	0.82500	0.00000	0.00000	0.80215
mode_chatter	0.25	CVaR guard	3	0.90625	0.00000	0.00000	0.88357
mode_chatter	0.25	CVaR guard	4	0.81250	0.00625	0.00000	0.78360
mode_chatter	0.25	CVaR guard	5	0.83125	0.00000	0.00000	0.80906
mode_chatter	0.25	CVaR guard	6	0.77500	0.00000	0.00000	0.75201
mode_chatter	0.25	CVaR guard	7	0.85625	0.00000	0.00000	0.83396
mode_chatter	0.25	CVaR guard	8	0.81875	0.00000	0.00000	0.79628
mode_chatter	0.25	CVaR guard	9	0.84375	0.00000	0.00000	0.82094
mode_chatter	0.25	Half-life v5	0	0.79375	0.00000	0.00000	0.78083
mode_chatter	0.25	Half-life v5	1	0.81875	0.00000	0.00000	0.80608
mode_chatter	0.25	Half-life v5	2	0.81250	0.00000	0.00000	0.79935
mode_chatter	0.25	Half-life v5	3	0.81250	0.00625	0.00000	0.79356
mode_chatter	0.25	Half-life v5	4	0.80000	0.00625	0.00000	0.78123
mode_chatter	0.25	Half-life v5	5	0.76875	0.00625	0.00000	0.74998
mode_chatter	0.25	Half-life v5	6	0.76250	0.00000	0.00000	0.74941
mode_chatter	0.25	Half-life v5	7	0.83750	0.00000	0.00000	0.82456
mode_chatter	0.25	Half-life v5	8	0.86250	0.00000	0.00000	0.84964
mode_chatter	0.25	Half-life v5	9	0.78750	0.00625	0.00000	0.76850
mode_chatter	0.25	Oracle	0	0.88750	0.00000	0.00000	0.87330
mode_chatter	0.25	Oracle	1	0.80625	0.00000	0.00000	0.79242
mode_chatter	0.25	Oracle	2	0.82500	0.00000	0.00000	0.81100
mode_chatter	0.25	Oracle	3	0.86875	0.00000	0.00000	0.85519
mode_chatter	0.25	Oracle	4	0.81875	0.00625	0.00000	0.79841
mode_chatter	0.25	Oracle	5	0.80000	0.00000	0.00000	0.78605
mode_chatter	0.25	Oracle	6	0.88750	0.00000	0.00000	0.87369
mode_chatter	0.25	Oracle	7	0.83750	0.00000	0.00000	0.82384
mode_chatter	0.25	Oracle	8	0.80625	0.00000	0.00000	0.79219
mode_chatter	0.25	Oracle	9	0.84375	0.00000	0.00000	0.82986
mode_chatter	0.50	Domain rand.	0	0.46875	0.25000	0.57500	0.02350
mode_chatter	0.50	Domain rand.	1	0.47500	0.24375	0.58125	0.03387
mode_chatter	0.50	Domain rand.	2	0.49375	0.21875	0.58125	0.07759
mode_chatter	0.50	Domain rand.	3	0.47500	0.26250	0.59375	0.01089
mode_chatter	0.50	Domain rand.	4	0.50625	0.18125	0.58750	0.12543
mode_chatter	0.50	Domain rand.	5	0.53125	0.20625	0.59375	0.12333
mode_chatter	0.50	Domain rand.	6	0.53125	0.18750	0.60625	0.13781
mode_chatter	0.50	Domain rand.	7	0.46875	0.20625	0.57500	0.06721
mode_chatter	0.50	Domain rand.	8	0.50625	0.23750	0.57500	0.07349
mode_chatter	0.50	Domain rand.	9	0.45625	0.25000	0.57500	0.01100
mode_chatter	0.50	Conformal	0	0.76875	0.00000	0.00000	0.75552
mode_chatter	0.50	Conformal	1	0.84375	0.00000	0.00000	0.83048
mode_chatter	0.50	Conformal	2	0.78750	0.01250	0.00000	0.76166
mode_chatter	0.50	Conformal	3	0.74375	0.00000	0.00000	0.73034
mode_chatter	0.50	Conformal	4	0.85000	0.00625	0.00000	0.83040
mode_chatter	0.50	Conformal	5	0.75625	0.00000	0.00000	0.74288
mode_chatter	0.50	Conformal	6	0.78750	0.00000	0.00000	0.77431
mode_chatter	0.50	Conformal	7	0.83750	0.00000	0.00000	0.82409
mode_chatter	0.50	Conformal	8	0.76250	0.00625	0.00000	0.74288
mode_chatter	0.50	Conformal	9	0.78125	0.00625	0.00000	0.76192
mode_chatter	0.50	Ensemble	0	0.79375	0.00000	0.00000	0.78025
mode_chatter	0.50	Ensemble	1	0.78750	0.00625	0.00000	0.76774
mode_chatter	0.50	Ensemble	2	0.77500	0.00000	0.00000	0.76134
mode_chatter	0.50	Ensemble	3	0.71875	0.00000	0.00000	0.70548
mode_chatter	0.50	Ensemble	4	0.80000	0.00625	0.00000	0.78024
mode_chatter	0.50	Ensemble	5	0.83125	0.00625	0.00000	0.81153
mode_chatter	0.50	Ensemble	6	0.77500	0.00625	0.00000	0.75530
mode_chatter	0.50	Ensemble	7	0.81875	0.00625	0.00000	0.79918
mode_chatter	0.50	Ensemble	8	0.79375	0.00000	0.00000	0.78027
mode_chatter	0.50	Ensemble	9	0.80625	0.00625	0.00000	0.78644
mode_chatter	0.50	Bayes survival	0	0.76875	0.01250	0.00000	0.74071
mode_chatter	0.50	Bayes survival	1	0.82500	0.00625	0.00000	0.80352

Table 10: Seed-level stress metrics. (continued)

Axis	Level	Method	Seed	Success	Unsafe	Stale	Safety utility
mode_chatter	0.50	Bayes survival	2	0.86250	0.00625	0.00000	0.84118
mode_chatter	0.50	Bayes survival	3	0.80625	0.00000	0.00000	0.79093
mode_chatter	0.50	Bayes survival	4	0.76875	0.00000	0.00000	0.75376
mode_chatter	0.50	Bayes survival	5	0.85625	0.01250	0.00000	0.82846
mode_chatter	0.50	Bayes survival	6	0.80625	0.00000	0.00000	0.79107
mode_chatter	0.50	Bayes survival	7	0.85625	0.00000	0.00000	0.84095
mode_chatter	0.50	Bayes survival	8	0.78125	0.00000	0.00000	0.76616
mode_chatter	0.50	Bayes survival	9	0.80000	0.00000	0.00000	0.78460
mode_chatter	0.50	CVaR guard	0	0.75000	0.00000	0.00000	0.72616
mode_chatter	0.50	CVaR guard	1	0.87500	0.00000	0.00000	0.85195
mode_chatter	0.50	CVaR guard	2	0.80625	0.00000	0.00000	0.78305
mode_chatter	0.50	CVaR guard	3	0.88125	0.00000	0.00000	0.85789
mode_chatter	0.50	CVaR guard	4	0.82500	0.00000	0.00000	0.80166
mode_chatter	0.50	CVaR guard	5	0.87500	0.00625	0.00000	0.84558
mode_chatter	0.50	CVaR guard	6	0.85625	0.00000	0.00000	0.83219
mode_chatter	0.50	CVaR guard	7	0.80000	0.00000	0.00000	0.77695
mode_chatter	0.50	CVaR guard	8	0.86250	0.00000	0.00000	0.83897
mode_chatter	0.50	CVaR guard	9	0.86250	0.00625	0.00000	0.83226
mode_chatter	0.50	Half-life v5	0	0.80000	0.01875	0.00000	0.76789
mode_chatter	0.50	Half-life v5	1	0.79375	0.00000	0.00000	0.78056
mode_chatter	0.50	Half-life v5	2	0.82500	0.00625	0.00000	0.80559
mode_chatter	0.50	Half-life v5	3	0.86250	0.00625	0.00000	0.84317
mode_chatter	0.50	Half-life v5	4	0.86875	0.00000	0.00000	0.85538
mode_chatter	0.50	Half-life v5	5	0.76250	0.00000	0.00000	0.74904
mode_chatter	0.50	Half-life v5	6	0.83125	0.00625	0.00000	0.81194
mode_chatter	0.50	Half-life v5	7	0.83125	0.00000	0.00000	0.81800
mode_chatter	0.50	Half-life v5	8	0.83750	0.00625	0.00000	0.81823
mode_chatter	0.50	Half-life v5	9	0.78125	0.00625	0.00000	0.76200
mode_chatter	0.50	Oracle	0	0.84375	0.00625	0.00000	0.82268
mode_chatter	0.50	Oracle	1	0.83125	0.00000	0.00000	0.81705
mode_chatter	0.50	Oracle	2	0.90000	0.00000	0.00000	0.88574
mode_chatter	0.50	Oracle	3	0.81875	0.00000	0.00000	0.80457
mode_chatter	0.50	Oracle	4	0.83125	0.00000	0.00000	0.81651
mode_chatter	0.50	Oracle	5	0.84375	0.00000	0.00000	0.82949
mode_chatter	0.50	Oracle	6	0.86875	0.00000	0.00000	0.85463
mode_chatter	0.50	Oracle	7	0.87500	0.00000	0.00000	0.86088
mode_chatter	0.50	Oracle	8	0.85000	0.00625	0.00000	0.82968
mode_chatter	0.50	Oracle	9	0.85000	0.00000	0.00000	0.83580
mode_chatter	0.75	Domain rand.	0	0.50625	0.18125	0.57500	0.12968
mode_chatter	0.75	Domain rand.	1	0.53125	0.20000	0.58125	0.13382
mode_chatter	0.75	Domain rand.	2	0.54375	0.25625	0.58750	0.08801
mode_chatter	0.75	Domain rand.	3	0.48750	0.23125	0.60000	0.05248
mode_chatter	0.75	Domain rand.	4	0.51875	0.19375	0.60000	0.12119
mode_chatter	0.75	Domain rand.	5	0.43125	0.23125	0.59375	-0.00164
mode_chatter	0.75	Domain rand.	6	0.46250	0.26250	0.61875	-0.01011
mode_chatter	0.75	Domain rand.	7	0.49375	0.24375	0.58125	0.05262
mode_chatter	0.75	Domain rand.	8	0.53750	0.20625	0.57500	0.13596
mode_chatter	0.75	Domain rand.	9	0.46875	0.27500	0.58125	-0.00360
mode_chatter	0.75	Conformal	0	0.79375	0.02500	0.00000	0.75553
mode_chatter	0.75	Conformal	1	0.83125	0.00000	0.00000	0.81709
mode_chatter	0.75	Conformal	2	0.78125	0.00000	0.00000	0.76811
mode_chatter	0.75	Conformal	3	0.78125	0.00000	0.00000	0.76779
mode_chatter	0.75	Conformal	4	0.81250	0.00000	0.00000	0.79909
mode_chatter	0.75	Conformal	5	0.79375	0.01875	0.00000	0.76160
mode_chatter	0.75	Conformal	6	0.78125	0.01250	0.00000	0.75543
mode_chatter	0.75	Conformal	7	0.73750	0.00000	0.00000	0.72423
mode_chatter	0.75	Conformal	8	0.73125	0.01875	0.00000	0.69894
mode_chatter	0.75	Conformal	9	0.76875	0.00000	0.00000	0.75548
mode_chatter	0.75	Ensemble	0	0.77500	0.01250	0.00000	0.74908
mode_chatter	0.75	Ensemble	1	0.85000	0.00000	0.00000	0.83588
mode_chatter	0.75	Ensemble	2	0.85000	0.00000	0.00000	0.83644
mode_chatter	0.75	Ensemble	3	0.77500	0.00000	0.00000	0.76129
mode_chatter	0.75	Ensemble	4	0.85000	0.01250	0.00000	0.82385
mode_chatter	0.75	Ensemble	5	0.76250	0.00625	0.00000	0.74274
mode_chatter	0.75	Ensemble	6	0.81250	0.00000	0.00000	0.79894
mode_chatter	0.75	Ensemble	7	0.81250	0.00000	0.00000	0.79882
mode_chatter	0.75	Ensemble	8	0.81875	0.00625	0.00000	0.79876
mode_chatter	0.75	Ensemble	9	0.80625	0.01250	0.00000	0.78002
mode_chatter	0.75	Bayes survival	0	0.84375	0.00000	0.00000	0.82820
mode_chatter	0.75	Bayes survival	1	0.80000	0.00625	0.00000	0.77833
mode_chatter	0.75	Bayes survival	2	0.82500	0.00625	0.00000	0.80331
mode_chatter	0.75	Bayes survival	3	0.78750	0.00625	0.00000	0.76579
mode_chatter	0.75	Bayes survival	4	0.84375	0.00625	0.00000	0.82210
mode_chatter	0.75	Bayes survival	5	0.82500	0.00000	0.00000	0.80964
mode_chatter	0.75	Bayes survival	6	0.84375	0.00000	0.00000	0.82769
mode_chatter	0.75	Bayes survival	7	0.72500	0.00000	0.00000	0.70962
mode_chatter	0.75	Bayes survival	8	0.80625	0.00000	0.00000	0.79076

Table 10: Seed-level stress metrics. (continued)

Axis	Level	Method	Seed	Success	Unsafe	Stale	Safety utility
mode_chatter	0.75	Bayes survival	9	0.82500	0.01250	0.00000	0.79699
mode_chatter	0.75	CVaR guard	0	0.80000	0.00000	0.00000	0.77571
mode_chatter	0.75	CVaR guard	1	0.90000	0.00000	0.00000	0.87573
mode_chatter	0.75	CVaR guard	2	0.82500	0.00000	0.00000	0.80057
mode_chatter	0.75	CVaR guard	3	0.83750	0.00000	0.00000	0.81300
mode_chatter	0.75	CVaR guard	4	0.81875	0.00000	0.00000	0.79477
mode_chatter	0.75	CVaR guard	5	0.87500	0.00000	0.00000	0.85067
mode_chatter	0.75	CVaR guard	6	0.81250	0.00000	0.00000	0.78870
mode_chatter	0.75	CVaR guard	7	0.84375	0.00000	0.00000	0.81987
mode_chatter	0.75	CVaR guard	8	0.86875	0.00000	0.00000	0.84467
mode_chatter	0.75	CVaR guard	9	0.75000	0.00000	0.00000	0.72592
mode_chatter	0.75	Half-life v5	0	0.79375	0.00000	0.00000	0.78044
mode_chatter	0.75	Half-life v5	1	0.78125	0.00625	0.00000	0.76128
mode_chatter	0.75	Half-life v5	2	0.77500	0.00625	0.00000	0.75594
mode_chatter	0.75	Half-life v5	3	0.81250	0.00625	0.00000	0.79259
mode_chatter	0.75	Half-life v5	4	0.82500	0.00000	0.00000	0.81194
mode_chatter	0.75	Half-life v5	5	0.78750	0.00000	0.00000	0.77429
mode_chatter	0.75	Half-life v5	6	0.76875	0.00000	0.00000	0.75538
mode_chatter	0.75	Half-life v5	7	0.78125	0.00625	0.00000	0.76200
mode_chatter	0.75	Half-life v5	8	0.83125	0.00625	0.00000	0.81124
mode_chatter	0.75	Half-life v5	9	0.80000	0.01250	0.00000	0.77430
mode_chatter	0.75	Oracle	0	0.82500	0.00000	0.00000	0.81055
mode_chatter	0.75	Oracle	1	0.86875	0.00000	0.00000	0.85399
mode_chatter	0.75	Oracle	2	0.86250	0.00000	0.00000	0.84809
mode_chatter	0.75	Oracle	3	0.82500	0.00000	0.00000	0.81040
mode_chatter	0.75	Oracle	4	0.81875	0.00000	0.00000	0.80415
mode_chatter	0.75	Oracle	5	0.81875	0.00625	0.00000	0.79762
mode_chatter	0.75	Oracle	6	0.80625	0.00000	0.00000	0.79153
mode_chatter	0.75	Oracle	7	0.86250	0.00000	0.00000	0.84834
mode_chatter	0.75	Oracle	8	0.78750	0.00000	0.00000	0.77270
mode_chatter	0.75	Oracle	9	0.84375	0.00000	0.00000	0.82930
mode_chatter	1.00	Domain rand.	0	0.48125	0.28125	0.56875	0.00691
mode_chatter	1.00	Domain rand.	1	0.50625	0.23125	0.58125	0.07761
mode_chatter	1.00	Domain rand.	2	0.50000	0.24375	0.59375	0.05462
mode_chatter	1.00	Domain rand.	3	0.46250	0.24375	0.59375	0.01712
mode_chatter	1.00	Domain rand.	4	0.50625	0.25625	0.60000	0.04626
mode_chatter	1.00	Domain rand.	5	0.49375	0.22500	0.60000	0.06498
mode_chatter	1.00	Domain rand.	6	0.51875	0.26875	0.60000	0.04627
mode_chatter	1.00	Domain rand.	7	0.48750	0.22500	0.61250	0.05447
mode_chatter	1.00	Domain rand.	8	0.45000	0.25000	0.58125	0.00262
mode_chatter	1.00	Domain rand.	9	0.48750	0.21875	0.58125	0.07134
mode_chatter	1.00	Conformal	0	0.83125	0.00000	0.00000	0.81740
mode_chatter	1.00	Conformal	1	0.77500	0.00000	0.00000	0.76128
mode_chatter	1.00	Conformal	2	0.78750	0.00625	0.00000	0.76789
mode_chatter	1.00	Conformal	3	0.76250	0.00625	0.00000	0.74251
mode_chatter	1.00	Conformal	4	0.75000	0.00625	0.00000	0.72953
mode_chatter	1.00	Conformal	5	0.81875	0.00000	0.00000	0.80469
mode_chatter	1.00	Conformal	6	0.78125	0.00000	0.00000	0.76761
mode_chatter	1.00	Conformal	7	0.75000	0.00625	0.00000	0.72976
mode_chatter	1.00	Conformal	8	0.78125	0.00625	0.00000	0.76114
mode_chatter	1.00	Conformal	9	0.76250	0.00000	0.00000	0.74890
mode_chatter	1.00	Ensemble	0	0.75625	0.00000	0.00000	0.74194
mode_chatter	1.00	Ensemble	1	0.78750	0.00625	0.00000	0.76734
mode_chatter	1.00	Ensemble	2	0.78750	0.01250	0.00000	0.76106
mode_chatter	1.00	Ensemble	3	0.82500	0.00625	0.00000	0.80490
mode_chatter	1.00	Ensemble	4	0.78125	0.00625	0.00000	0.76039
mode_chatter	1.00	Ensemble	5	0.76250	0.00000	0.00000	0.74834
mode_chatter	1.00	Ensemble	6	0.84375	0.00000	0.00000	0.82967
mode_chatter	1.00	Ensemble	7	0.75625	0.00000	0.00000	0.74194
mode_chatter	1.00	Ensemble	8	0.83750	0.00000	0.00000	0.82344
mode_chatter	1.00	Ensemble	9	0.78750	0.01250	0.00000	0.76153
mode_chatter	1.00	Bayes survival	0	0.80000	0.01250	0.00000	0.77099
mode_chatter	1.00	Bayes survival	1	0.78750	0.00000	0.00000	0.77177
mode_chatter	1.00	Bayes survival	2	0.78750	0.00000	0.00000	0.77176
mode_chatter	1.00	Bayes survival	3	0.83125	0.00000	0.00000	0.81515
mode_chatter	1.00	Bayes survival	4	0.76875	0.00625	0.00000	0.74650
mode_chatter	1.00	Bayes survival	5	0.83750	0.00625	0.00000	0.81496
mode_chatter	1.00	Bayes survival	6	0.81250	0.00000	0.00000	0.79675
mode_chatter	1.00	Bayes survival	7	0.79375	0.00000	0.00000	0.77754
mode_chatter	1.00	Bayes survival	8	0.81875	0.00000	0.00000	0.80320
mode_chatter	1.00	Bayes survival	9	0.85625	0.00625	0.00000	0.83423
mode_chatter	1.00	CVaR guard	0	0.84375	0.00000	0.00000	0.81903
mode_chatter	1.00	CVaR guard	1	0.79375	0.00625	0.00000	0.76332
mode_chatter	1.00	CVaR guard	2	0.77500	0.00000	0.00000	0.75022
mode_chatter	1.00	CVaR guard	3	0.77500	0.00000	0.00000	0.75001
mode_chatter	1.00	CVaR guard	4	0.86875	0.00000	0.00000	0.84382
mode_chatter	1.00	CVaR guard	5	0.84375	0.00000	0.00000	0.81932

Table 10: Seed-level stress metrics. (continued)

Axis	Level	Method	Seed	Success	Unsafe	Stale	Safety utility
mode_chatter	1.00	CVaR guard	6	0.78125	0.00000	0.00000	0.75725
mode_chatter	1.00	CVaR guard	7	0.83750	0.00000	0.00000	0.81228
mode_chatter	1.00	CVaR guard	8	0.76875	0.00625	0.00000	0.73822
mode_chatter	1.00	CVaR guard	9	0.77500	0.00000	0.00000	0.75048
mode_chatter	1.00	Half-life v5	0	0.81250	0.00000	0.00000	0.79882
mode_chatter	1.00	Half-life v5	1	0.81250	0.00625	0.00000	0.79303
mode_chatter	1.00	Half-life v5	2	0.81875	0.00000	0.00000	0.80571
mode_chatter	1.00	Half-life v5	3	0.76875	0.00000	0.00000	0.75471
mode_chatter	1.00	Half-life v5	4	0.81875	0.00000	0.00000	0.80490
mode_chatter	1.00	Half-life v5	5	0.80625	0.00625	0.00000	0.78574
mode_chatter	1.00	Half-life v5	6	0.77500	0.00000	0.00000	0.76171
mode_chatter	1.00	Half-life v5	7	0.81250	0.00000	0.00000	0.79863
mode_chatter	1.00	Half-life v5	8	0.79375	0.00000	0.00000	0.77996
mode_chatter	1.00	Half-life v5	9	0.80625	0.01250	0.00000	0.78006
mode_chatter	1.00	Oracle	0	0.83125	0.00000	0.00000	0.81668
mode_chatter	1.00	Oracle	1	0.81250	0.00000	0.00000	0.79801
mode_chatter	1.00	Oracle	2	0.79375	0.00000	0.00000	0.77893
mode_chatter	1.00	Oracle	3	0.82500	0.00000	0.00000	0.81018
mode_chatter	1.00	Oracle	4	0.83125	0.00000	0.00000	0.81603
mode_chatter	1.00	Oracle	5	0.83750	0.00000	0.00000	0.82234
mode_chatter	1.00	Oracle	6	0.83125	0.00000	0.00000	0.81651
mode_chatter	1.00	Oracle	7	0.88750	0.00000	0.00000	0.87276
mode_chatter	1.00	Oracle	8	0.81250	0.00000	0.00000	0.79762
mode_chatter	1.00	Oracle	9	0.86875	0.00625	0.00000	0.84793
mode_chatter	1.25	Domain rand.	0	0.46875	0.26875	0.59375	-0.00161
mode_chatter	1.25	Domain rand.	1	0.48750	0.24375	0.58750	0.04424
mode_chatter	1.25	Domain rand.	2	0.48750	0.24375	0.59375	0.04212
mode_chatter	1.25	Domain rand.	3	0.45625	0.26875	0.60625	-0.01836
mode_chatter	1.25	Domain rand.	4	0.50000	0.23125	0.61250	0.06073
mode_chatter	1.25	Domain rand.	5	0.45625	0.30000	0.61250	-0.05170
mode_chatter	1.25	Domain rand.	6	0.48750	0.24375	0.60625	0.03787
mode_chatter	1.25	Domain rand.	7	0.42500	0.23750	0.60000	-0.01626
mode_chatter	1.25	Domain rand.	8	0.44375	0.25000	0.59375	-0.00788
mode_chatter	1.25	Domain rand.	9	0.45000	0.23125	0.59375	0.01711
mode_chatter	1.25	Conformal	0	0.78750	0.00625	0.00000	0.76730
mode_chatter	1.25	Conformal	1	0.77500	0.00625	0.00000	0.75464
mode_chatter	1.25	Conformal	2	0.78750	0.00625	0.00000	0.76741
mode_chatter	1.25	Conformal	3	0.81250	0.00625	0.00000	0.79197
mode_chatter	1.25	Conformal	4	0.80625	0.00000	0.00000	0.79225
mode_chatter	1.25	Conformal	5	0.86250	0.00000	0.00000	0.84832
mode_chatter	1.25	Conformal	6	0.78750	0.00000	0.00000	0.77363
mode_chatter	1.25	Conformal	7	0.74375	0.00000	0.00000	0.72924
mode_chatter	1.25	Conformal	8	0.80000	0.00000	0.00000	0.78551
mode_chatter	1.25	Conformal	9	0.75625	0.00625	0.00000	0.73589
mode_chatter	1.25	Ensemble	0	0.81875	0.00000	0.00000	0.80438
mode_chatter	1.25	Ensemble	1	0.76875	0.00000	0.00000	0.75438
mode_chatter	1.25	Ensemble	2	0.75625	0.01875	0.00000	0.72359
mode_chatter	1.25	Ensemble	3	0.76875	0.00000	0.00000	0.75430
mode_chatter	1.25	Ensemble	4	0.78125	0.01250	0.00000	0.75446
mode_chatter	1.25	Ensemble	5	0.78125	0.00000	0.00000	0.76705
mode_chatter	1.25	Ensemble	6	0.80625	0.01250	0.00000	0.77958
mode_chatter	1.25	Ensemble	7	0.78750	0.00625	0.00000	0.76658
mode_chatter	1.25	Ensemble	8	0.80625	0.00625	0.00000	0.78552
mode_chatter	1.25	Ensemble	9	0.81250	0.01250	0.00000	0.78562
mode_chatter	1.25	Bayes survival	0	0.78125	0.00000	0.00000	0.76521
mode_chatter	1.25	Bayes survival	1	0.82500	0.00000	0.00000	0.80850
mode_chatter	1.25	Bayes survival	2	0.81250	0.00000	0.00000	0.79677
mode_chatter	1.25	Bayes survival	3	0.79375	0.00000	0.00000	0.77785
mode_chatter	1.25	Bayes survival	4	0.78750	0.00000	0.00000	0.77189
mode_chatter	1.25	Bayes survival	5	0.82500	0.00000	0.00000	0.80910
mode_chatter	1.25	Bayes survival	6	0.78125	0.00000	0.00000	0.76471
mode_chatter	1.25	Bayes survival	7	0.84375	0.00000	0.00000	0.82717
mode_chatter	1.25	Bayes survival	8	0.85000	0.00625	0.00000	0.82740
mode_chatter	1.25	Bayes survival	9	0.86250	0.00625	0.00000	0.83994
mode_chatter	1.25	CVaR guard	0	0.82500	0.00625	0.00000	0.79314
mode_chatter	1.25	CVaR guard	1	0.81250	0.00000	0.00000	0.78712
mode_chatter	1.25	CVaR guard	2	0.81875	0.00000	0.00000	0.79405
mode_chatter	1.25	CVaR guard	3	0.86875	0.00000	0.00000	0.84347
mode_chatter	1.25	CVaR guard	4	0.82500	0.00625	0.00000	0.79389
mode_chatter	1.25	CVaR guard	5	0.81250	0.00000	0.00000	0.78838
mode_chatter	1.25	CVaR guard	6	0.86875	0.00000	0.00000	0.84345
mode_chatter	1.25	CVaR guard	7	0.79375	0.00000	0.00000	0.76868
mode_chatter	1.25	CVaR guard	8	0.86250	0.00000	0.00000	0.83757
mode_chatter	1.25	CVaR guard	9	0.81250	0.00000	0.00000	0.78697
mode_chatter	1.25	Half-life v5	0	0.81875	0.00000	0.00000	0.80504
mode_chatter	1.25	Half-life v5	1	0.78750	0.00000	0.00000	0.77388
mode_chatter	1.25	Half-life v5	2	0.82500	0.00000	0.00000	0.81142

Table 10: Seed-level stress metrics. (continued)

Axis	Level	Method	Seed	Success	Unsafe	Stale	Safety utility
mode_chatter	1.25	Half-life v5	3	0.77500	0.00625	0.00000	0.75443
mode_chatter	1.25	Half-life v5	4	0.83750	0.00000	0.00000	0.82369
mode_chatter	1.25	Half-life v5	5	0.82500	0.00625	0.00000	0.80501
mode_chatter	1.25	Half-life v5	6	0.83125	0.00000	0.00000	0.81738
mode_chatter	1.25	Half-life v5	7	0.80000	0.00625	0.00000	0.77930
mode_chatter	1.25	Half-life v5	8	0.80625	0.00000	0.00000	0.79248
mode_chatter	1.25	Half-life v5	9	0.80000	0.00625	0.00000	0.77947
mode_chatter	1.25	Oracle	0	0.83125	0.01250	0.00000	0.80396
mode_chatter	1.25	Oracle	1	0.81250	0.00000	0.00000	0.79745
mode_chatter	1.25	Oracle	2	0.86250	0.00000	0.00000	0.84778
mode_chatter	1.25	Oracle	3	0.84375	0.00000	0.00000	0.82853
mode_chatter	1.25	Oracle	4	0.81875	0.00625	0.00000	0.79762
mode_chatter	1.25	Oracle	5	0.82500	0.00000	0.00000	0.80981
mode_chatter	1.25	Oracle	6	0.80000	0.00000	0.00000	0.78514
mode_chatter	1.25	Oracle	7	0.80625	0.00000	0.00000	0.79060
mode_chatter	1.25	Oracle	8	0.83750	0.00000	0.00000	0.82231
mode_chatter	1.25	Oracle	9	0.86250	0.00000	0.00000	0.84708
mode_chatter	1.50	Domain rand.	0	0.46875	0.27500	0.59375	-0.00785
mode_chatter	1.50	Domain rand.	1	0.48125	0.25000	0.58750	0.03175
mode_chatter	1.50	Domain rand.	2	0.46875	0.30625	0.59375	-0.03907
mode_chatter	1.50	Domain rand.	3	0.41875	0.28125	0.60625	-0.06834
mode_chatter	1.50	Domain rand.	4	0.45625	0.27500	0.63125	-0.03310
mode_chatter	1.50	Domain rand.	5	0.48125	0.21250	0.61875	0.05859
mode_chatter	1.50	Domain rand.	6	0.45000	0.28125	0.61875	-0.04134
mode_chatter	1.50	Domain rand.	7	0.51250	0.25625	0.58750	0.05676
mode_chatter	1.50	Domain rand.	8	0.46250	0.28750	0.60000	-0.02871
mode_chatter	1.50	Domain rand.	9	0.53750	0.23750	0.60000	0.09624
mode_chatter	1.50	Conformal	0	0.86250	0.00000	0.00000	0.84770
mode_chatter	1.50	Conformal	1	0.77500	0.00000	0.00000	0.76020
mode_chatter	1.50	Conformal	2	0.80625	0.00000	0.00000	0.79197
mode_chatter	1.50	Conformal	3	0.75625	0.00000	0.00000	0.74162
mode_chatter	1.50	Conformal	4	0.75625	0.00625	0.00000	0.73543
mode_chatter	1.50	Conformal	5	0.82500	0.00000	0.00000	0.81045
mode_chatter	1.50	Conformal	6	0.80625	0.00625	0.00000	0.78560
mode_chatter	1.50	Conformal	7	0.77500	0.00625	0.00000	0.75456
mode_chatter	1.50	Conformal	8	0.77500	0.00625	0.00000	0.75414
mode_chatter	1.50	Conformal	9	0.75000	0.01875	0.00000	0.71661
mode_chatter	1.50	Ensemble	0	0.84375	0.01250	0.00000	0.81607
mode_chatter	1.50	Ensemble	1	0.77500	0.00625	0.00000	0.75377
mode_chatter	1.50	Ensemble	2	0.78750	0.00625	0.00000	0.76666
mode_chatter	1.50	Ensemble	3	0.76250	0.00000	0.00000	0.74745
mode_chatter	1.50	Ensemble	4	0.75625	0.00000	0.00000	0.74151
mode_chatter	1.50	Ensemble	5	0.76250	0.00000	0.00000	0.74770
mode_chatter	1.50	Ensemble	6	0.76875	0.00625	0.00000	0.74785
mode_chatter	1.50	Ensemble	7	0.80000	0.00625	0.00000	0.77959
mode_chatter	1.50	Ensemble	8	0.80625	0.00625	0.00000	0.78506
mode_chatter	1.50	Ensemble	9	0.71875	0.01875	0.00000	0.68570
mode_chatter	1.50	Bayes survival	0	0.81875	0.00000	0.00000	0.80215
mode_chatter	1.50	Bayes survival	1	0.78750	0.00625	0.00000	0.76422
mode_chatter	1.50	Bayes survival	2	0.81250	0.00625	0.00000	0.78992
mode_chatter	1.50	Bayes survival	3	0.79375	0.00625	0.00000	0.77078
mode_chatter	1.50	Bayes survival	4	0.84375	0.00000	0.00000	0.82657
mode_chatter	1.50	Bayes survival	5	0.80000	0.00625	0.00000	0.77734
mode_chatter	1.50	Bayes survival	6	0.80000	0.00000	0.00000	0.78331
mode_chatter	1.50	Bayes survival	7	0.82500	0.00000	0.00000	0.80885
mode_chatter	1.50	Bayes survival	8	0.78750	0.00000	0.00000	0.77065
mode_chatter	1.50	Bayes survival	9	0.81250	0.00625	0.00000	0.78901
mode_chatter	1.50	CVaR guard	0	0.84375	0.00000	0.00000	0.81756
mode_chatter	1.50	CVaR guard	1	0.83750	0.00000	0.00000	0.81160
mode_chatter	1.50	CVaR guard	2	0.83125	0.00000	0.00000	0.80572
mode_chatter	1.50	CVaR guard	3	0.83125	0.00000	0.00000	0.80562
mode_chatter	1.50	CVaR guard	4	0.81875	0.00000	0.00000	0.79281
mode_chatter	1.50	CVaR guard	5	0.81250	0.00000	0.00000	0.78636
mode_chatter	1.50	CVaR guard	6	0.79375	0.00000	0.00000	0.76793
mode_chatter	1.50	CVaR guard	7	0.83750	0.01875	0.00000	0.79324
mode_chatter	1.50	CVaR guard	8	0.78750	0.00000	0.00000	0.76152
mode_chatter	1.50	CVaR guard	9	0.79375	0.00000	0.00000	0.76795
mode_chatter	1.50	Half-life v5	0	0.83125	0.00625	0.00000	0.81082
mode_chatter	1.50	Half-life v5	1	0.76875	0.00000	0.00000	0.75442
mode_chatter	1.50	Half-life v5	2	0.76250	0.00000	0.00000	0.74836
mode_chatter	1.50	Half-life v5	3	0.84375	0.00000	0.00000	0.82959
mode_chatter	1.50	Half-life v5	4	0.81250	0.00000	0.00000	0.79823
mode_chatter	1.50	Half-life v5	5	0.77500	0.00000	0.00000	0.76073
mode_chatter	1.50	Half-life v5	6	0.83125	0.00000	0.00000	0.81732
mode_chatter	1.50	Half-life v5	7	0.79375	0.00000	0.00000	0.77967
mode_chatter	1.50	Half-life v5	8	0.83125	0.00625	0.00000	0.81080
mode_chatter	1.50	Half-life v5	9	0.76875	0.00625	0.00000	0.74874

Table 10: Seed-level stress metrics. (continued)

Axis	Level	Method	Seed	Success	Unsafe	Stale	Safety utility
mode_chatter	1.50	Oracle	0	0.88750	0.00000	0.00000	0.87195
mode_chatter	1.50	Oracle	1	0.80000	0.00000	0.00000	0.78506
mode_chatter	1.50	Oracle	2	0.80000	0.00625	0.00000	0.77817
mode_chatter	1.50	Oracle	3	0.81875	0.00000	0.00000	0.80341
mode_chatter	1.50	Oracle	4	0.75000	0.00000	0.00000	0.73474
mode_chatter	1.50	Oracle	5	0.81875	0.00000	0.00000	0.80341
mode_chatter	1.50	Oracle	6	0.83125	0.00000	0.00000	0.81556
mode_chatter	1.50	Oracle	7	0.83750	0.00000	0.00000	0.82228
mode_chatter	1.50	Oracle	8	0.86250	0.00000	0.00000	0.84716
mode_chatter	1.50	Oracle	9	0.84375	0.00625	0.00000	0.82239
probe_cost	0.00	Domain rand.	0	0.52500	0.19375	0.55625	0.14232
probe_cost	0.00	Domain rand.	1	0.44375	0.23125	0.56250	0.02148
probe_cost	0.00	Domain rand.	2	0.54375	0.18750	0.57500	0.16094
probe_cost	0.00	Domain rand.	3	0.48125	0.20625	0.57500	0.07971
probe_cost	0.00	Domain rand.	4	0.56250	0.18125	0.58125	0.18381
probe_cost	0.00	Domain rand.	5	0.50000	0.26875	0.58125	0.03389
probe_cost	0.00	Domain rand.	6	0.53125	0.16250	0.56875	0.17554
probe_cost	0.00	Domain rand.	7	0.39375	0.24375	0.57500	-0.04526
probe_cost	0.00	Domain rand.	8	0.48750	0.21250	0.56875	0.08184
probe_cost	0.00	Domain rand.	9	0.53750	0.20000	0.56875	0.14432
probe_cost	0.00	Conformal	0	0.78750	0.00000	0.00000	0.77888
probe_cost	0.00	Conformal	1	0.83750	0.00000	0.00000	0.82895
probe_cost	0.00	Conformal	2	0.79375	0.01250	0.00000	0.77273
probe_cost	0.00	Conformal	3	0.79375	0.00000	0.00000	0.78502
probe_cost	0.00	Conformal	4	0.78125	0.00625	0.00000	0.76652
probe_cost	0.00	Conformal	5	0.76875	0.00625	0.00000	0.75393
probe_cost	0.00	Conformal	6	0.75000	0.00625	0.00000	0.73533
probe_cost	0.00	Conformal	7	0.81875	0.00625	0.00000	0.80391
probe_cost	0.00	Conformal	8	0.77500	0.01250	0.00000	0.75377
probe_cost	0.00	Conformal	9	0.80000	0.01250	0.00000	0.77877
probe_cost	0.00	Ensemble	0	0.80000	0.00625	0.00000	0.78512
probe_cost	0.00	Ensemble	1	0.76875	0.01875	0.00000	0.74180
probe_cost	0.00	Ensemble	2	0.75625	0.00625	0.00000	0.74155
probe_cost	0.00	Ensemble	3	0.81250	0.00000	0.00000	0.80362
probe_cost	0.00	Ensemble	4	0.83125	0.00625	0.00000	0.81634
probe_cost	0.00	Ensemble	5	0.85000	0.01250	0.00000	0.82904
probe_cost	0.00	Ensemble	6	0.78750	0.00000	0.00000	0.77892
probe_cost	0.00	Ensemble	7	0.82500	0.00000	0.00000	0.81604
probe_cost	0.00	Ensemble	8	0.86250	0.00000	0.00000	0.85366
probe_cost	0.00	Ensemble	9	0.78750	0.00625	0.00000	0.77254
probe_cost	0.00	Bayes survival	0	0.80000	0.00000	0.00000	0.79024
probe_cost	0.00	Bayes survival	1	0.83750	0.00000	0.00000	0.82751
probe_cost	0.00	Bayes survival	2	0.81875	0.00000	0.00000	0.80898
probe_cost	0.00	Bayes survival	3	0.79375	0.00000	0.00000	0.78336
probe_cost	0.00	Bayes survival	4	0.82500	0.01250	0.00000	0.80237
probe_cost	0.00	Bayes survival	5	0.82500	0.00000	0.00000	0.81527
probe_cost	0.00	Bayes survival	6	0.81250	0.00000	0.00000	0.80240
probe_cost	0.00	Bayes survival	7	0.81875	0.00000	0.00000	0.80862
probe_cost	0.00	Bayes survival	8	0.83125	0.00000	0.00000	0.82103
probe_cost	0.00	Bayes survival	9	0.85625	0.00000	0.00000	0.84615
probe_cost	0.00	CVaR guard	0	0.76875	0.00000	0.00000	0.75025
probe_cost	0.00	CVaR guard	1	0.80625	0.00000	0.00000	0.78798
probe_cost	0.00	CVaR guard	2	0.77500	0.00000	0.00000	0.75643
probe_cost	0.00	CVaR guard	3	0.86250	0.00000	0.00000	0.84429
probe_cost	0.00	CVaR guard	4	0.83750	0.00000	0.00000	0.81985
probe_cost	0.00	CVaR guard	5	0.86250	0.00625	0.00000	0.83897
probe_cost	0.00	CVaR guard	6	0.78125	0.00000	0.00000	0.76393
probe_cost	0.00	CVaR guard	7	0.86250	0.00000	0.00000	0.84378
probe_cost	0.00	CVaR guard	8	0.86875	0.00000	0.00000	0.85003
probe_cost	0.00	CVaR guard	9	0.83750	0.00625	0.00000	0.81335
probe_cost	0.00	Half-life v5	0	0.76875	0.00000	0.00000	0.76056
probe_cost	0.00	Half-life v5	1	0.85000	0.01250	0.00000	0.82942
probe_cost	0.00	Half-life v5	2	0.83125	0.00625	0.00000	0.81682
probe_cost	0.00	Half-life v5	3	0.83125	0.01250	0.00000	0.81038
probe_cost	0.00	Half-life v5	4	0.75625	0.00625	0.00000	0.74191
probe_cost	0.00	Half-life v5	5	0.88750	0.00625	0.00000	0.87287
probe_cost	0.00	Half-life v5	6	0.76875	0.00625	0.00000	0.75447
probe_cost	0.00	Half-life v5	7	0.80000	0.00000	0.00000	0.79162
probe_cost	0.00	Half-life v5	8	0.80000	0.00625	0.00000	0.78537
probe_cost	0.00	Half-life v5	9	0.85000	0.00000	0.00000	0.84149
probe_cost	0.00	Oracle	0	0.88125	0.00000	0.00000	0.87179
probe_cost	0.00	Oracle	1	0.85000	0.00625	0.00000	0.83419
probe_cost	0.00	Oracle	2	0.81250	0.00000	0.00000	0.80316
probe_cost	0.00	Oracle	3	0.81250	0.00000	0.00000	0.80294
probe_cost	0.00	Oracle	4	0.81250	0.00000	0.00000	0.80301
probe_cost	0.00	Oracle	5	0.86875	0.00000	0.00000	0.85955
probe_cost	0.00	Oracle	6	0.86250	0.00000	0.00000	0.85310

Table 10: Seed-level stress metrics. (continued)

Axis	Level	Method	Seed	Success	Unsafe	Stale	Safety utility
probe_cost	0.00	Oracle	7	0.88125	0.00000	0.00000	0.87176
probe_cost	0.00	Oracle	8	0.84375	0.00000	0.00000	0.83420
probe_cost	0.00	Oracle	9	0.78750	0.00000	0.00000	0.77792
probe_cost	0.25	Domain rand.	0	0.60625	0.18125	0.55000	0.23818
probe_cost	0.25	Domain rand.	1	0.46875	0.22500	0.57500	0.04847
probe_cost	0.25	Domain rand.	2	0.46875	0.24375	0.57500	0.02974
probe_cost	0.25	Domain rand.	3	0.49375	0.20625	0.58125	0.09008
probe_cost	0.25	Domain rand.	4	0.53125	0.21250	0.59375	0.11709
probe_cost	0.25	Domain rand.	5	0.46875	0.20625	0.58125	0.06508
probe_cost	0.25	Domain rand.	6	0.50000	0.23750	0.56875	0.06936
probe_cost	0.25	Domain rand.	7	0.50000	0.19375	0.55625	0.11732
probe_cost	0.25	Domain rand.	8	0.50000	0.21875	0.56250	0.09022
probe_cost	0.25	Domain rand.	9	0.48125	0.16875	0.55625	0.12354
probe_cost	0.25	Conformal	0	0.79375	0.00625	0.00000	0.77564
probe_cost	0.25	Conformal	1	0.83750	0.00625	0.00000	0.81908
probe_cost	0.25	Conformal	2	0.78750	0.00000	0.00000	0.77572
probe_cost	0.25	Conformal	3	0.80000	0.00000	0.00000	0.78843
probe_cost	0.25	Conformal	4	0.79375	0.00625	0.00000	0.77575
probe_cost	0.25	Conformal	5	0.87500	0.00000	0.00000	0.86347
probe_cost	0.25	Conformal	6	0.75625	0.00000	0.00000	0.74393
probe_cost	0.25	Conformal	7	0.87500	0.00000	0.00000	0.86341
probe_cost	0.25	Conformal	8	0.76250	0.00625	0.00000	0.74411
probe_cost	0.25	Conformal	9	0.76875	0.00000	0.00000	0.75694
probe_cost	0.25	Ensemble	0	0.76875	0.00000	0.00000	0.75661
probe_cost	0.25	Ensemble	1	0.78125	0.00625	0.00000	0.76283
probe_cost	0.25	Ensemble	2	0.80625	0.01250	0.00000	0.78163
probe_cost	0.25	Ensemble	3	0.84375	0.00000	0.00000	0.83167
probe_cost	0.25	Ensemble	4	0.81875	0.00000	0.00000	0.80694
probe_cost	0.25	Ensemble	5	0.80000	0.00625	0.00000	0.78182
probe_cost	0.25	Ensemble	6	0.78750	0.00000	0.00000	0.77527
probe_cost	0.25	Ensemble	7	0.79375	0.00625	0.00000	0.77555
probe_cost	0.25	Ensemble	8	0.82500	0.01250	0.00000	0.80010
probe_cost	0.25	Ensemble	9	0.80000	0.00000	0.00000	0.78787
probe_cost	0.25	Bayes survival	0	0.84375	0.00000	0.00000	0.83001
probe_cost	0.25	Bayes survival	1	0.84375	0.00000	0.00000	0.82937
probe_cost	0.25	Bayes survival	2	0.80625	0.00000	0.00000	0.79243
probe_cost	0.25	Bayes survival	3	0.80625	0.00625	0.00000	0.78630
probe_cost	0.25	Bayes survival	4	0.82500	0.00000	0.00000	0.81096
probe_cost	0.25	Bayes survival	5	0.81250	0.00000	0.00000	0.79878
probe_cost	0.25	Bayes survival	6	0.83750	0.00000	0.00000	0.82343
probe_cost	0.25	Bayes survival	7	0.78750	0.00625	0.00000	0.76768
probe_cost	0.25	Bayes survival	8	0.80000	0.00625	0.00000	0.77995
probe_cost	0.25	Bayes survival	9	0.80625	0.00625	0.00000	0.78644
probe_cost	0.25	CVaR guard	0	0.85000	0.00000	0.00000	0.82869
probe_cost	0.25	CVaR guard	1	0.88750	0.00000	0.00000	0.86506
probe_cost	0.25	CVaR guard	2	0.82500	0.00000	0.00000	0.80281
probe_cost	0.25	CVaR guard	3	0.83750	0.00000	0.00000	0.81547
probe_cost	0.25	CVaR guard	4	0.86875	0.00625	0.00000	0.84061
probe_cost	0.25	CVaR guard	5	0.86250	0.00000	0.00000	0.84094
probe_cost	0.25	CVaR guard	6	0.85625	0.00000	0.00000	0.83397
probe_cost	0.25	CVaR guard	7	0.80000	0.00000	0.00000	0.77869
probe_cost	0.25	CVaR guard	8	0.79375	0.00000	0.00000	0.77250
probe_cost	0.25	CVaR guard	9	0.81250	0.00625	0.00000	0.78345
probe_cost	0.25	Half-life v5	0	0.78750	0.00000	0.00000	0.77586
probe_cost	0.25	Half-life v5	1	0.85000	0.00000	0.00000	0.83817
probe_cost	0.25	Half-life v5	2	0.83125	0.01250	0.00000	0.80712
probe_cost	0.25	Half-life v5	3	0.84375	0.01250	0.00000	0.81964
probe_cost	0.25	Half-life v5	4	0.74375	0.00000	0.00000	0.73192
probe_cost	0.25	Half-life v5	5	0.81875	0.00000	0.00000	0.80704
probe_cost	0.25	Half-life v5	6	0.81875	0.00625	0.00000	0.80066
probe_cost	0.25	Half-life v5	7	0.80625	0.00000	0.00000	0.79447
probe_cost	0.25	Half-life v5	8	0.78750	0.00000	0.00000	0.77579
probe_cost	0.25	Half-life v5	9	0.75625	0.00625	0.00000	0.73825
probe_cost	0.25	Oracle	0	0.83750	0.00000	0.00000	0.82441
probe_cost	0.25	Oracle	1	0.83125	0.00000	0.00000	0.81842
probe_cost	0.25	Oracle	2	0.82500	0.00625	0.00000	0.80600
probe_cost	0.25	Oracle	3	0.81875	0.00000	0.00000	0.80583
probe_cost	0.25	Oracle	4	0.80000	0.00000	0.00000	0.78693
probe_cost	0.25	Oracle	5	0.81250	0.00625	0.00000	0.79341
probe_cost	0.25	Oracle	6	0.83750	0.00000	0.00000	0.82459
probe_cost	0.25	Oracle	7	0.81250	0.00000	0.00000	0.79988
probe_cost	0.25	Oracle	8	0.84375	0.00000	0.00000	0.83096
probe_cost	0.25	Oracle	9	0.88125	0.00000	0.00000	0.86868
probe_cost	0.50	Domain rand.	0	0.48750	0.21250	0.55000	0.08821
probe_cost	0.50	Domain rand.	1	0.50000	0.20625	0.57500	0.09846
probe_cost	0.50	Domain rand.	2	0.47500	0.19375	0.56875	0.08807
probe_cost	0.50	Domain rand.	3	0.42500	0.21250	0.56875	0.01934

Table 10: Seed-level stress metrics. (continued)

Axis	Level	Method	Seed	Success	Unsafe	Stale	Safety utility
probe_cost	0.50	Domain rand.	4	0.48750	0.23125	0.57500	0.06098
probe_cost	0.50	Domain rand.	5	0.53750	0.17500	0.58125	0.16505
probe_cost	0.50	Domain rand.	6	0.55625	0.17500	0.57500	0.18592
probe_cost	0.50	Domain rand.	7	0.48750	0.16875	0.56875	0.12554
probe_cost	0.50	Domain rand.	8	0.47500	0.21250	0.56250	0.07146
probe_cost	0.50	Domain rand.	9	0.49375	0.18750	0.56250	0.11519
probe_cost	0.50	Conformal	0	0.73125	0.00625	0.00000	0.70979
probe_cost	0.50	Conformal	1	0.86250	0.00625	0.00000	0.84114
probe_cost	0.50	Conformal	2	0.78750	0.01250	0.00000	0.75968
probe_cost	0.50	Conformal	3	0.82500	0.00000	0.00000	0.80979
probe_cost	0.50	Conformal	4	0.78125	0.00000	0.00000	0.76604
probe_cost	0.50	Conformal	5	0.79375	0.00625	0.00000	0.77229
probe_cost	0.50	Conformal	6	0.81250	0.00000	0.00000	0.79749
probe_cost	0.50	Conformal	7	0.80625	0.00625	0.00000	0.78436
probe_cost	0.50	Conformal	8	0.79375	0.00000	0.00000	0.77843
probe_cost	0.50	Conformal	9	0.71875	0.01250	0.00000	0.69126
probe_cost	0.50	Ensemble	0	0.75000	0.00625	0.00000	0.72799
probe_cost	0.50	Ensemble	1	0.78125	0.00000	0.00000	0.76559
probe_cost	0.50	Ensemble	2	0.83125	0.00625	0.00000	0.80936
probe_cost	0.50	Ensemble	3	0.79375	0.00625	0.00000	0.77208
probe_cost	0.50	Ensemble	4	0.81250	0.00000	0.00000	0.79685
probe_cost	0.50	Ensemble	5	0.84375	0.00000	0.00000	0.82787
probe_cost	0.50	Ensemble	6	0.82500	0.00000	0.00000	0.81000
probe_cost	0.50	Ensemble	7	0.83750	0.00000	0.00000	0.82142
probe_cost	0.50	Ensemble	8	0.81875	0.00625	0.00000	0.79687
probe_cost	0.50	Ensemble	9	0.82500	0.00000	0.00000	0.80912
probe_cost	0.50	Bayes survival	0	0.80625	0.00000	0.00000	0.78867
probe_cost	0.50	Bayes survival	1	0.85625	0.00000	0.00000	0.83833
probe_cost	0.50	Bayes survival	2	0.81250	0.01875	0.00000	0.77565
probe_cost	0.50	Bayes survival	3	0.84375	0.00625	0.00000	0.81995
probe_cost	0.50	Bayes survival	4	0.83750	0.00000	0.00000	0.81971
probe_cost	0.50	Bayes survival	5	0.79375	0.00000	0.00000	0.77661
probe_cost	0.50	Bayes survival	6	0.75625	0.00625	0.00000	0.73266
probe_cost	0.50	Bayes survival	7	0.87500	0.00000	0.00000	0.85787
probe_cost	0.50	Bayes survival	8	0.86875	0.00000	0.00000	0.85108
probe_cost	0.50	Bayes survival	9	0.84375	0.00000	0.00000	0.82564
probe_cost	0.50	CVaR guard	0	0.80625	0.00000	0.00000	0.78011
probe_cost	0.50	CVaR guard	1	0.84375	0.00000	0.00000	0.81839
probe_cost	0.50	CVaR guard	2	0.78750	0.00625	0.00000	0.75511
probe_cost	0.50	CVaR guard	3	0.85625	0.00000	0.00000	0.83056
probe_cost	0.50	CVaR guard	4	0.86250	0.00000	0.00000	0.83702
probe_cost	0.50	CVaR guard	5	0.83125	0.00000	0.00000	0.80627
probe_cost	0.50	CVaR guard	6	0.85000	0.00000	0.00000	0.82501
probe_cost	0.50	CVaR guard	7	0.82500	0.00000	0.00000	0.79932
probe_cost	0.50	CVaR guard	8	0.88750	0.00000	0.00000	0.86205
probe_cost	0.50	CVaR guard	9	0.83125	0.00000	0.00000	0.80523
probe_cost	0.50	Half-life v5	0	0.81250	0.00000	0.00000	0.79674
probe_cost	0.50	Half-life v5	1	0.80625	0.00000	0.00000	0.79126
probe_cost	0.50	Half-life v5	2	0.83125	0.00625	0.00000	0.81001
probe_cost	0.50	Half-life v5	3	0.78750	0.00625	0.00000	0.76616
probe_cost	0.50	Half-life v5	4	0.86250	0.00625	0.00000	0.84083
probe_cost	0.50	Half-life v5	5	0.78750	0.00625	0.00000	0.76618
probe_cost	0.50	Half-life v5	6	0.80000	0.00625	0.00000	0.77942
probe_cost	0.50	Half-life v5	7	0.79375	0.00625	0.00000	0.77231
probe_cost	0.50	Half-life v5	8	0.79375	0.01250	0.00000	0.76617
probe_cost	0.50	Half-life v5	9	0.80000	0.01875	0.00000	0.76595
probe_cost	0.50	Oracle	0	0.80000	0.00000	0.00000	0.78365
probe_cost	0.50	Oracle	1	0.81250	0.00000	0.00000	0.79603
probe_cost	0.50	Oracle	2	0.81250	0.00000	0.00000	0.79604
probe_cost	0.50	Oracle	3	0.78125	0.00000	0.00000	0.76512
probe_cost	0.50	Oracle	4	0.83125	0.00625	0.00000	0.80856
probe_cost	0.50	Oracle	5	0.82500	0.00000	0.00000	0.80867
probe_cost	0.50	Oracle	6	0.82500	0.00000	0.00000	0.80921
probe_cost	0.50	Oracle	7	0.83125	0.00000	0.00000	0.81514
probe_cost	0.50	Oracle	8	0.83125	0.00625	0.00000	0.80833
probe_cost	0.50	Oracle	9	0.83125	0.00000	0.00000	0.81480
probe_cost	0.75	Domain rand.	0	0.48750	0.23125	0.55625	0.06736
probe_cost	0.75	Domain rand.	1	0.51250	0.18125	0.55625	0.14231
probe_cost	0.75	Domain rand.	2	0.45625	0.20625	0.56875	0.05683
probe_cost	0.75	Domain rand.	3	0.58125	0.18750	0.56875	0.20056
probe_cost	0.75	Domain rand.	4	0.47500	0.25000	0.58750	0.02550
probe_cost	0.75	Domain rand.	5	0.51875	0.17500	0.60000	0.13993
probe_cost	0.75	Domain rand.	6	0.53125	0.20000	0.58125	0.13382
probe_cost	0.75	Domain rand.	7	0.60625	0.15000	0.56875	0.26302
probe_cost	0.75	Domain rand.	8	0.50625	0.22500	0.55625	0.09235
probe_cost	0.75	Domain rand.	9	0.56250	0.20625	0.56250	0.16521
probe_cost	0.75	Conformal	0	0.78125	0.00625	0.00000	0.75577

Table 10: Seed-level stress metrics. (continued)

	Axis	Level	Method	Seed	Success	Unsafe	Stale	Safety utility
1998								
1999								
2000								
2001	probe_cost	0.75	Conformal	1	0.73125	0.01250	0.00000	0.70034
2002	probe_cost	0.75	Conformal	2	0.79375	0.00000	0.00000	0.77534
2003	probe_cost	0.75	Conformal	3	0.75000	0.00000	0.00000	0.73151
2004	probe_cost	0.75	Conformal	4	0.77500	0.00625	0.00000	0.74968
2005	probe_cost	0.75	Conformal	5	0.85000	0.00000	0.00000	0.83167
2006	probe_cost	0.75	Conformal	6	0.81250	0.00000	0.00000	0.79415
2007	probe_cost	0.75	Conformal	7	0.76875	0.00625	0.00000	0.74453
2008	probe_cost	0.75	Conformal	8	0.86875	0.00625	0.00000	0.84327
2009	probe_cost	0.75	Conformal	9	0.82500	0.00000	0.00000	0.80666
2010	probe_cost	0.75	Ensemble	0	0.80625	0.01250	0.00000	0.77431
2011	probe_cost	0.75	Ensemble	1	0.85625	0.00000	0.00000	0.83761
2012	probe_cost	0.75	Ensemble	2	0.80625	0.01250	0.00000	0.77527
2013	probe_cost	0.75	Ensemble	3	0.79375	0.00625	0.00000	0.76812
2014	probe_cost	0.75	Ensemble	4	0.85625	0.00000	0.00000	0.83754
2015	probe_cost	0.75	Ensemble	5	0.85000	0.00000	0.00000	0.83122
2016	probe_cost	0.75	Ensemble	6	0.77500	0.01250	0.00000	0.74365
2017	probe_cost	0.75	Ensemble	7	0.82500	0.00000	0.00000	0.80592
2018	probe_cost	0.75	Ensemble	8	0.77500	0.00000	0.00000	0.75591
2019	probe_cost	0.75	Ensemble	9	0.80000	0.00625	0.00000	0.77497
2020	probe_cost	0.75	Bayes survival	0	0.83750	0.00625	0.00000	0.80999
2021	probe_cost	0.75	Bayes survival	1	0.82500	0.00000	0.00000	0.80359
2022	probe_cost	0.75	Bayes survival	2	0.81875	0.01875	0.00000	0.77861
2023	probe_cost	0.75	Bayes survival	3	0.80000	0.01250	0.00000	0.76572
2024	probe_cost	0.75	Bayes survival	4	0.80000	0.00625	0.00000	0.77265
2025	probe_cost	0.75	Bayes survival	5	0.85000	0.00000	0.00000	0.82845
2026	probe_cost	0.75	Bayes survival	6	0.75625	0.01875	0.00000	0.71588
2027	probe_cost	0.75	Bayes survival	7	0.83125	0.00625	0.00000	0.80382
2028	probe_cost	0.75	Bayes survival	8	0.79375	0.00000	0.00000	0.77234
2029	probe_cost	0.75	Bayes survival	9	0.80000	0.00000	0.00000	0.77866
2030	probe_cost	0.75	CVaR guard	0	0.85000	0.00000	0.00000	0.82073
2031	probe_cost	0.75	CVaR guard	1	0.85625	0.00000	0.00000	0.82691
2032	probe_cost	0.75	CVaR guard	2	0.86875	0.00000	0.00000	0.83994
2033	probe_cost	0.75	CVaR guard	3	0.82500	0.00000	0.00000	0.79533
2034	probe_cost	0.75	CVaR guard	4	0.83125	0.00000	0.00000	0.80236
2035	probe_cost	0.75	CVaR guard	5	0.83750	0.01250	0.00000	0.79576
2036	probe_cost	0.75	CVaR guard	6	0.81875	0.00000	0.00000	0.78955
2037	probe_cost	0.75	CVaR guard	7	0.86875	0.00000	0.00000	0.83981
2038	probe_cost	0.75	CVaR guard	8	0.80000	0.00625	0.00000	0.76451
2039	probe_cost	0.75	CVaR guard	9	0.85625	0.00000	0.00000	0.82723
2040	probe_cost	0.75	Half-life v5	0	0.81250	0.00000	0.00000	0.79291
2041	probe_cost	0.75	Half-life v5	1	0.83750	0.00000	0.00000	0.81924
2042	probe_cost	0.75	Half-life v5	2	0.82500	0.00000	0.00000	0.80622
2043	probe_cost	0.75	Half-life v5	3	0.80000	0.00625	0.00000	0.77535
2044	probe_cost	0.75	Half-life v5	4	0.78125	0.00000	0.00000	0.76225
2045	probe_cost	0.75	Half-life v5	5	0.76875	0.00000	0.00000	0.74997
2046	probe_cost	0.75	Half-life v5	6	0.83125	0.00000	0.00000	0.81284
2047	probe_cost	0.75	Half-life v5	7	0.78750	0.00000	0.00000	0.76894
2048	probe_cost	0.75	Half-life v5	8	0.86875	0.01250	0.00000	0.83793
2049	probe_cost	0.75	Half-life v5	9	0.81875	0.00000	0.00000	0.79961
2050	probe_cost	0.75	Oracle	0	0.83125	0.00000	0.00000	0.81141
2051	probe_cost	0.75	Oracle	1	0.89375	0.00000	0.00000	0.87435
	probe_cost	0.75	Oracle	2	0.78750	0.00625	0.00000	0.76134
	probe_cost	0.75	Oracle	3	0.86250	0.00625	0.00000	0.83679
	probe_cost	0.75	Oracle	4	0.78125	0.00000	0.00000	0.76163
	probe_cost	0.75	Oracle	5	0.84375	0.00000	0.00000	0.82405
	probe_cost	0.75	Oracle	6	0.81250	0.00000	0.00000	0.79339
	probe_cost	0.75	Oracle	7	0.83750	0.00625	0.00000	0.81119
	probe_cost	0.75	Oracle	8	0.81875	0.00000	0.00000	0.79920
	probe_cost	0.75	Oracle	9	0.84375	0.00000	0.00000	0.82413
	probe_cost	1.00	Domain rand.	0	0.53125	0.18750	0.55625	0.15481
	probe_cost	1.00	Domain rand.	1	0.58125	0.20000	0.56250	0.19020
	probe_cost	1.00	Domain rand.	2	0.53125	0.20000	0.56875	0.13807
	probe_cost	1.00	Domain rand.	3	0.46875	0.24375	0.58125	0.02762
	probe_cost	1.00	Domain rand.	4	0.52500	0.21250	0.60000	0.10871
	probe_cost	1.00	Domain rand.	5	0.51250	0.21250	0.58750	0.10046
	probe_cost	1.00	Domain rand.	6	0.47500	0.19375	0.58125	0.08382
	probe_cost	1.00	Domain rand.	7	0.52500	0.16250	0.56250	0.17141
	probe_cost	1.00	Domain rand.	8	0.47500	0.22500	0.55625	0.06110
	probe_cost	1.00	Domain rand.	9	0.54375	0.19375	0.55625	0.16107
	probe_cost	1.00	Conformal	0	0.78125	0.00625	0.00000	0.75269
	probe_cost	1.00	Conformal	1	0.79375	0.00625	0.00000	0.76567
	probe_cost	1.00	Conformal	2	0.80000	0.00000	0.00000	0.77828
	probe_cost	1.00	Conformal	3	0.74375	0.00000	0.00000	0.72154
	probe_cost	1.00	Conformal	4	0.84375	0.00000	0.00000	0.82214
	probe_cost	1.00	Conformal	5	0.82500	0.00000	0.00000	0.80309
	probe_cost	1.00	Conformal	6	0.76250	0.00000	0.00000	0.74123
	probe_cost	1.00	Conformal	7	0.79375	0.00625	0.00000	0.76526

Table 10: Seed-level stress metrics. (continued)

Axis	Level	Method	Seed	Success	Unsafe	Stale	Safety utility
probe_cost	1.00	Conformal	8	0.83125	0.00000	0.00000	0.80964
probe_cost	1.00	Conformal	9	0.82500	0.00625	0.00000	0.79606
probe_cost	1.00	Ensemble	0	0.71250	0.01250	0.00000	0.67769
probe_cost	1.00	Ensemble	1	0.85000	0.01250	0.00000	0.81511
probe_cost	1.00	Ensemble	2	0.85000	0.00625	0.00000	0.82110
probe_cost	1.00	Ensemble	3	0.78125	0.00000	0.00000	0.75836
probe_cost	1.00	Ensemble	4	0.78750	0.00000	0.00000	0.76537
probe_cost	1.00	Ensemble	5	0.79375	0.00625	0.00000	0.76560
probe_cost	1.00	Ensemble	6	0.81250	0.00000	0.00000	0.79074
probe_cost	1.00	Ensemble	7	0.78750	0.00000	0.00000	0.76484
probe_cost	1.00	Ensemble	8	0.85000	0.00000	0.00000	0.82824
probe_cost	1.00	Ensemble	9	0.77500	0.00000	0.00000	0.75193
probe_cost	1.00	Bayes survival	0	0.81875	0.00000	0.00000	0.79397
probe_cost	1.00	Bayes survival	1	0.84375	0.00000	0.00000	0.81927
probe_cost	1.00	Bayes survival	2	0.81250	0.00000	0.00000	0.78765
probe_cost	1.00	Bayes survival	3	0.81875	0.00000	0.00000	0.79419
probe_cost	1.00	Bayes survival	4	0.80625	0.00000	0.00000	0.78072
probe_cost	1.00	Bayes survival	5	0.80625	0.00000	0.00000	0.78114
probe_cost	1.00	Bayes survival	6	0.85000	0.00000	0.00000	0.82537
probe_cost	1.00	Bayes survival	7	0.85625	0.00000	0.00000	0.83177
probe_cost	1.00	Bayes survival	8	0.81250	0.00000	0.00000	0.78674
probe_cost	1.00	Bayes survival	9	0.80625	0.00000	0.00000	0.78042
probe_cost	1.00	CVaR guard	0	0.81875	0.00000	0.00000	0.78599
probe_cost	1.00	CVaR guard	1	0.85625	0.00625	0.00000	0.81706
probe_cost	1.00	CVaR guard	2	0.86875	0.00000	0.00000	0.83610
probe_cost	1.00	CVaR guard	3	0.85000	0.00000	0.00000	0.81757
probe_cost	1.00	CVaR guard	4	0.83125	0.00000	0.00000	0.79875
probe_cost	1.00	CVaR guard	5	0.81875	0.00625	0.00000	0.77979
probe_cost	1.00	CVaR guard	6	0.87500	0.00000	0.00000	0.84277
probe_cost	1.00	CVaR guard	7	0.84375	0.00000	0.00000	0.81100
probe_cost	1.00	CVaR guard	8	0.80000	0.00000	0.00000	0.76755
probe_cost	1.00	CVaR guard	9	0.90625	0.00000	0.00000	0.87360
probe_cost	1.00	Half-life v5	0	0.75625	0.00000	0.00000	0.73532
probe_cost	1.00	Half-life v5	1	0.76875	0.00625	0.00000	0.74102
probe_cost	1.00	Half-life v5	2	0.82500	0.00000	0.00000	0.80257
probe_cost	1.00	Half-life v5	3	0.77500	0.00000	0.00000	0.75295
probe_cost	1.00	Half-life v5	4	0.83125	0.00625	0.00000	0.80258
probe_cost	1.00	Half-life v5	5	0.79375	0.00000	0.00000	0.77136
probe_cost	1.00	Half-life v5	6	0.80625	0.00000	0.00000	0.78453
probe_cost	1.00	Half-life v5	7	0.80625	0.00000	0.00000	0.78348
probe_cost	1.00	Half-life v5	8	0.78750	0.00625	0.00000	0.75916
probe_cost	1.00	Half-life v5	9	0.83750	0.00625	0.00000	0.80807
probe_cost	1.00	Oracle	0	0.84375	0.00000	0.00000	0.82029
probe_cost	1.00	Oracle	1	0.81875	0.00000	0.00000	0.79575
probe_cost	1.00	Oracle	2	0.83750	0.00625	0.00000	0.80821
probe_cost	1.00	Oracle	3	0.85000	0.00000	0.00000	0.82752
probe_cost	1.00	Oracle	4	0.86250	0.00000	0.00000	0.84021
probe_cost	1.00	Oracle	5	0.86875	0.00000	0.00000	0.84612
probe_cost	1.00	Oracle	6	0.81250	0.00000	0.00000	0.78987
probe_cost	1.00	Oracle	7	0.80000	0.00000	0.00000	0.77662
probe_cost	1.00	Oracle	8	0.83750	0.00000	0.00000	0.81389
probe_cost	1.00	Oracle	9	0.81875	0.01250	0.00000	0.78326
probe_cost	1.25	Domain rand.	0	0.51875	0.21250	0.56875	0.11309
probe_cost	1.25	Domain rand.	1	0.56875	0.22500	0.56875	0.15060
probe_cost	1.25	Domain rand.	2	0.48125	0.23750	0.58750	0.04424
probe_cost	1.25	Domain rand.	3	0.50625	0.25000	0.57500	0.06100
probe_cost	1.25	Domain rand.	4	0.46875	0.18125	0.57500	0.09218
probe_cost	1.25	Domain rand.	5	0.53125	0.21250	0.57500	0.12346
probe_cost	1.25	Domain rand.	6	0.47500	0.20000	0.58125	0.07757
probe_cost	1.25	Domain rand.	7	0.51875	0.22500	0.55625	0.10485
probe_cost	1.25	Domain rand.	8	0.50625	0.19375	0.56250	0.12144
probe_cost	1.25	Domain rand.	9	0.48125	0.23125	0.56250	0.05898
probe_cost	1.25	Conformal	0	0.80625	0.00000	0.00000	0.77963
probe_cost	1.25	Conformal	1	0.82500	0.00000	0.00000	0.80043
probe_cost	1.25	Conformal	2	0.80625	0.00625	0.00000	0.77362
probe_cost	1.25	Conformal	3	0.86250	0.00000	0.00000	0.83657
probe_cost	1.25	Conformal	4	0.81875	0.01250	0.00000	0.78101
probe_cost	1.25	Conformal	5	0.79375	0.00000	0.00000	0.76804
probe_cost	1.25	Conformal	6	0.78750	0.00625	0.00000	0.75600
probe_cost	1.25	Conformal	7	0.80000	0.00000	0.00000	0.77407
probe_cost	1.25	Conformal	8	0.85625	0.00625	0.00000	0.82475
probe_cost	1.25	Conformal	9	0.79375	0.00000	0.00000	0.76827
probe_cost	1.25	Ensemble	0	0.81875	0.00625	0.00000	0.78543
probe_cost	1.25	Ensemble	1	0.82500	0.00000	0.00000	0.79884
probe_cost	1.25	Ensemble	2	0.81250	0.00000	0.00000	0.78588
probe_cost	1.25	Ensemble	3	0.82500	0.00000	0.00000	0.79861
probe_cost	1.25	Ensemble	4	0.80000	0.00000	0.00000	0.77361

Table 10: Seed-level stress metrics. (continued)

Axis	Level	Method	Seed	Success	Unsafe	Stale	Safety utility
probe_cost	1.25	Ensemble	5	0.78125	0.00625	0.00000	0.74862
probe_cost	1.25	Ensemble	6	0.81250	0.00625	0.00000	0.78009
probe_cost	1.25	Ensemble	7	0.76875	0.00000	0.00000	0.74259
probe_cost	1.25	Ensemble	8	0.87500	0.00625	0.00000	0.84282
probe_cost	1.25	Ensemble	9	0.77500	0.00625	0.00000	0.74214
probe_cost	1.25	Bayes survival	0	0.80000	0.00000	0.00000	0.77088
probe_cost	1.25	Bayes survival	1	0.86875	0.00000	0.00000	0.83918
probe_cost	1.25	Bayes survival	2	0.83750	0.00000	0.00000	0.80747
probe_cost	1.25	Bayes survival	3	0.78750	0.01250	0.00000	0.74521
probe_cost	1.25	Bayes survival	4	0.83125	0.01250	0.00000	0.78805
probe_cost	1.25	Bayes survival	5	0.81250	0.00000	0.00000	0.78292
probe_cost	1.25	Bayes survival	6	0.85000	0.00000	0.00000	0.82179
probe_cost	1.25	Bayes survival	7	0.85000	0.00000	0.00000	0.82111
probe_cost	1.25	Bayes survival	8	0.83750	0.00000	0.00000	0.80975
probe_cost	1.25	Bayes survival	9	0.84375	0.00625	0.00000	0.80952
probe_cost	1.25	CVaR guard	0	0.81875	0.00000	0.00000	0.78303
probe_cost	1.25	CVaR guard	1	0.83750	0.00625	0.00000	0.79554
probe_cost	1.25	CVaR guard	2	0.83750	0.00000	0.00000	0.80178
probe_cost	1.25	CVaR guard	3	0.80625	0.00625	0.00000	0.76474
probe_cost	1.25	CVaR guard	4	0.84375	0.00000	0.00000	0.80894
probe_cost	1.25	CVaR guard	5	0.78750	0.00000	0.00000	0.75224
probe_cost	1.25	CVaR guard	6	0.83125	0.00625	0.00000	0.78929
probe_cost	1.25	CVaR guard	7	0.88750	0.00000	0.00000	0.85133
probe_cost	1.25	CVaR guard	8	0.85000	0.00000	0.00000	0.81383
probe_cost	1.25	CVaR guard	9	0.84375	0.00000	0.00000	0.80780
probe_cost	1.25	Half-life v5	0	0.75625	0.00000	0.00000	0.72940
probe_cost	1.25	Half-life v5	1	0.80000	0.00625	0.00000	0.76737
probe_cost	1.25	Half-life v5	2	0.85625	0.00000	0.00000	0.83077
probe_cost	1.25	Half-life v5	3	0.85625	0.00625	0.00000	0.82339
probe_cost	1.25	Half-life v5	4	0.78750	0.00000	0.00000	0.76088
probe_cost	1.25	Half-life v5	5	0.79375	0.01250	0.00000	0.75578
probe_cost	1.25	Half-life v5	6	0.81250	0.00625	0.00000	0.78078
probe_cost	1.25	Half-life v5	7	0.83750	0.00000	0.00000	0.81179
probe_cost	1.25	Half-life v5	8	0.80625	0.00000	0.00000	0.78054
probe_cost	1.25	Half-life v5	9	0.82500	0.00000	0.00000	0.79861
probe_cost	1.25	Oracle	0	0.81250	0.00000	0.00000	0.78634
probe_cost	1.25	Oracle	1	0.86250	0.00000	0.00000	0.83611
probe_cost	1.25	Oracle	2	0.86875	0.00000	0.00000	0.84213
probe_cost	1.25	Oracle	3	0.81875	0.00625	0.00000	0.78612
probe_cost	1.25	Oracle	4	0.86250	0.00000	0.00000	0.83634
probe_cost	1.25	Oracle	5	0.82500	0.00000	0.00000	0.79906
probe_cost	1.25	Oracle	6	0.82500	0.00000	0.00000	0.79906
probe_cost	1.25	Oracle	7	0.81250	0.00625	0.00000	0.77987
probe_cost	1.25	Oracle	8	0.84375	0.00000	0.00000	0.81759
probe_cost	1.25	Oracle	9	0.86875	0.00000	0.00000	0.84304
probe_cost	1.50	Domain rand.	0	0.54375	0.15000	0.55625	0.20477
probe_cost	1.50	Domain rand.	1	0.45625	0.21875	0.58750	0.03797
probe_cost	1.50	Domain rand.	2	0.58125	0.17500	0.56875	0.21305
probe_cost	1.50	Domain rand.	3	0.48125	0.17500	0.57500	0.11093
probe_cost	1.50	Domain rand.	4	0.50000	0.17500	0.58125	0.12755
probe_cost	1.50	Domain rand.	5	0.49375	0.20625	0.59375	0.08583
probe_cost	1.50	Domain rand.	6	0.53125	0.19375	0.57500	0.14219
probe_cost	1.50	Domain rand.	7	0.54375	0.15625	0.56250	0.19641
probe_cost	1.50	Domain rand.	8	0.48125	0.16250	0.56250	0.12766
probe_cost	1.50	Domain rand.	9	0.45000	0.25000	0.55625	0.01113
probe_cost	1.50	Conformal	0	0.76875	0.01250	0.00000	0.72881
probe_cost	1.50	Conformal	1	0.72500	0.00625	0.00000	0.69054
probe_cost	1.50	Conformal	2	0.75625	0.00000	0.00000	0.72746
probe_cost	1.50	Conformal	3	0.83125	0.00625	0.00000	0.79542
probe_cost	1.50	Conformal	4	0.75625	0.00000	0.00000	0.72910
probe_cost	1.50	Conformal	5	0.77500	0.00000	0.00000	0.74709
probe_cost	1.50	Conformal	6	0.80625	0.01250	0.00000	0.76501
probe_cost	1.50	Conformal	7	0.80000	0.01250	0.00000	0.75926
probe_cost	1.50	Conformal	8	0.80625	0.00000	0.00000	0.77746
probe_cost	1.50	Conformal	9	0.73125	0.00000	0.00000	0.70201
probe_cost	1.50	Ensemble	0	0.76250	0.00000	0.00000	0.73261
probe_cost	1.50	Ensemble	1	0.80000	0.01250	0.00000	0.75899
probe_cost	1.50	Ensemble	2	0.80000	0.00000	0.00000	0.77033
probe_cost	1.50	Ensemble	3	0.75000	0.00625	0.00000	0.71447
probe_cost	1.50	Ensemble	4	0.76875	0.00625	0.00000	0.73421
probe_cost	1.50	Ensemble	5	0.80000	0.00000	0.00000	0.77113
probe_cost	1.50	Ensemble	6	0.79375	0.00625	0.00000	0.75769
probe_cost	1.50	Ensemble	7	0.79375	0.01875	0.00000	0.74531
probe_cost	1.50	Ensemble	8	0.82500	0.00000	0.00000	0.79484
probe_cost	1.50	Ensemble	9	0.80000	0.00000	0.00000	0.77102
probe_cost	1.50	Bayes survival	0	0.88125	0.00000	0.00000	0.85068
probe_cost	1.50	Bayes survival	1	0.85000	0.00000	0.00000	0.81646

Table 10: Seed-level stress metrics. (continued)

Axis	Level	Method	Seed	Success	Unsafe	Stale	Safety utility
probe.cost	1.50	Bayes survival	2	0.80625	0.00625	0.00000	0.76745
probe.cost	1.50	Bayes survival	3	0.79375	0.00000	0.00000	0.76097
probe.cost	1.50	Bayes survival	4	0.85625	0.00000	0.00000	0.82125
probe.cost	1.50	Bayes survival	5	0.81250	0.00625	0.00000	0.77424
probe.cost	1.50	Bayes survival	6	0.80625	0.02500	0.00000	0.74708
probe.cost	1.50	Bayes survival	7	0.84375	0.00000	0.00000	0.80933
probe.cost	1.50	Bayes survival	8	0.83125	0.00625	0.00000	0.79264
probe.cost	1.50	Bayes survival	9	0.84375	0.00000	0.00000	0.81017
probe.cost	1.50	CVaR guard	0	0.84375	0.00000	0.00000	0.80447
probe.cost	1.50	CVaR guard	1	0.81875	0.00000	0.00000	0.77993
probe.cost	1.50	CVaR guard	2	0.86250	0.00000	0.00000	0.82437
probe.cost	1.50	CVaR guard	3	0.85000	0.00000	0.00000	0.81107
probe.cost	1.50	CVaR guard	4	0.86250	0.00000	0.00000	0.82395
probe.cost	1.50	CVaR guard	5	0.83125	0.00625	0.00000	0.78648
probe.cost	1.50	CVaR guard	6	0.80000	0.00000	0.00000	0.76133
probe.cost	1.50	CVaR guard	7	0.86250	0.00000	0.00000	0.82314
probe.cost	1.50	CVaR guard	8	0.86250	0.00625	0.00000	0.81732
probe.cost	1.50	CVaR guard	9	0.77500	0.00000	0.00000	0.73568
probe.cost	1.50	Half-life v5	0	0.88750	0.00625	0.00000	0.85315
probe.cost	1.50	Half-life v5	1	0.76875	0.00625	0.00000	0.73315
probe.cost	1.50	Half-life v5	2	0.76250	0.00625	0.00000	0.72735
probe.cost	1.50	Half-life v5	3	0.82500	0.00000	0.00000	0.79613
probe.cost	1.50	Half-life v5	4	0.83750	0.00000	0.00000	0.80859
probe.cost	1.50	Half-life v5	5	0.83750	0.00000	0.00000	0.80753
probe.cost	1.50	Half-life v5	6	0.82500	0.00000	0.00000	0.79587
probe.cost	1.50	Half-life v5	7	0.81250	0.01250	0.00000	0.77027
probe.cost	1.50	Half-life v5	8	0.79375	0.00625	0.00000	0.75895
probe.cost	1.50	Half-life v5	9	0.73750	0.00000	0.00000	0.70783
probe.cost	1.50	Oracle	0	0.86875	0.00000	0.00000	0.83936
probe.cost	1.50	Oracle	1	0.79375	0.00000	0.00000	0.76508
probe.cost	1.50	Oracle	2	0.85000	0.00000	0.00000	0.82129
probe.cost	1.50	Oracle	3	0.84375	0.00000	0.00000	0.81424
probe.cost	1.50	Oracle	4	0.80625	0.00625	0.00000	0.77076
probe.cost	1.50	Oracle	5	0.78750	0.00625	0.00000	0.75068
probe.cost	1.50	Oracle	6	0.87500	0.00000	0.00000	0.84446
probe.cost	1.50	Oracle	7	0.85000	0.00000	0.00000	0.81969
probe.cost	1.50	Oracle	8	0.86875	0.00000	0.00000	0.83875
probe.cost	1.50	Oracle	9	0.86250	0.00000	0.00000	0.83276
combined	0.00	Domain rand.	0	0.62500	0.04375	0.38125	0.45167
combined	0.00	Domain rand.	1	0.59375	0.04375	0.38750	0.41829
combined	0.00	Domain rand.	2	0.66250	0.03750	0.41250	0.48479
combined	0.00	Domain rand.	3	0.55000	0.06875	0.41250	0.34107
combined	0.00	Domain rand.	4	0.60625	0.02500	0.40000	0.44527
combined	0.00	Domain rand.	5	0.55000	0.06250	0.36875	0.36219
combined	0.00	Domain rand.	6	0.62500	0.04375	0.36250	0.45804
combined	0.00	Domain rand.	7	0.63125	0.03750	0.36250	0.47054
combined	0.00	Domain rand.	8	0.56250	0.03750	0.35625	0.40391
combined	0.00	Domain rand.	9	0.61875	0.08125	0.38125	0.40796
combined	0.00	Conformal	0	0.80000	0.00000	0.00000	0.79340
combined	0.00	Conformal	1	0.85000	0.00625	0.00000	0.83720
combined	0.00	Conformal	2	0.77500	0.01250	0.00000	0.75588
combined	0.00	Conformal	3	0.74375	0.00000	0.00000	0.73727
combined	0.00	Conformal	4	0.78750	0.00000	0.00000	0.78083
combined	0.00	Conformal	5	0.79375	0.00000	0.00000	0.78700
combined	0.00	Conformal	6	0.80000	0.00625	0.00000	0.78701
combined	0.00	Conformal	7	0.80625	0.00000	0.00000	0.79942
combined	0.00	Conformal	8	0.85000	0.00000	0.00000	0.84340
combined	0.00	Conformal	9	0.81875	0.00000	0.00000	0.81224
combined	0.00	Ensemble	0	0.83125	0.00625	0.00000	0.81833
combined	0.00	Ensemble	1	0.80000	0.00000	0.00000	0.79317
combined	0.00	Ensemble	2	0.79375	0.00000	0.00000	0.78707
combined	0.00	Ensemble	3	0.78750	0.00000	0.00000	0.78113
combined	0.00	Ensemble	4	0.76875	0.00000	0.00000	0.76207
combined	0.00	Ensemble	5	0.85000	0.00000	0.00000	0.84324
combined	0.00	Ensemble	6	0.88125	0.00625	0.00000	0.86847
combined	0.00	Ensemble	7	0.77500	0.00000	0.00000	0.76832
combined	0.00	Ensemble	8	0.80000	0.00625	0.00000	0.78700
combined	0.00	Ensemble	9	0.80000	0.00000	0.00000	0.79343
combined	0.00	Bayes survival	0	0.83125	0.00000	0.00000	0.82418
combined	0.00	Bayes survival	1	0.81875	0.00000	0.00000	0.81172
combined	0.00	Bayes survival	2	0.77500	0.00625	0.00000	0.76181
combined	0.00	Bayes survival	3	0.82500	0.00000	0.00000	0.81771
combined	0.00	Bayes survival	4	0.81875	0.00000	0.00000	0.81168
combined	0.00	Bayes survival	5	0.89375	0.00000	0.00000	0.88665
combined	0.00	Bayes survival	6	0.80000	0.00000	0.00000	0.79297
combined	0.00	Bayes survival	7	0.83750	0.00000	0.00000	0.83043
combined	0.00	Bayes survival	8	0.81875	0.00000	0.00000	0.81164

Table 10: Seed-level stress metrics. (continued)

Axis	Level	Method	Seed	Success	Unsafe	Stale	Safety utility
combined	0.00	Bayes survival	9	0.85000	0.00000	0.00000	0.84275
combined	0.00	CVaR guard	0	0.81875	0.00000	0.00000	0.81067
combined	0.00	CVaR guard	1	0.78125	0.00625	0.00000	0.76720
combined	0.00	CVaR guard	2	0.85625	0.00000	0.00000	0.84809
combined	0.00	CVaR guard	3	0.78125	0.00000	0.00000	0.77294
combined	0.00	CVaR guard	4	0.80000	0.00000	0.00000	0.79192
combined	0.00	CVaR guard	5	0.74375	0.00625	0.00000	0.72927
combined	0.00	CVaR guard	6	0.86250	0.00000	0.00000	0.85411
combined	0.00	CVaR guard	7	0.82500	0.00000	0.00000	0.81714
combined	0.00	CVaR guard	8	0.81875	0.00625	0.00000	0.80427
combined	0.00	CVaR guard	9	0.83125	0.00000	0.00000	0.82302
combined	0.00	Half-life v5	0	0.80000	0.00000	0.00000	0.79308
combined	0.00	Half-life v5	1	0.82500	0.01250	0.00000	0.80617
combined	0.00	Half-life v5	2	0.88125	0.00625	0.00000	0.86829
combined	0.00	Half-life v5	3	0.83125	0.00625	0.00000	0.81847
combined	0.00	Half-life v5	4	0.83750	0.00000	0.00000	0.83090
combined	0.00	Half-life v5	5	0.81875	0.00000	0.00000	0.81195
combined	0.00	Half-life v5	6	0.81250	0.00000	0.00000	0.80551
combined	0.00	Half-life v5	7	0.84375	0.00000	0.00000	0.83710
combined	0.00	Half-life v5	8	0.78750	0.00000	0.00000	0.78093
combined	0.00	Half-life v5	9	0.83125	0.00000	0.00000	0.82445
combined	0.00	Oracle	0	0.88125	0.00000	0.00000	0.87438
combined	0.00	Oracle	1	0.84375	0.00625	0.00000	0.83064
combined	0.00	Oracle	2	0.86250	0.00000	0.00000	0.85555
combined	0.00	Oracle	3	0.84375	0.00625	0.00000	0.83037
combined	0.00	Oracle	4	0.89375	0.00000	0.00000	0.88688
combined	0.00	Oracle	5	0.84375	0.00000	0.00000	0.83700
combined	0.00	Oracle	6	0.84375	0.00625	0.00000	0.83068
combined	0.00	Oracle	7	0.87500	0.00625	0.00000	0.86189
combined	0.00	Oracle	8	0.82500	0.00000	0.00000	0.81813
combined	0.00	Oracle	9	0.82500	0.00000	0.00000	0.81790
combined	0.25	Domain rand.	0	0.60625	0.08125	0.46250	0.36783
combined	0.25	Domain rand.	1	0.54375	0.07500	0.47500	0.30732
combined	0.25	Domain rand.	2	0.64375	0.09375	0.50000	0.38009
combined	0.25	Domain rand.	3	0.60000	0.10625	0.48750	0.32811
combined	0.25	Domain rand.	4	0.58750	0.08125	0.50000	0.33633
combined	0.25	Domain rand.	5	0.56875	0.12500	0.48750	0.27813
combined	0.25	Domain rand.	6	0.58750	0.03125	0.46875	0.39691
combined	0.25	Domain rand.	7	0.55625	0.12500	0.46250	0.27413
combined	0.25	Domain rand.	8	0.66250	0.07500	0.45625	0.43245
combined	0.25	Domain rand.	9	0.61250	0.11875	0.46875	0.33449
combined	0.25	Conformal	0	0.83750	0.00000	0.00000	0.82766
combined	0.25	Conformal	1	0.79375	0.00000	0.00000	0.78362
combined	0.25	Conformal	2	0.81250	0.00625	0.00000	0.79615
combined	0.25	Conformal	3	0.78750	0.00000	0.00000	0.77753
combined	0.25	Conformal	4	0.78125	0.00000	0.00000	0.77150
combined	0.25	Conformal	5	0.82500	0.01250	0.00000	0.80276
combined	0.25	Conformal	6	0.81250	0.00000	0.00000	0.80266
combined	0.25	Conformal	7	0.83125	0.00000	0.00000	0.82141
combined	0.25	Conformal	8	0.76875	0.00000	0.00000	0.75871
combined	0.25	Conformal	9	0.85000	0.00000	0.00000	0.84016
combined	0.25	Ensemble	0	0.81875	0.00000	0.00000	0.80857
combined	0.25	Ensemble	1	0.80000	0.00000	0.00000	0.78955
combined	0.25	Ensemble	2	0.83125	0.00000	0.00000	0.82073
combined	0.25	Ensemble	3	0.85000	0.00625	0.00000	0.83331
combined	0.25	Ensemble	4	0.85625	0.00625	0.00000	0.83990
combined	0.25	Ensemble	5	0.82500	0.00000	0.00000	0.81505
combined	0.25	Ensemble	6	0.78750	0.00625	0.00000	0.77101
combined	0.25	Ensemble	7	0.75625	0.00000	0.00000	0.74594
combined	0.25	Ensemble	8	0.77500	0.00625	0.00000	0.75883
combined	0.25	Ensemble	9	0.85625	0.00000	0.00000	0.84589
combined	0.25	Bayes survival	0	0.83750	0.00000	0.00000	0.82666
combined	0.25	Bayes survival	1	0.86250	0.00000	0.00000	0.85180
combined	0.25	Bayes survival	2	0.82500	0.00000	0.00000	0.81389
combined	0.25	Bayes survival	3	0.78125	0.00000	0.00000	0.77039
combined	0.25	Bayes survival	4	0.78750	0.00000	0.00000	0.77675
combined	0.25	Bayes survival	5	0.83125	0.00000	0.00000	0.82051
combined	0.25	Bayes survival	6	0.75625	0.00000	0.00000	0.74548
combined	0.25	Bayes survival	7	0.82500	0.00000	0.00000	0.81410
combined	0.25	Bayes survival	8	0.84375	0.00000	0.00000	0.83271
combined	0.25	Bayes survival	9	0.80000	0.01250	0.00000	0.77629
combined	0.25	CVaR guard	0	0.81250	0.00000	0.00000	0.79778
combined	0.25	CVaR guard	1	0.81250	0.00000	0.00000	0.79819
combined	0.25	CVaR guard	2	0.77500	0.00000	0.00000	0.76042
combined	0.25	CVaR guard	3	0.80000	0.00000	0.00000	0.78590
combined	0.25	CVaR guard	4	0.80625	0.00000	0.00000	0.79169
combined	0.25	CVaR guard	5	0.84375	0.00625	0.00000	0.82327

Table 10: Seed-level stress metrics. (continued)

Axis	Level	Method	Seed	Success	Unsafe	Stale	Safety utility
combined	0.25	CVaR guard	6	0.80625	0.00000	0.00000	0.79179
combined	0.25	CVaR guard	7	0.85000	0.01250	0.00000	0.82334
combined	0.25	CVaR guard	8	0.80000	0.00000	0.00000	0.78551
combined	0.25	CVaR guard	9	0.81250	0.00000	0.00000	0.79731
combined	0.25	Half-life v5	0	0.81250	0.00000	0.00000	0.80232
combined	0.25	Half-life v5	1	0.77500	0.01250	0.00000	0.75238
combined	0.25	Half-life v5	2	0.89375	0.00000	0.00000	0.88348
combined	0.25	Half-life v5	3	0.78125	0.00625	0.00000	0.76474
combined	0.25	Half-life v5	4	0.87500	0.00000	0.00000	0.86501
combined	0.25	Half-life v5	5	0.81250	0.00000	0.00000	0.80251
combined	0.25	Half-life v5	6	0.80625	0.00000	0.00000	0.79623
combined	0.25	Half-life v5	7	0.85625	0.00000	0.00000	0.84626
combined	0.25	Half-life v5	8	0.91250	0.00625	0.00000	0.89619
combined	0.25	Half-life v5	9	0.82500	0.00000	0.00000	0.81464
combined	0.25	Oracle	0	0.92500	0.00000	0.00000	0.91434
combined	0.25	Oracle	1	0.84375	0.00000	0.00000	0.83325
combined	0.25	Oracle	2	0.87500	0.00000	0.00000	0.86414
combined	0.25	Oracle	3	0.81875	0.00000	0.00000	0.80834
combined	0.25	Oracle	4	0.83125	0.00000	0.00000	0.82098
combined	0.25	Oracle	5	0.81875	0.00000	0.00000	0.80828
combined	0.25	Oracle	6	0.78125	0.00000	0.00000	0.77068
combined	0.25	Oracle	7	0.83750	0.00000	0.00000	0.82707
combined	0.25	Oracle	8	0.79375	0.00000	0.00000	0.78316
combined	0.25	Oracle	9	0.86250	0.00625	0.00000	0.84569
combined	0.50	Domain rand.	0	0.49375	0.17500	0.53125	0.13830
combined	0.50	Domain rand.	1	0.55625	0.15625	0.55000	0.21316
combined	0.50	Domain rand.	2	0.54375	0.14375	0.53750	0.21739
combined	0.50	Domain rand.	3	0.52500	0.13750	0.54375	0.20276
combined	0.50	Domain rand.	4	0.53125	0.19375	0.54375	0.15282
combined	0.50	Domain rand.	5	0.45000	0.13750	0.54375	0.12776
combined	0.50	Domain rand.	6	0.53125	0.17500	0.53750	0.17367
combined	0.50	Domain rand.	7	0.53750	0.19375	0.52500	0.16544
combined	0.50	Domain rand.	8	0.47500	0.23125	0.53750	0.06123
combined	0.50	Domain rand.	9	0.48125	0.18125	0.53750	0.11743
combined	0.50	Conformal	0	0.83125	0.00625	0.00000	0.81194
combined	0.50	Conformal	1	0.78125	0.00000	0.00000	0.76790
combined	0.50	Conformal	2	0.81875	0.00000	0.00000	0.80591
combined	0.50	Conformal	3	0.80000	0.00000	0.00000	0.78691
combined	0.50	Conformal	4	0.83125	0.00625	0.00000	0.81169
combined	0.50	Conformal	5	0.82500	0.00625	0.00000	0.80588
combined	0.50	Conformal	6	0.76250	0.00625	0.00000	0.74262
combined	0.50	Conformal	7	0.80000	0.00000	0.00000	0.78671
combined	0.50	Conformal	8	0.79375	0.00625	0.00000	0.77431
combined	0.50	Conformal	9	0.80000	0.00000	0.00000	0.78694
combined	0.50	Ensemble	0	0.82500	0.00625	0.00000	0.80525
combined	0.50	Ensemble	1	0.81875	0.01250	0.00000	0.79256
combined	0.50	Ensemble	2	0.81250	0.00000	0.00000	0.79928
combined	0.50	Ensemble	3	0.76250	0.00625	0.00000	0.74291
combined	0.50	Ensemble	4	0.81250	0.00625	0.00000	0.79290
combined	0.50	Ensemble	5	0.76875	0.00000	0.00000	0.75549
combined	0.50	Ensemble	6	0.77500	0.00000	0.00000	0.76120
combined	0.50	Ensemble	7	0.81250	0.00625	0.00000	0.79255
combined	0.50	Ensemble	8	0.81875	0.00625	0.00000	0.79851
combined	0.50	Ensemble	9	0.81250	0.01250	0.00000	0.78637
combined	0.50	Bayes survival	0	0.81875	0.00000	0.00000	0.80381
combined	0.50	Bayes survival	1	0.83125	0.00000	0.00000	0.81570
combined	0.50	Bayes survival	2	0.80000	0.00000	0.00000	0.78445
combined	0.50	Bayes survival	3	0.79375	0.00000	0.00000	0.77910
combined	0.50	Bayes survival	4	0.82500	0.00000	0.00000	0.80968
combined	0.50	Bayes survival	5	0.81875	0.00000	0.00000	0.80410
combined	0.50	Bayes survival	6	0.81250	0.00000	0.00000	0.79724
combined	0.50	Bayes survival	7	0.82500	0.00000	0.00000	0.80984
combined	0.50	Bayes survival	8	0.83750	0.00000	0.00000	0.82230
combined	0.50	Bayes survival	9	0.84375	0.00000	0.00000	0.82852
combined	0.50	CVaR guard	0	0.81250	0.00000	0.00000	0.79089
combined	0.50	CVaR guard	1	0.86875	0.00000	0.00000	0.84641
combined	0.50	CVaR guard	2	0.86250	0.00000	0.00000	0.84039
combined	0.50	CVaR guard	3	0.76250	0.00000	0.00000	0.74046
combined	0.50	CVaR guard	4	0.85625	0.00000	0.00000	0.83460
combined	0.50	CVaR guard	5	0.84375	0.01250	0.00000	0.80991
combined	0.50	CVaR guard	6	0.83750	0.00000	0.00000	0.81644
combined	0.50	CVaR guard	7	0.82500	0.00000	0.00000	0.80319
combined	0.50	CVaR guard	8	0.77500	0.00000	0.00000	0.75319
combined	0.50	CVaR guard	9	0.85625	0.00000	0.00000	0.83415
combined	0.50	Half-life v5	0	0.75625	0.00000	0.00000	0.74306
combined	0.50	Half-life v5	1	0.83125	0.00625	0.00000	0.81153
combined	0.50	Half-life v5	2	0.81250	0.00625	0.00000	0.79326

Table 10: Seed-level stress metrics. (continued)

Axis	Level	Method	Seed	Success	Unsafe	Stale	Safety utility
combined	0.50	Half-life v5	3	0.79375	0.00000	0.00000	0.78069
combined	0.50	Half-life v5	4	0.80625	0.00000	0.00000	0.79297
combined	0.50	Half-life v5	5	0.81875	0.01250	0.00000	0.79336
combined	0.50	Half-life v5	6	0.81875	0.01250	0.00000	0.79221
combined	0.50	Half-life v5	7	0.82500	0.00000	0.00000	0.81139
combined	0.50	Half-life v5	8	0.81875	0.00625	0.00000	0.79836
combined	0.50	Half-life v5	9	0.80625	0.00625	0.00000	0.78672
combined	0.50	Oracle	0	0.85625	0.00000	0.00000	0.84212
combined	0.50	Oracle	1	0.82500	0.00000	0.00000	0.81048
combined	0.50	Oracle	2	0.82500	0.00000	0.00000	0.81048
combined	0.50	Oracle	3	0.81875	0.00000	0.00000	0.80465
combined	0.50	Oracle	4	0.88125	0.00000	0.00000	0.86754
combined	0.50	Oracle	5	0.83750	0.00000	0.00000	0.82353
combined	0.50	Oracle	6	0.84375	0.00625	0.00000	0.82318
combined	0.50	Oracle	7	0.83750	0.00000	0.00000	0.82327
combined	0.50	Oracle	8	0.82500	0.00000	0.00000	0.81083
combined	0.50	Oracle	9	0.85000	0.00000	0.00000	0.83564
combined	0.75	Domain rand.	0	0.51875	0.21250	0.56875	0.11309
combined	0.75	Domain rand.	1	0.48750	0.24375	0.57500	0.04849
combined	0.75	Domain rand.	2	0.58125	0.17500	0.58750	0.20668
combined	0.75	Domain rand.	3	0.50625	0.21250	0.58750	0.09421
combined	0.75	Domain rand.	4	0.53125	0.25625	0.60000	0.07126
combined	0.75	Domain rand.	5	0.48125	0.23750	0.60625	0.03786
combined	0.75	Domain rand.	6	0.53750	0.22500	0.58750	0.11297
combined	0.75	Domain rand.	7	0.40625	0.25000	0.58125	-0.04113
combined	0.75	Domain rand.	8	0.45625	0.30625	0.58750	-0.04944
combined	0.75	Domain rand.	9	0.50625	0.21250	0.58750	0.09421
combined	0.75	Conformal	0	0.76875	0.00000	0.00000	0.75133
combined	0.75	Conformal	1	0.78125	0.00000	0.00000	0.76331
combined	0.75	Conformal	2	0.83125	0.00000	0.00000	0.81353
combined	0.75	Conformal	3	0.78750	0.00000	0.00000	0.76956
combined	0.75	Conformal	4	0.81250	0.00000	0.00000	0.79510
combined	0.75	Conformal	5	0.81875	0.00000	0.00000	0.80085
combined	0.75	Conformal	6	0.83125	0.00000	0.00000	0.81378
combined	0.75	Conformal	7	0.78750	0.00625	0.00000	0.76369
combined	0.75	Conformal	8	0.76875	0.01250	0.00000	0.73967
combined	0.75	Conformal	9	0.78750	0.00000	0.00000	0.76950
combined	0.75	Ensemble	0	0.79375	0.01250	0.00000	0.76369
combined	0.75	Ensemble	1	0.79375	0.00000	0.00000	0.77568
combined	0.75	Ensemble	2	0.81250	0.00000	0.00000	0.79468
combined	0.75	Ensemble	3	0.85625	0.00000	0.00000	0.83793
combined	0.75	Ensemble	4	0.82500	0.01250	0.00000	0.79469
combined	0.75	Ensemble	5	0.78125	0.00625	0.00000	0.75725
combined	0.75	Ensemble	6	0.76875	0.00000	0.00000	0.75114
combined	0.75	Ensemble	7	0.77500	0.00000	0.00000	0.75714
combined	0.75	Ensemble	8	0.78750	0.01875	0.00000	0.75143
combined	0.75	Ensemble	9	0.79375	0.00000	0.00000	0.77587
combined	0.75	Bayes survival	0	0.80000	0.00000	0.00000	0.77938
combined	0.75	Bayes survival	1	0.87500	0.00000	0.00000	0.85472
combined	0.75	Bayes survival	2	0.85000	0.00000	0.00000	0.83026
combined	0.75	Bayes survival	3	0.84375	0.00000	0.00000	0.82363
combined	0.75	Bayes survival	4	0.85000	0.00000	0.00000	0.83007
combined	0.75	Bayes survival	5	0.83125	0.00000	0.00000	0.81091
combined	0.75	Bayes survival	6	0.80000	0.00000	0.00000	0.77996
combined	0.75	Bayes survival	7	0.80000	0.00000	0.00000	0.77992
combined	0.75	Bayes survival	8	0.83750	0.00000	0.00000	0.81717
combined	0.75	Bayes survival	9	0.78750	0.00625	0.00000	0.76044
combined	0.75	CVaR guard	0	0.84375	0.00000	0.00000	0.81434
combined	0.75	CVaR guard	1	0.86250	0.00000	0.00000	0.83344
combined	0.75	CVaR guard	2	0.83125	0.00625	0.00000	0.79566
combined	0.75	CVaR guard	3	0.83750	0.00000	0.00000	0.80785
combined	0.75	CVaR guard	4	0.86250	0.00000	0.00000	0.83301
combined	0.75	CVaR guard	5	0.85625	0.00000	0.00000	0.82633
combined	0.75	CVaR guard	6	0.83750	0.00000	0.00000	0.80832
combined	0.75	CVaR guard	7	0.89375	0.00000	0.00000	0.86477
combined	0.75	CVaR guard	8	0.84375	0.00000	0.00000	0.81508
combined	0.75	CVaR guard	9	0.85625	0.00625	0.00000	0.82072
combined	0.75	Half-life v5	0	0.82500	0.00000	0.00000	0.80801
combined	0.75	Half-life v5	1	0.76250	0.01250	0.00000	0.73255
combined	0.75	Half-life v5	2	0.79375	0.00000	0.00000	0.77674
combined	0.75	Half-life v5	3	0.81875	0.00000	0.00000	0.80147
combined	0.75	Half-life v5	4	0.79375	0.00000	0.00000	0.77649
combined	0.75	Half-life v5	5	0.78125	0.00000	0.00000	0.76359
combined	0.75	Half-life v5	6	0.78750	0.00000	0.00000	0.77049
combined	0.75	Half-life v5	7	0.80625	0.00625	0.00000	0.78289
combined	0.75	Half-life v5	8	0.80625	0.00000	0.00000	0.78974
combined	0.75	Half-life v5	9	0.82500	0.00000	0.00000	0.80766

Table 10: Seed-level stress metrics. (continued)

Axis	Level	Method	Seed	Success	Unsafe	Stale	Safety utility
combined	0.75	Oracle	0	0.85625	0.00000	0.00000	0.83822
combined	0.75	Oracle	1	0.80000	0.00000	0.00000	0.78212
combined	0.75	Oracle	2	0.92500	0.00000	0.00000	0.90714
combined	0.75	Oracle	3	0.85000	0.00000	0.00000	0.83189
combined	0.75	Oracle	4	0.84375	0.00000	0.00000	0.82625
combined	0.75	Oracle	5	0.84375	0.00625	0.00000	0.81930
combined	0.75	Oracle	6	0.77500	0.00000	0.00000	0.75677
combined	0.75	Oracle	7	0.86875	0.00000	0.00000	0.85075
combined	0.75	Oracle	8	0.86250	0.00000	0.00000	0.84485
combined	0.75	Oracle	9	0.83125	0.00000	0.00000	0.81317
combined	1.00	Domain rand.	0	0.48125	0.25000	0.63125	0.01687
combined	1.00	Domain rand.	1	0.41250	0.31875	0.61875	-0.11631
combined	1.00	Domain rand.	2	0.45000	0.34375	0.61250	-0.10166
combined	1.00	Domain rand.	3	0.44375	0.28125	0.61875	-0.04759
combined	1.00	Domain rand.	4	0.44375	0.31250	0.62500	-0.08094
combined	1.00	Domain rand.	5	0.48125	0.26250	0.63125	0.00439
combined	1.00	Domain rand.	6	0.48125	0.30625	0.61250	-0.03294
combined	1.00	Domain rand.	7	0.50000	0.26250	0.60625	0.03164
combined	1.00	Domain rand.	8	0.47500	0.25000	0.62500	0.01275
combined	1.00	Domain rand.	9	0.53125	0.25625	0.61250	0.06701
combined	1.00	Conformal	0	0.80625	0.00000	0.00000	0.78352
combined	1.00	Conformal	1	0.72500	0.01875	0.00000	0.68369
combined	1.00	Conformal	2	0.79375	0.00625	0.00000	0.76477
combined	1.00	Conformal	3	0.79375	0.00000	0.00000	0.77087
combined	1.00	Conformal	4	0.83750	0.01250	0.00000	0.80243
combined	1.00	Conformal	5	0.76875	0.00000	0.00000	0.74624
combined	1.00	Conformal	6	0.80625	0.00000	0.00000	0.78336
combined	1.00	Conformal	7	0.85000	0.00000	0.00000	0.82750
combined	1.00	Conformal	8	0.77500	0.00000	0.00000	0.75242
combined	1.00	Conformal	9	0.81250	0.00000	0.00000	0.79038
combined	1.00	Ensemble	0	0.80000	0.00000	0.00000	0.77773
combined	1.00	Ensemble	1	0.73750	0.00000	0.00000	0.71531
combined	1.00	Ensemble	2	0.77500	0.00625	0.00000	0.74579
combined	1.00	Ensemble	3	0.76250	0.00625	0.00000	0.73306
combined	1.00	Ensemble	4	0.81875	0.00625	0.00000	0.78908
combined	1.00	Ensemble	5	0.71875	0.00000	0.00000	0.69655
combined	1.00	Ensemble	6	0.77500	0.01250	0.00000	0.73931
combined	1.00	Ensemble	7	0.81875	0.00000	0.00000	0.79625
combined	1.00	Ensemble	8	0.81875	0.00625	0.00000	0.79015
combined	1.00	Ensemble	9	0.81875	0.00000	0.00000	0.79640
combined	1.00	Bayes survival	0	0.83750	0.00000	0.00000	0.81250
combined	1.00	Bayes survival	1	0.84375	0.00625	0.00000	0.81159
combined	1.00	Bayes survival	2	0.85625	0.01250	0.00000	0.81784
combined	1.00	Bayes survival	3	0.82500	0.00000	0.00000	0.79955
combined	1.00	Bayes survival	4	0.85000	0.00000	0.00000	0.82438
combined	1.00	Bayes survival	5	0.82500	0.00625	0.00000	0.79344
combined	1.00	Bayes survival	6	0.82500	0.00000	0.00000	0.79938
combined	1.00	Bayes survival	7	0.81875	0.00000	0.00000	0.79375
combined	1.00	Bayes survival	8	0.81875	0.00000	0.00000	0.79276
combined	1.00	Bayes survival	9	0.91250	0.00000	0.00000	0.88734
combined	1.00	CVaR guard	0	0.88125	0.00000	0.00000	0.84797
combined	1.00	CVaR guard	1	0.83750	0.00000	0.00000	0.80407
combined	1.00	CVaR guard	2	0.84375	0.00000	0.00000	0.81003
combined	1.00	CVaR guard	3	0.85000	0.00625	0.00000	0.81026
combined	1.00	CVaR guard	4	0.87500	0.00000	0.00000	0.84120
combined	1.00	CVaR guard	5	0.86875	0.00000	0.00000	0.83555
combined	1.00	CVaR guard	6	0.85000	0.00000	0.00000	0.81620
combined	1.00	CVaR guard	7	0.88750	0.00625	0.00000	0.84776
combined	1.00	CVaR guard	8	0.86875	0.00000	0.00000	0.83503
combined	1.00	CVaR guard	9	0.80625	0.00000	0.00000	0.77305
combined	1.00	Half-life v5	0	0.77500	0.00625	0.00000	0.74815
combined	1.00	Half-life v5	1	0.80625	0.00625	0.00000	0.77871
combined	1.00	Half-life v5	2	0.81875	0.00000	0.00000	0.79776
combined	1.00	Half-life v5	3	0.78125	0.00625	0.00000	0.75371
combined	1.00	Half-life v5	4	0.77500	0.00000	0.00000	0.75348
combined	1.00	Half-life v5	5	0.80625	0.00625	0.00000	0.77925
combined	1.00	Half-life v5	6	0.87500	0.00000	0.00000	0.85417
combined	1.00	Half-life v5	7	0.80625	0.01250	0.00000	0.77300
combined	1.00	Half-life v5	8	0.77500	0.00000	0.00000	0.75340
combined	1.00	Half-life v5	9	0.82500	0.00000	0.00000	0.80401
combined	1.00	Oracle	0	0.84375	0.00000	0.00000	0.82208
combined	1.00	Oracle	1	0.75625	0.00000	0.00000	0.73520
combined	1.00	Oracle	2	0.82500	0.00625	0.00000	0.79687
combined	1.00	Oracle	3	0.78125	0.00000	0.00000	0.75943
combined	1.00	Oracle	4	0.81250	0.00000	0.00000	0.79061
combined	1.00	Oracle	5	0.82500	0.00000	0.00000	0.80334
combined	1.00	Oracle	6	0.88125	0.00625	0.00000	0.85342

Table 10: Seed-level stress metrics. (continued)

Axis	Level	Method	Seed	Success	Unsafe	Stale	Safety utility
combined	1.00	Oracle	7	0.88750	0.00000	0.00000	0.86637
combined	1.00	Oracle	8	0.83125	0.00000	0.00000	0.80951
combined	1.00	Oracle	9	0.80625	0.00000	0.00000	0.78504
combined	1.25	Domain rand.	0	0.46875	0.33125	0.65625	-0.08529
combined	1.25	Domain rand.	1	0.42500	0.28750	0.63750	-0.07896
combined	1.25	Domain rand.	2	0.46875	0.33750	0.65000	-0.08941
combined	1.25	Domain rand.	3	0.45000	0.31250	0.64375	-0.08106
combined	1.25	Domain rand.	4	0.43125	0.33125	0.66875	-0.12704
combined	1.25	Domain rand.	5	0.43750	0.36250	0.65625	-0.14776
combined	1.25	Domain rand.	6	0.43750	0.31250	0.64375	-0.09356
combined	1.25	Domain rand.	7	0.45625	0.31250	0.64375	-0.07481
combined	1.25	Domain rand.	8	0.46250	0.28750	0.65000	-0.04571
combined	1.25	Domain rand.	9	0.45625	0.35625	0.62500	-0.11214
combined	1.25	Conformal	0	0.76250	0.00625	0.00000	0.72886
combined	1.25	Conformal	1	0.77500	0.00000	0.00000	0.74615
combined	1.25	Conformal	2	0.79375	0.00000	0.00000	0.76635
combined	1.25	Conformal	3	0.83750	0.00000	0.00000	0.80942
combined	1.25	Conformal	4	0.79375	0.00625	0.00000	0.75938
combined	1.25	Conformal	5	0.78750	0.00625	0.00000	0.75331
combined	1.25	Conformal	6	0.79375	0.00625	0.00000	0.76024
combined	1.25	Conformal	7	0.75625	0.00625	0.00000	0.72270
combined	1.25	Conformal	8	0.80000	0.00000	0.00000	0.77215
combined	1.25	Conformal	9	0.85000	0.00625	0.00000	0.81572
combined	1.25	Ensemble	0	0.84375	0.00000	0.00000	0.81631
combined	1.25	Ensemble	1	0.80625	0.00000	0.00000	0.77749
combined	1.25	Ensemble	2	0.75625	0.00000	0.00000	0.72863
combined	1.25	Ensemble	3	0.85000	0.00625	0.00000	0.81604
combined	1.25	Ensemble	4	0.83125	0.00000	0.00000	0.80340
combined	1.25	Ensemble	5	0.79375	0.00625	0.00000	0.75956
combined	1.25	Ensemble	6	0.80000	0.01250	0.00000	0.76002
combined	1.25	Ensemble	7	0.84375	0.00625	0.00000	0.81056
combined	1.25	Ensemble	8	0.81250	0.00000	0.00000	0.78506
combined	1.25	Ensemble	9	0.81250	0.00000	0.00000	0.78442
combined	1.25	Bayes survival	0	0.86250	0.00000	0.00000	0.83214
combined	1.25	Bayes survival	1	0.81250	0.00000	0.00000	0.78287
combined	1.25	Bayes survival	2	0.79375	0.00000	0.00000	0.76321
combined	1.25	Bayes survival	3	0.85625	0.00000	0.00000	0.82630
combined	1.25	Bayes survival	4	0.83750	0.00000	0.00000	0.80746
combined	1.25	Bayes survival	5	0.84375	0.00000	0.00000	0.81393
combined	1.25	Bayes survival	6	0.83125	0.00000	0.00000	0.80117
combined	1.25	Bayes survival	7	0.78750	0.00000	0.00000	0.75750
combined	1.25	Bayes survival	8	0.86250	0.00000	0.00000	0.83228
combined	1.25	Bayes survival	9	0.82500	0.00625	0.00000	0.78844
combined	1.25	CVaR guard	0	0.86250	0.00000	0.00000	0.82706
combined	1.25	CVaR guard	1	0.78750	0.00625	0.00000	0.74582
combined	1.25	CVaR guard	2	0.88125	0.00000	0.00000	0.84595
combined	1.25	CVaR guard	3	0.82500	0.00000	0.00000	0.78961
combined	1.25	CVaR guard	4	0.86875	0.00000	0.00000	0.83350
combined	1.25	CVaR guard	5	0.84375	0.00000	0.00000	0.80841
combined	1.25	CVaR guard	6	0.81250	0.00000	0.00000	0.77706
combined	1.25	CVaR guard	7	0.84375	0.00000	0.00000	0.80831
combined	1.25	CVaR guard	8	0.83125	0.00625	0.00000	0.78948
combined	1.25	CVaR guard	9	0.78125	0.00000	0.00000	0.74581
combined	1.25	Half-life v5	0	0.76250	0.00625	0.00000	0.73286
combined	1.25	Half-life v5	1	0.78750	0.01250	0.00000	0.74967
combined	1.25	Half-life v5	2	0.75000	0.01250	0.00000	0.71221
combined	1.25	Half-life v5	3	0.78750	0.00625	0.00000	0.75659
combined	1.25	Half-life v5	4	0.80000	0.00625	0.00000	0.76963
combined	1.25	Half-life v5	5	0.80625	0.00625	0.00000	0.77420
combined	1.25	Half-life v5	6	0.71250	0.00000	0.00000	0.68697
combined	1.25	Half-life v5	7	0.85625	0.01250	0.00000	0.81932
combined	1.25	Half-life v5	8	0.83750	0.00000	0.00000	0.81252
combined	1.25	Half-life v5	9	0.76250	0.00000	0.00000	0.73584
combined	1.25	Oracle	0	0.83750	0.00000	0.00000	0.81174
combined	1.25	Oracle	1	0.83125	0.00000	0.00000	0.80517
combined	1.25	Oracle	2	0.82500	0.00000	0.00000	0.79960
combined	1.25	Oracle	3	0.83750	0.00000	0.00000	0.81210
combined	1.25	Oracle	4	0.80000	0.00000	0.00000	0.77406
combined	1.25	Oracle	5	0.88750	0.00000	0.00000	0.86215
combined	1.25	Oracle	6	0.80625	0.00000	0.00000	0.78103
combined	1.25	Oracle	7	0.82500	0.00000	0.00000	0.79951
combined	1.25	Oracle	8	0.81875	0.00000	0.00000	0.79308
combined	1.25	Oracle	9	0.82500	0.00000	0.00000	0.79978
combined	1.50	Domain rand.	0	0.38125	0.36875	0.65625	-0.21026
combined	1.50	Domain rand.	1	0.39375	0.38125	0.65625	-0.21024
combined	1.50	Domain rand.	2	0.37500	0.34375	0.65625	-0.19153
combined	1.50	Domain rand.	3	0.40000	0.38750	0.66250	-0.21236

Table 10: Seed-level stress metrics. (continued)

Axis	Level	Method	Seed	Success	Unsafe	Stale	Safety utility
combined	1.50	Domain rand.	4	0.38750	0.40000	0.67500	-0.24160
combined	1.50	Domain rand.	5	0.41250	0.35000	0.66875	-0.16453
combined	1.50	Domain rand.	6	0.45000	0.34375	0.66250	-0.11866
combined	1.50	Domain rand.	7	0.46250	0.39375	0.66875	-0.15823
combined	1.50	Domain rand.	8	0.33750	0.40625	0.68125	-0.29997
combined	1.50	Domain rand.	9	0.37500	0.40000	0.65000	-0.24560
combined	1.50	Conformal	0	0.80625	0.00000	0.00000	0.77424
combined	1.50	Conformal	1	0.81875	0.00625	0.00000	0.78172
combined	1.50	Conformal	2	0.85000	0.00000	0.00000	0.81836
combined	1.50	Conformal	3	0.77500	0.00000	0.00000	0.74347
combined	1.50	Conformal	4	0.81250	0.00000	0.00000	0.78034
combined	1.50	Conformal	5	0.83750	0.00000	0.00000	0.80626
combined	1.50	Conformal	6	0.79375	0.00625	0.00000	0.75499
combined	1.50	Conformal	7	0.83750	0.00625	0.00000	0.79998
combined	1.50	Conformal	8	0.81250	0.00625	0.00000	0.77414
combined	1.50	Conformal	9	0.78125	0.00625	0.00000	0.74270
combined	1.50	Ensemble	0	0.81250	0.00000	0.00000	0.78095
combined	1.50	Ensemble	1	0.79375	0.00000	0.00000	0.76321
combined	1.50	Ensemble	2	0.80625	0.00000	0.00000	0.77412
combined	1.50	Ensemble	3	0.78750	0.00625	0.00000	0.74974
combined	1.50	Ensemble	4	0.84375	0.00000	0.00000	0.81204
combined	1.50	Ensemble	5	0.81875	0.00625	0.00000	0.78174
combined	1.50	Ensemble	6	0.82500	0.00000	0.00000	0.79381
combined	1.50	Ensemble	7	0.80625	0.00625	0.00000	0.76900
combined	1.50	Ensemble	8	0.78125	0.00625	0.00000	0.74316
combined	1.50	Ensemble	9	0.82500	0.00000	0.00000	0.79397
combined	1.50	Bayes survival	0	0.90625	0.00625	0.00000	0.86604
combined	1.50	Bayes survival	1	0.84375	0.00000	0.00000	0.81010
combined	1.50	Bayes survival	2	0.80625	0.00000	0.00000	0.77242
combined	1.50	Bayes survival	3	0.82500	0.00000	0.00000	0.79150
combined	1.50	Bayes survival	4	0.81875	0.00000	0.00000	0.78520
combined	1.50	Bayes survival	5	0.86250	0.00000	0.00000	0.82925
combined	1.50	Bayes survival	6	0.82500	0.00000	0.00000	0.79145
combined	1.50	Bayes survival	7	0.84375	0.00000	0.00000	0.81025
combined	1.50	Bayes survival	8	0.81875	0.00000	0.00000	0.78480
combined	1.50	Bayes survival	9	0.83750	0.00000	0.00000	0.80402
combined	1.50	CVaR guard	0	0.85625	0.01250	0.00000	0.80750
combined	1.50	CVaR guard	1	0.80000	0.00000	0.00000	0.76383
combined	1.50	CVaR guard	2	0.80000	0.00000	0.00000	0.76379
combined	1.50	CVaR guard	3	0.81250	0.00000	0.00000	0.77629
combined	1.50	CVaR guard	4	0.85625	0.00000	0.00000	0.82003
combined	1.50	CVaR guard	5	0.80000	0.00000	0.00000	0.76381
combined	1.50	CVaR guard	6	0.82500	0.00000	0.00000	0.78874
combined	1.50	CVaR guard	7	0.83125	0.00000	0.00000	0.79503
combined	1.50	CVaR guard	8	0.82500	0.00000	0.00000	0.78879
combined	1.50	CVaR guard	9	0.81250	0.00000	0.00000	0.77628
combined	1.50	Half-life v5	0	0.79375	0.00625	0.00000	0.75804
combined	1.50	Half-life v5	1	0.75625	0.00625	0.00000	0.72244
combined	1.50	Half-life v5	2	0.81875	0.02500	0.00000	0.76492
combined	1.50	Half-life v5	3	0.79375	0.00000	0.00000	0.76454
combined	1.50	Half-life v5	4	0.78125	0.00625	0.00000	0.74786
combined	1.50	Half-life v5	5	0.83125	0.00000	0.00000	0.80287
combined	1.50	Half-life v5	6	0.73750	0.00000	0.00000	0.70703
combined	1.50	Half-life v5	7	0.82500	0.00625	0.00000	0.79013
combined	1.50	Half-life v5	8	0.78125	0.00625	0.00000	0.74492
combined	1.50	Half-life v5	9	0.80000	0.01250	0.00000	0.76018
combined	1.50	Oracle	0	0.83125	0.00000	0.00000	0.80168
combined	1.50	Oracle	1	0.84375	0.00000	0.00000	0.81502
combined	1.50	Oracle	2	0.82500	0.00000	0.00000	0.79579
combined	1.50	Oracle	3	0.81875	0.00000	0.00000	0.78959
combined	1.50	Oracle	4	0.85000	0.00000	0.00000	0.82121
combined	1.50	Oracle	5	0.83125	0.00000	0.00000	0.80329
combined	1.50	Oracle	6	0.83125	0.00000	0.00000	0.80206
combined	1.50	Oracle	7	0.81250	0.00625	0.00000	0.77689
combined	1.50	Oracle	8	0.82500	0.00000	0.00000	0.79627
combined	1.50	Oracle	9	0.79375	0.00000	0.00000	0.76542

E ABLATION SEED-LEVEL METRICS

Table 11: Seed-level ablation metrics.

Split	Ablation	Seed	Success	Unsafe	HLE	Cost	Safety utility
combined_micro.shift	full_skill_half_life_scheduler_v5	0	0.82188	0.00625	0.11792	1.10078	0.80227
combined_micro.shift	full_skill_half_life_scheduler_v5	1	0.81250	0.00000	0.12626	1.09725	0.79888
combined_micro.shift	full_skill_half_life_scheduler_v5	2	0.78750	0.00313	0.11945	1.09842	0.77097
combined_micro.shift	full_skill_half_life_scheduler_v5	3	0.84375	0.00313	0.11897	1.09834	0.82723
combined_micro.shift	full_skill_half_life_scheduler_v5	4	0.80937	0.00000	0.12795	1.09636	0.79588
combined_micro.shift	full_skill_half_life_scheduler_v5	5	0.82812	0.00313	0.12940	1.09702	0.81179
combined_micro.shift	full_skill_half_life_scheduler_v5	6	0.85313	0.00000	0.12890	1.09347	0.84004
combined_micro.shift	full_skill_half_life_scheduler_v5	7	0.80937	0.00000	0.12414	1.09525	0.79604
combined_micro.shift	full_skill_half_life_scheduler_v5	8	0.82812	0.00000	0.12015	1.09428	0.81493
combined_micro.shift	full_skill_half_life_scheduler_v5	9	0.81250	0.00625	0.12435	1.10189	0.79274
combined_micro.shift	minus_per_skill_survival	0	0.81250	0.00313	0.13077	1.10241	0.79541
combined_micro.shift	minus_per_skill_survival	1	0.87813	0.00000	0.13448	1.09850	0.86434
combined_micro.shift	minus_per_skill_survival	2	0.80937	0.00000	0.13324	1.09969	0.79542
combined_micro.shift	minus_per_skill_survival	3	0.80937	0.00000	0.13251	1.09864	0.79557
combined_micro.shift	minus_per_skill_survival	4	0.79688	0.00313	0.13933	1.09937	0.78021
combined_micro.shift	minus_per_skill_survival	5	0.81875	0.00625	0.13548	1.10300	0.79883
combined_micro.shift	minus_per_skill_survival	6	0.83437	0.00000	0.13798	1.09680	0.82082
combined_micro.shift	minus_per_skill_survival	7	0.80937	0.00313	0.12747	1.10116	0.79246
combined_micro.shift	minus_per_skill_survival	8	0.76562	0.00625	0.12955	1.10314	0.74569
combined_micro.shift	minus_per_skill_survival	9	0.80000	0.00000	0.13953	1.09931	0.78610
combined_micro.shift	minus_probe_updates	0	0.79063	0.00313	0.18382	1.05059	0.78079
combined_micro.shift	minus_probe_updates	1	0.79688	0.00313	0.19382	1.05059	0.78704
combined_micro.shift	minus_probe_updates	2	0.82188	0.00313	0.19545	1.04897	0.81227
combined_micro.shift	minus_probe_updates	3	0.83125	0.00313	0.18287	1.04978	0.82153
combined_micro.shift	minus_probe_updates	4	0.79688	0.00000	0.19654	1.04794	0.79016
combined_micro.shift	minus_probe_updates	5	0.74687	0.00937	0.19126	1.05509	0.73091
combined_micro.shift	minus_probe_updates	6	0.83750	0.00625	0.17815	1.05163	0.82477
combined_micro.shift	minus_probe_updates	7	0.81563	0.00000	0.18361	1.04794	0.80891
combined_micro.shift	minus_probe_updates	8	0.85313	0.00625	0.19959	1.05406	0.84006
combined_micro.shift	minus_probe_updates	9	0.80937	0.00625	0.18456	1.05244	0.79653
combined_micro.shift	minus_hazard_margin	0	0.70937	0.00625	0.20169	1.09797	0.69016
combined_micro.shift	minus_hazard_margin	1	0.75938	0.00937	0.20928	1.09870	0.73731
combined_micro.shift	minus_hazard_margin	2	0.77187	0.00625	0.19705	1.09597	0.75294
combined_micro.shift	minus_hazard_margin	3	0.76562	0.00625	0.19588	1.09537	0.74677
combined_micro.shift	minus_hazard_margin	4	0.74375	0.01250	0.21234	1.10269	0.71837
combined_micro.shift	minus_hazard_margin	5	0.72813	0.00937	0.21697	1.09981	0.70590
combined_micro.shift	minus_hazard_margin	6	0.75313	0.00000	0.21717	1.08947	0.74060
combined_micro.shift	minus_hazard_margin	7	0.77500	0.00625	0.21313	1.09686	0.75594
combined_micro.shift	minus_hazard_margin	8	0.73750	0.00937	0.20566	1.09847	0.71546
combined_micro.shift	minus_hazard_margin	9	0.76562	0.01250	0.20887	1.10291	0.74022
combined_micro.shift	minus_shift_decomposition	0	0.84062	0.00937	0.10128	1.10484	0.81770
combined_micro.shift	minus_shift_decomposition	1	0.77187	0.00625	0.10727	1.10286	0.75197
combined_micro.shift	minus_shift_decomposition	2	0.82500	0.00313	0.10477	1.09783	0.80855
combined_micro.shift	minus_shift_decomposition	3	0.78125	0.00313	0.10092	1.09953	0.76457
combined_micro.shift	minus_shift_decomposition	4	0.81250	0.00313	0.10868	1.09783	0.79605
combined_micro.shift	minus_shift_decomposition	5	0.80312	0.00625	0.10783	1.10019	0.78360
combined_micro.shift	minus_shift_decomposition	6	0.82500	0.00313	0.11094	1.09545	0.80889
combined_micro.shift	minus_shift_decomposition	7	0.79375	0.00000	0.10421	1.09428	0.78055
combined_micro.shift	minus_shift_decomposition	8	0.80937	0.00313	0.10240	1.09834	0.79286
combined_micro.shift	minus_shift_decomposition	9	0.80312	0.00625	0.10735	1.10256	0.78327
combined_micro.shift	minus_uncertainty_calibration	0	0.84062	0.00000	0.16234	1.09592	0.82720
combined_micro.shift	minus_uncertainty_calibration	1	0.84062	0.00000	0.16954	1.09452	0.82739
combined_micro.shift	minus_uncertainty_calibration	2	0.80000	0.00313	0.15926	1.09747	0.78360
combined_micro.shift	minus_uncertainty_calibration	3	0.85000	0.00000	0.15896	1.09266	0.83703
combined_micro.shift	minus_uncertainty_calibration	4	0.76875	0.00000	0.17227	1.09489	0.75547
combined_micro.shift	minus_uncertainty_calibration	5	0.79063	0.00000	0.17594	1.09303	0.77760
combined_micro.shift	minus_uncertainty_calibration	6	0.78125	0.00000	0.17606	1.09222	0.76834
combined_micro.shift	minus_uncertainty_calibration	7	0.83437	0.00000	0.17133	1.09503	0.82107
combined_micro.shift	minus_uncertainty_calibration	8	0.77812	0.00313	0.16554	1.09613	0.76192
combined_micro.shift	minus_uncertainty_calibration	9	0.80312	0.00000	0.16886	1.09481	0.78985
combined_micro.shift	minus_skill_age_state	0	0.78125	0.00000	0.11792	1.09547	0.76788
combined_micro.shift	minus_skill_age_state	1	0.82188	0.00313	0.12626	1.09991	0.80514
combined_micro.shift	minus_skill_age_state	2	0.80000	0.00313	0.11945	1.09842	0.78347
combined_micro.shift	minus_skill_age_state	3	0.80625	0.00000	0.11897	1.09569	0.79285
combined_micro.shift	minus_skill_age_state	4	0.77500	0.00937	0.12795	1.10433	0.75214
combined_micro.shift	minus_skill_age_state	5	0.80000	0.00313	0.12940	1.09702	0.78367
combined_micro.shift	minus_skill_age_state	6	0.77812	0.00000	0.12890	1.09347	0.76504
combined_micro.shift	minus_skill_age_state	7	0.79063	0.01250	0.12414	1.10587	0.76480
combined_micro.shift	minus_skill_age_state	8	0.83437	0.00313	0.12015	1.09694	0.81805
combined_micro.shift	minus_skill_age_state	9	0.79063	0.00313	0.12435	1.09923	0.77398
combined_micro.shift	fixed_global_half_life	0	0.75625	0.00625	0.53147	1.09175	0.73578
combined_micro.shift	fixed_global_half_life	1	0.70625	0.00937	0.53746	1.09381	0.68168
combined_micro.shift	fixed_global_half_life	2	0.68437	0.00313	0.53282	1.08850	0.66711
combined_micro.shift	fixed_global_half_life	3	0.79688	0.00625	0.53352	1.09294	0.77411
combined_micro.shift	fixed_global_half_life	4	0.74062	0.00313	0.54988	1.09058	0.72201
combined_micro.shift	fixed_global_half_life	5	0.75625	0.00625	0.54257	1.09234	0.73463
combined_micro.shift	fixed_global_half_life	6	0.70000	0.00313	0.52437	1.08991	0.68360

Table 11: Seed-level ablation metrics. (continued)

Split	Ablation	Seed	Success	Unsafe	HLE	Cost	Safety utility
combined_micro.shift	fixed_global_half_life	7	0.77187	0.00937	0.53280	1.09589	0.74489
combined_micro.shift	fixed_global_half_life	8	0.71875	0.00937	0.51960	1.09500	0.69295
combined_micro.shift	fixed_global_half_life	9	0.75938	0.00000	0.54244	1.08881	0.74375
combined_micro.shift	threshold_only_risk_gate	0	0.77500	0.00937	0.14676	1.06364	0.75784
combined_micro.shift	threshold_only_risk_gate	1	0.75938	0.00313	0.15542	1.05603	0.74878
combined_micro.shift	threshold_only_risk_gate	2	0.77500	0.00937	0.14654	1.06045	0.75829
combined_micro.shift	threshold_only_risk_gate	3	0.75625	0.00313	0.14297	1.05395	0.74595
combined_micro.shift	threshold_only_risk_gate	4	0.80312	0.00625	0.15699	1.06055	0.78915
combined_micro.shift	threshold_only_risk_gate	5	0.76875	0.00937	0.15626	1.06372	0.75158
combined_micro.shift	threshold_only_risk_gate	6	0.75625	0.00313	0.16052	1.05759	0.74544
combined_micro.shift	threshold_only_risk_gate	7	0.78750	0.00313	0.15421	1.06019	0.77632
combined_micro.shift	threshold_only_risk_gate	8	0.72500	0.00313	0.15030	1.05908	0.71398
combined_micro.shift	threshold_only_risk_gate	9	0.78438	0.00313	0.15514	1.05803	0.77350
combined_micro.shift	expected_success_only	0	0.73125	0.00313	0.15291	1.05842	0.72032
combined_micro.shift	expected_success_only	1	0.77500	0.00000	0.16022	1.05137	0.76781
combined_micro.shift	expected_success_only	2	0.70000	0.00625	0.15043	1.05803	0.68638
combined_micro.shift	expected_success_only	3	0.75313	0.01875	0.15024	1.06591	0.72740
combined_micro.shift	expected_success_only	4	0.72500	0.01250	0.16274	1.06669	0.70466
combined_micro.shift	expected_success_only	5	0.71562	0.00937	0.16610	1.05995	0.69898
combined_micro.shift	expected_success_only	6	0.73438	0.00937	0.16601	1.06217	0.71742
combined_micro.shift	expected_success_only	7	0.75938	0.01250	0.16134	1.06594	0.73914
combined_micro.shift	expected_success_only	8	0.72500	0.00937	0.15589	1.06188	0.70809
combined_micro.shift	expected_success_only	9	0.70937	0.00625	0.15928	1.06189	0.69521
contact_mode.chatter	full_skill_half_life_scheduler.v5	0	0.84688	0.00000	0.10458	1.08277	0.83529
contact_mode.chatter	full_skill_half_life_scheduler.v5	1	0.83125	0.00313	0.10979	1.08544	0.81654
contact_mode.chatter	full_skill_half_life_scheduler.v5	2	0.80312	0.00000	0.11132	1.08252	0.79157
contact_mode.chatter	full_skill_half_life_scheduler.v5	3	0.77812	0.00625	0.10728	1.08916	0.76014
contact_mode.chatter	full_skill_half_life_scheduler.v5	4	0.80937	0.00625	0.10701	1.08728	0.79166
contact_mode.chatter	full_skill_half_life_scheduler.v5	5	0.82812	0.00000	0.10639	1.08487	0.81624
contact_mode.chatter	full_skill_half_life_scheduler.v5	6	0.80937	0.00313	0.11583	1.08808	0.79429
contact_mode.chatter	full_skill_half_life_scheduler.v5	7	0.80625	0.00313	0.10624	1.08647	0.79139
contact_mode.chatter	full_skill_half_life_scheduler.v5	8	0.79063	0.00000	0.10556	1.08302	0.77900
contact_mode.chatter	full_skill_half_life_scheduler.v5	9	0.79688	0.00937	0.10969	1.09100	0.77589
contact_mode.chatter	minus_per_skill_survival	0	0.80937	0.00000	0.13863	1.08708	0.79718
contact_mode.chatter	minus_per_skill_survival	1	0.84688	0.00000	0.13591	1.08733	0.83465
contact_mode.chatter	minus_per_skill_survival	2	0.82812	0.00625	0.14164	1.09183	0.80977
contact_mode.chatter	minus_per_skill_survival	3	0.83437	0.00000	0.14277	1.08491	0.82249
contact_mode.chatter	minus_per_skill_survival	4	0.80312	0.00000	0.13345	1.08442	0.79131
contact_mode.chatter	minus_per_skill_survival	5	0.80937	0.00000	0.13787	1.08813	0.79704
contact_mode.chatter	minus_per_skill_survival	6	0.83750	0.00313	0.13625	1.08972	0.82219
contact_mode.chatter	minus_per_skill_survival	7	0.83125	0.00313	0.13833	1.08892	0.81605
contact_mode.chatter	minus_per_skill_survival	8	0.85000	0.00000	0.14403	1.08598	0.83796
contact_mode.chatter	minus_per_skill_survival	9	0.81563	0.00000	0.13997	1.08573	0.80362
contact_mode.chatter	minus_probe_updates	0	0.81563	0.00625	0.17892	1.04594	0.80369
contact_mode.chatter	minus_probe_updates	1	0.79063	0.00937	0.18325	1.04859	0.77557
contact_mode.chatter	minus_probe_updates	2	0.80312	0.00625	0.18542	1.04431	0.79142
contact_mode.chatter	minus_probe_updates	3	0.82188	0.00000	0.18381	1.04062	0.81619
contact_mode.chatter	minus_probe_updates	4	0.80000	0.00625	0.17666	1.04756	0.78784
contact_mode.chatter	minus_probe_updates	5	0.81875	0.00313	0.17238	1.04166	0.81017
contact_mode.chatter	minus_probe_updates	6	0.79063	0.00313	0.17903	1.04247	0.78193
contact_mode.chatter	minus_probe_updates	7	0.84375	0.00000	0.16274	1.04225	0.83783
contact_mode.chatter	minus_probe_updates	8	0.79375	0.00625	0.19855	1.04513	0.78193
contact_mode.chatter	minus_probe_updates	9	0.81250	0.00313	0.17199	1.04328	0.80369
contact_mode.chatter	minus_hazard_margin	0	0.78438	0.00625	0.18363	1.08592	0.76685
contact_mode.chatter	minus_hazard_margin	1	0.74062	0.00000	0.18392	1.08114	0.72927
contact_mode.chatter	minus_hazard_margin	2	0.74687	0.00000	0.18506	1.08088	0.73555
contact_mode.chatter	minus_hazard_margin	3	0.77812	0.00313	0.18452	1.08383	0.76364
contact_mode.chatter	minus_hazard_margin	4	0.76875	0.00625	0.18504	1.08406	0.75148
contact_mode.chatter	minus_hazard_margin	5	0.70937	0.01250	0.18327	1.09019	0.68575
contact_mode.chatter	minus_hazard_margin	6	0.77812	0.00313	0.19567	1.08353	0.76368
contact_mode.chatter	minus_hazard_margin	7	0.81563	0.00625	0.18357	1.08620	0.79806
contact_mode.chatter	minus_hazard_margin	8	0.77187	0.01562	0.18582	1.09334	0.74506
contact_mode.chatter	minus_hazard_margin	9	0.75625	0.00625	0.18819	1.08539	0.73880
contact_mode.chatter	minus_shift_decomposition	0	0.84062	0.00625	0.09030	1.08916	0.82264
contact_mode.chatter	minus_shift_decomposition	1	0.83125	0.00000	0.09160	1.08277	0.81966
contact_mode.chatter	minus_shift_decomposition	2	0.78125	0.00625	0.09923	1.08944	0.76323
contact_mode.chatter	minus_shift_decomposition	3	0.81250	0.00313	0.09402	1.08569	0.79775
contact_mode.chatter	minus_shift_decomposition	4	0.80312	0.00313	0.09063	1.08619	0.78831
contact_mode.chatter	minus_shift_decomposition	5	0.85000	0.00000	0.09043	1.08595	0.83797
contact_mode.chatter	minus_shift_decomposition	6	0.79063	0.00000	0.09702	1.08569	0.77863
contact_mode.chatter	minus_shift_decomposition	7	0.81875	0.00937	0.08705	1.09259	0.79754
contact_mode.chatter	minus_shift_decomposition	8	0.82500	0.00000	0.09183	1.08409	0.81323
contact_mode.chatter	minus_shift_decomposition	9	0.83125	0.00313	0.09592	1.08728	0.81628
contact_mode.chatter	minus_uncertainty_calibration	0	0.78125	0.00313	0.14481	1.08380	0.76677
contact_mode.chatter	minus_uncertainty_calibration	1	0.79688	0.00313	0.14714	1.08328	0.78247
contact_mode.chatter	minus_uncertainty_calibration	2	0.79688	0.00313	0.14944	1.08406	0.78236
contact_mode.chatter	minus_uncertainty_calibration	3	0.81250	0.00313	0.14678	1.08594	0.79772

Table 11: Seed-level ablation metrics. (continued)

Split	Ablation	Seed	Success	Unsafe	HLE	Cost	Safety utility
contact_mode_chatter	minus_uncertainty_calibration	4	0.81875	0.00313	0.14634	1.08300	0.80438
contact_mode_chatter	minus_uncertainty_calibration	5	0.76562	0.00313	0.14585	1.08538	0.75092
contact_mode_chatter	minus_uncertainty_calibration	6	0.80937	0.00313	0.15769	1.08484	0.79475
contact_mode_chatter	minus_uncertainty_calibration	7	0.80312	0.00000	0.14563	1.08169	0.79169
contact_mode_chatter	minus_uncertainty_calibration	8	0.83750	0.00313	0.14683	1.08461	0.82290
contact_mode_chatter	minus_uncertainty_calibration	9	0.82188	0.00000	0.14911	1.08222	0.81036
contact_mode_chatter	minus_skill_age_state	0	0.87500	0.00000	0.10458	1.08277	0.86341
contact_mode_chatter	minus_skill_age_state	1	0.83437	0.00313	0.10979	1.08544	0.81966
contact_mode_chatter	minus_skill_age_state	2	0.78438	0.01250	0.11132	1.09314	0.76034
contact_mode_chatter	minus_skill_age_state	3	0.81875	0.00313	0.10728	1.08650	0.80389
contact_mode_chatter	minus_skill_age_state	4	0.83437	0.00313	0.10701	1.08463	0.81978
contact_mode_chatter	minus_skill_age_state	5	0.83750	0.00000	0.10639	1.08487	0.82562
contact_mode_chatter	minus_skill_age_state	6	0.83437	0.00000	0.11583	1.08542	0.82242
contact_mode_chatter	minus_skill_age_state	7	0.81875	0.00313	0.10624	1.08647	0.80389
contact_mode_chatter	minus_skill_age_state	8	0.80625	0.00313	0.10556	1.08567	0.79151
contact_mode_chatter	minus_skill_age_state	9	0.84375	0.00000	0.10969	1.08303	0.83213
contact_mode_chatter	fixed_global_half_life	0	0.73750	0.00937	0.35921	1.08672	0.71498
contact_mode_chatter	fixed_global_half_life	1	0.78438	0.00313	0.35299	1.08034	0.77038
contact_mode_chatter	fixed_global_half_life	2	0.74687	0.00313	0.36565	1.08139	0.73167
contact_mode_chatter	fixed_global_half_life	3	0.71250	0.00625	0.36192	1.08352	0.69318
contact_mode_chatter	fixed_global_half_life	4	0.79375	0.00625	0.35328	1.08433	0.77644
contact_mode_chatter	fixed_global_half_life	5	0.75313	0.00313	0.34789	1.08166	0.73576
contact_mode_chatter	fixed_global_half_life	6	0.77187	0.00937	0.35270	1.08752	0.75031
contact_mode_chatter	fixed_global_half_life	7	0.78750	0.00000	0.33640	1.07820	0.77443
contact_mode_chatter	fixed_global_half_life	8	0.75938	0.00625	0.33585	1.08486	0.74093
contact_mode_chatter	fixed_global_half_life	9	0.76562	0.00625	0.36800	1.08566	0.74495
contact_mode_chatter	threshold_only_risk_gate	0	0.74062	0.01562	0.12839	1.06264	0.71811
contact_mode_chatter	threshold_only_risk_gate	1	0.77500	0.00313	0.12774	1.05173	0.76501
contact_mode_chatter	threshold_only_risk_gate	2	0.79375	0.00000	0.13535	1.04694	0.78718
contact_mode_chatter	threshold_only_risk_gate	3	0.74062	0.00313	0.13158	1.04908	0.73100
contact_mode_chatter	threshold_only_risk_gate	4	0.78438	0.00000	0.12913	1.04855	0.77758
contact_mode_chatter	threshold_only_risk_gate	5	0.79375	0.00313	0.12842	1.05014	0.78398
contact_mode_chatter	threshold_only_risk_gate	6	0.78438	0.00000	0.13893	1.04909	0.77750
contact_mode_chatter	threshold_only_risk_gate	7	0.79375	0.00313	0.12481	1.05069	0.78390
contact_mode_chatter	threshold_only_risk_gate	8	0.75313	0.01875	0.13115	1.06342	0.72775
contact_mode_chatter	threshold_only_risk_gate	9	0.72188	0.01562	0.13488	1.06131	0.69954
contact_mode_chatter	expected_success_only	0	0.77812	0.00625	0.13584	1.05089	0.76550
contact_mode_chatter	expected_success_only	1	0.77812	0.01250	0.13872	1.06047	0.75866
contact_mode_chatter	expected_success_only	2	0.77187	0.00937	0.14106	1.05409	0.75605
contact_mode_chatter	expected_success_only	3	0.79375	0.00313	0.13799	1.04852	0.78421
contact_mode_chatter	expected_success_only	4	0.80000	0.00937	0.13754	1.05275	0.78437
contact_mode_chatter	expected_success_only	5	0.77812	0.00937	0.13726	1.05463	0.76223
contact_mode_chatter	expected_success_only	6	0.77500	0.00000	0.14866	1.05064	0.76791
contact_mode_chatter	expected_success_only	7	0.78750	0.00937	0.13710	1.05330	0.77179
contact_mode_chatter	expected_success_only	8	0.76562	0.01562	0.13770	1.05886	0.74363
contact_mode_chatter	expected_success_only	9	0.75625	0.00625	0.14024	1.05437	0.74314

F FIXED-RISK SEED-LEVEL METRICS

Table 12: Seed-level fixed-risk metrics.

Split	Budget	Method	Seed	Coverage	Fixed success	Executed success	False-safe
combined_micro_shift	0.02	Conformal	0	0.00000	0.00000	0.00000	0.00000
combined_micro_shift	0.02	Conformal	1	0.00000	0.00000	0.00000	0.00000
combined_micro_shift	0.02	Conformal	2	0.00000	0.00000	0.00000	0.00000
combined_micro_shift	0.02	Conformal	3	0.00000	0.00000	0.00000	0.00000
combined_micro_shift	0.02	Conformal	4	0.00000	0.00000	0.00000	0.00000
combined_micro_shift	0.02	Conformal	5	0.00000	0.00000	0.00000	0.00000
combined_micro_shift	0.02	Conformal	6	0.00000	0.00000	0.00000	0.00000
combined_micro_shift	0.02	Conformal	7	0.00000	0.00000	0.00000	0.00000
combined_micro_shift	0.02	Conformal	8	0.00000	0.00000	0.00000	0.00000
combined_micro_shift	0.02	Conformal	9	0.00000	0.00000	0.00000	0.00000
combined_micro_shift	0.02	Ensemble	0	0.00000	0.00000	0.00000	0.00000
combined_micro_shift	0.02	Ensemble	1	0.00000	0.00000	0.00000	0.00000
combined_micro_shift	0.02	Ensemble	2	0.00000	0.00000	0.00000	0.00000
combined_micro_shift	0.02	Ensemble	3	0.00000	0.00000	0.00000	0.00000
combined_micro_shift	0.02	Ensemble	4	0.00000	0.00000	0.00000	0.00000
combined_micro_shift	0.02	Ensemble	5	0.00000	0.00000	0.00000	0.00000
combined_micro_shift	0.02	Ensemble	6	0.00000	0.00000	0.00000	0.00000

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Table 12: Seed-level fixed-risk metrics. (continued)

Split	Budget	Method	Seed	Coverage	Fixed success	Executed success	False-safe
combined_micro.shift	0.10	Oracle	1	0.08500	0.08000	0.94118	0.00000
combined_micro.shift	0.10	Oracle	2	0.09500	0.07500	0.78947	0.00000
combined_micro.shift	0.10	Oracle	3	0.09500	0.07500	0.78947	0.00000
combined_micro.shift	0.10	Oracle	4	0.08500	0.08000	0.94118	0.00000
combined_micro.shift	0.10	Oracle	5	0.09000	0.07500	0.83333	0.00000
combined_micro.shift	0.10	Oracle	6	0.07500	0.07000	0.93333	0.00000
combined_micro.shift	0.10	Oracle	7	0.07000	0.06000	0.85714	0.00000
combined_micro.shift	0.10	Oracle	8	0.07500	0.06000	0.80000	0.00000
combined_micro.shift	0.10	Oracle	9	0.08000	0.06500	0.81250	0.00000
combined_micro.shift	0.20	Conformal	0	0.04500	0.04000	0.88889	0.00000
combined_micro.shift	0.20	Conformal	1	0.06000	0.05000	0.83333	0.00000
combined_micro.shift	0.20	Conformal	2	0.05500	0.05000	0.90909	0.00000
combined_micro.shift	0.20	Conformal	3	0.05000	0.04000	0.80000	0.00000
combined_micro.shift	0.20	Conformal	4	0.05500	0.05000	0.90909	0.00000
combined_micro.shift	0.20	Conformal	5	0.05500	0.04500	0.81818	0.00000
combined_micro.shift	0.20	Conformal	6	0.05500	0.04500	0.81818	0.00000
combined_micro.shift	0.20	Conformal	7	0.04500	0.04500	1.00000	0.00000
combined_micro.shift	0.20	Conformal	8	0.05500	0.04500	0.81818	0.00000
combined_micro.shift	0.20	Conformal	9	0.05500	0.05000	0.90909	0.00000
combined_micro.shift	0.20	Ensemble	0	0.08500	0.07500	0.88235	0.00000
combined_micro.shift	0.20	Ensemble	1	0.08500	0.07000	0.82353	0.00000
combined_micro.shift	0.20	Ensemble	2	0.10000	0.06500	0.65000	0.00000
combined_micro.shift	0.20	Ensemble	3	0.09000	0.08500	0.94444	0.00000
combined_micro.shift	0.20	Ensemble	4	0.08500	0.07000	0.82353	0.00000
combined_micro.shift	0.20	Ensemble	5	0.09500	0.07500	0.78947	0.00000
combined_micro.shift	0.20	Ensemble	6	0.10500	0.09000	0.85714	0.00000
combined_micro.shift	0.20	Ensemble	7	0.08000	0.08000	1.00000	0.00000
combined_micro.shift	0.20	Ensemble	8	0.08000	0.07000	0.87500	0.00000
combined_micro.shift	0.20	Ensemble	9	0.07500	0.05500	0.73333	0.00000
combined_micro.shift	0.20	Bayes survival	0	0.05500	0.04500	0.81818	0.00000
combined_micro.shift	0.20	Bayes survival	1	0.06500	0.05500	0.84615	0.00000
combined_micro.shift	0.20	Bayes survival	2	0.07000	0.04000	0.57143	0.00000
combined_micro.shift	0.20	Bayes survival	3	0.05500	0.04000	0.72727	0.00000
combined_micro.shift	0.20	Bayes survival	4	0.06000	0.06000	1.00000	0.00000
combined_micro.shift	0.20	Bayes survival	5	0.06500	0.05500	0.84615	0.00000
combined_micro.shift	0.20	Bayes survival	6	0.07500	0.06500	0.86667	0.00000
combined_micro.shift	0.20	Bayes survival	7	0.06000	0.05500	0.91667	0.00000
combined_micro.shift	0.20	Bayes survival	8	0.06500	0.06000	0.92308	0.00000
combined_micro.shift	0.20	Bayes survival	9	0.06000	0.06000	1.00000	0.00000
combined_micro.shift	0.20	CVaR guard	0	0.00000	0.00000	0.00000	0.00000
combined_micro.shift	0.20	CVaR guard	1	0.00000	0.00000	0.00000	0.00000
combined_micro.shift	0.20	CVaR guard	2	0.00500	0.00500	1.00000	0.00000
combined_micro.shift	0.20	CVaR guard	3	0.00000	0.00000	0.00000	0.00000
combined_micro.shift	0.20	CVaR guard	4	0.00000	0.00000	0.00000	0.00000
combined_micro.shift	0.20	CVaR guard	5	0.00000	0.00000	0.00000	0.00000
combined_micro.shift	0.20	CVaR guard	6	0.00000	0.00000	0.00000	0.00000
combined_micro.shift	0.20	CVaR guard	7	0.00000	0.00000	0.00000	0.00000
combined_micro.shift	0.20	CVaR guard	8	0.00000	0.00000	0.00000	0.00000
combined_micro.shift	0.20	CVaR guard	9	0.00000	0.00000	0.00000	0.00000
combined_micro.shift	0.20	Half-life v5	0	0.08000	0.05000	0.62500	0.00000
combined_micro.shift	0.20	Half-life v5	1	0.07500	0.06500	0.86667	0.00000
combined_micro.shift	0.20	Half-life v5	2	0.08500	0.07000	0.82353	0.00000
combined_micro.shift	0.20	Half-life v5	3	0.07500	0.06000	0.80000	0.00000
combined_micro.shift	0.20	Half-life v5	4	0.07500	0.06000	0.80000	0.00000
combined_micro.shift	0.20	Half-life v5	5	0.08500	0.08500	1.00000	0.00000
combined_micro.shift	0.20	Half-life v5	6	0.09000	0.07500	0.83333	0.00000
combined_micro.shift	0.20	Half-life v5	7	0.06500	0.06500	1.00000	0.00000
combined_micro.shift	0.20	Half-life v5	8	0.07500	0.06000	0.80000	0.00000
combined_micro.shift	0.20	Half-life v5	9	0.07500	0.06500	0.86667	0.00000
combined_micro.shift	0.20	Oracle	0	0.40500	0.35000	0.86420	0.00000
combined_micro.shift	0.20	Oracle	1	0.40000	0.36000	0.90000	0.00000
combined_micro.shift	0.20	Oracle	2	0.35500	0.32000	0.90141	0.01408
combined_micro.shift	0.20	Oracle	3	0.41000	0.36500	0.89024	0.00000
combined_micro.shift	0.20	Oracle	4	0.37000	0.30500	0.82432	0.00000
combined_micro.shift	0.20	Oracle	5	0.35500	0.28500	0.80282	0.00000
combined_micro.shift	0.20	Oracle	6	0.33000	0.27000	0.81818	0.00000
combined_micro.shift	0.20	Oracle	7	0.33000	0.27000	0.81818	0.00000
combined_micro.shift	0.20	Oracle	8	0.41500	0.36000	0.86747	0.00000
combined_micro.shift	0.20	Oracle	9	0.45500	0.37500	0.82418	0.01099
contact_mode.chatter	0.02	Conformal	0	0.00000	0.00000	0.00000	0.00000
contact_mode.chatter	0.02	Conformal	1	0.00000	0.00000	0.00000	0.00000
contact_mode.chatter	0.02	Conformal	2	0.00000	0.00000	0.00000	0.00000
contact_mode.chatter	0.02	Conformal	3	0.00000	0.00000	0.00000	0.00000
contact_mode.chatter	0.02	Conformal	4	0.00000	0.00000	0.00000	0.00000
contact_mode.chatter	0.02	Conformal	5	0.00000	0.00000	0.00000	0.00000
contact_mode.chatter	0.02	Conformal	6	0.00000	0.00000	0.00000	0.00000
contact_mode.chatter	0.02	Conformal	7	0.00000	0.00000	0.00000	0.00000

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Table 12: Seed-level fixed-risk metrics. (continued)

Split	Budget	Method	Seed	Coverage	Fixed success	Executed success	False-safe
contact_mode_chatter	0.10	Half-life v5	2	0.00000	0.00000	0.00000	0.00000
contact_mode_chatter	0.10	Half-life v5	3	0.00000	0.00000	0.00000	0.00000
contact_mode_chatter	0.10	Half-life v5	4	0.00000	0.00000	0.00000	0.00000
contact_mode_chatter	0.10	Half-life v5	5	0.00000	0.00000	0.00000	0.00000
contact_mode_chatter	0.10	Half-life v5	6	0.00000	0.00000	0.00000	0.00000
contact_mode_chatter	0.10	Half-life v5	7	0.00000	0.00000	0.00000	0.00000
contact_mode_chatter	0.10	Half-life v5	8	0.00000	0.00000	0.00000	0.00000
contact_mode_chatter	0.10	Half-life v5	9	0.00000	0.00000	0.00000	0.00000
contact_mode_chatter	0.10	Oracle	0	0.10000	0.09500	0.95000	0.00000
contact_mode_chatter	0.10	Oracle	1	0.11500	0.09500	0.82609	0.00000
contact_mode_chatter	0.10	Oracle	2	0.11500	0.10500	0.91304	0.00000
contact_mode_chatter	0.10	Oracle	3	0.11000	0.10500	0.95455	0.00000
contact_mode_chatter	0.10	Oracle	4	0.12000	0.11000	0.91667	0.00000
contact_mode_chatter	0.10	Oracle	5	0.11500	0.10500	0.91304	0.00000
contact_mode_chatter	0.10	Oracle	6	0.09500	0.09500	1.00000	0.00000
contact_mode_chatter	0.10	Oracle	7	0.09500	0.08000	0.84211	0.00000
contact_mode_chatter	0.10	Oracle	8	0.09500	0.08000	0.84211	0.00000
contact_mode_chatter	0.10	Oracle	9	0.10000	0.08000	0.80000	0.00000
contact_mode_chatter	0.20	Conformal	0	0.06500	0.06000	0.92308	0.00000
contact_mode_chatter	0.20	Conformal	1	0.08000	0.06500	0.81250	0.00000
contact_mode_chatter	0.20	Conformal	2	0.09000	0.08000	0.88889	0.00000
contact_mode_chatter	0.20	Conformal	3	0.07500	0.06000	0.80000	0.00000
contact_mode_chatter	0.20	Conformal	4	0.08000	0.08000	1.00000	0.00000
contact_mode_chatter	0.20	Conformal	5	0.06500	0.06000	0.92308	0.00000
contact_mode_chatter	0.20	Conformal	6	0.05500	0.05000	0.90909	0.00000
contact_mode_chatter	0.20	Conformal	7	0.07500	0.06500	0.86667	0.00000
contact_mode_chatter	0.20	Conformal	8	0.06500	0.05500	0.84615	0.00000
contact_mode_chatter	0.20	Conformal	9	0.07000	0.04500	0.64286	0.00000
contact_mode_chatter	0.20	Ensemble	0	0.11500	0.11000	0.95652	0.00000
contact_mode_chatter	0.20	Ensemble	1	0.11000	0.08500	0.77273	0.00000
contact_mode_chatter	0.20	Ensemble	2	0.11000	0.09500	0.86364	0.00000
contact_mode_chatter	0.20	Ensemble	3	0.12500	0.10500	0.84000	0.00000
contact_mode_chatter	0.20	Ensemble	4	0.11000	0.09500	0.86364	0.00000
contact_mode_chatter	0.20	Ensemble	5	0.10000	0.08000	0.80000	0.00000
contact_mode_chatter	0.20	Ensemble	6	0.11500	0.10500	0.91304	0.00000
contact_mode_chatter	0.20	Ensemble	7	0.10500	0.10000	0.95238	0.00000
contact_mode_chatter	0.20	Ensemble	8	0.09500	0.08000	0.84211	0.00000
contact_mode_chatter	0.20	Ensemble	9	0.11000	0.09000	0.81818	0.00000
contact_mode_chatter	0.20	Bayes survival	0	0.07000	0.07000	1.00000	0.00000
contact_mode_chatter	0.20	Bayes survival	1	0.08000	0.07500	0.93750	0.00000
contact_mode_chatter	0.20	Bayes survival	2	0.09000	0.08500	0.94444	0.00000
contact_mode_chatter	0.20	Bayes survival	3	0.08000	0.06000	0.75000	0.00000
contact_mode_chatter	0.20	Bayes survival	4	0.09500	0.08500	0.89474	0.00000
contact_mode_chatter	0.20	Bayes survival	5	0.08000	0.07000	0.87500	0.00000
contact_mode_chatter	0.20	Bayes survival	6	0.07000	0.06000	0.85714	0.00000
contact_mode_chatter	0.20	Bayes survival	7	0.08500	0.06500	0.76471	0.00000
contact_mode_chatter	0.20	Bayes survival	8	0.07000	0.06500	0.92857	0.00000
contact_mode_chatter	0.20	Bayes survival	9	0.08000	0.07000	0.87500	0.00000
contact_mode_chatter	0.20	CVaR guard	0	0.00000	0.00000	0.00000	0.00000
contact_mode_chatter	0.20	CVaR guard	1	0.00000	0.00000	0.00000	0.00000
contact_mode_chatter	0.20	CVaR guard	2	0.00000	0.00000	0.00000	0.00000
contact_mode_chatter	0.20	CVaR guard	3	0.00500	0.00500	1.00000	0.00000
contact_mode_chatter	0.20	CVaR guard	4	0.00000	0.00000	0.00000	0.00000
contact_mode_chatter	0.20	CVaR guard	5	0.00500	0.00500	1.00000	0.00000
contact_mode_chatter	0.20	CVaR guard	6	0.00000	0.00000	0.00000	0.00000
contact_mode_chatter	0.20	CVaR guard	7	0.00000	0.00000	0.00000	0.00000
contact_mode_chatter	0.20	CVaR guard	8	0.00000	0.00000	0.00000	0.00000
contact_mode_chatter	0.20	CVaR guard	9	0.00000	0.00000	0.00000	0.00000
contact_mode_chatter	0.20	Half-life v5	0	0.09500	0.06500	0.68421	0.00000
contact_mode_chatter	0.20	Half-life v5	1	0.10000	0.08000	0.80000	0.00000
contact_mode_chatter	0.20	Half-life v5	2	0.11500	0.10500	0.91304	0.00000
contact_mode_chatter	0.20	Half-life v5	3	0.10500	0.08500	0.80952	0.00000
contact_mode_chatter	0.20	Half-life v5	4	0.11000	0.09500	0.86364	0.00000
contact_mode_chatter	0.20	Half-life v5	5	0.08500	0.07500	0.88235	0.00000
contact_mode_chatter	0.20	Half-life v5	6	0.10000	0.08000	0.80000	0.00000
contact_mode_chatter	0.20	Half-life v5	7	0.10000	0.08500	0.85000	0.00000
contact_mode_chatter	0.20	Half-life v5	8	0.08500	0.08000	0.94118	0.00000
contact_mode_chatter	0.20	Half-life v5	9	0.10000	0.09000	0.90000	0.00000
contact_mode_chatter	0.20	Oracle	0	0.58500	0.48000	0.82051	0.00000
contact_mode_chatter	0.20	Oracle	1	0.58500	0.50500	0.86325	0.00000
contact_mode_chatter	0.20	Oracle	2	0.58500	0.51000	0.87179	0.00000
contact_mode_chatter	0.20	Oracle	3	0.43500	0.36500	0.83908	0.00000
contact_mode_chatter	0.20	Oracle	4	0.47000	0.40000	0.85106	0.00000
contact_mode_chatter	0.20	Oracle	5	0.37000	0.33500	0.90541	0.00000
contact_mode_chatter	0.20	Oracle	6	0.66500	0.55000	0.82707	0.00000
contact_mode_chatter	0.20	Oracle	7	0.58500	0.44500	0.76068	0.00000
contact_mode_chatter	0.20	Oracle	8	0.58000	0.51000	0.87931	0.00000

Table 12: Seed-level fixed-risk metrics. (continued)

Split	Budget	Method	Seed	Coverage	Fixed success	Executed success	False-safe
contact_mode_chatter	0.20	Oracle	9	0.61500	0.51000	0.82927	0.00000

G FIXED-RISK PAIRWISE METRICS

Table 13: Fixed-risk pairwise differences for Half-life v5.

Split	Budget	Reference	Metric	Mean diff	CI95	Lower95
combined_micro_shift	0.02	Conformal	coverage	0.00000	0.00000	0.00000
combined_micro_shift	0.02	Conformal	fixed_risk_success	0.00000	0.00000	0.00000
combined_micro_shift	0.02	Conformal	executed_success	0.00000	0.00000	0.00000
combined_micro_shift	0.02	Conformal	false_safe_rate	0.00000	0.00000	0.00000
combined_micro_shift	0.02	Ensemble	coverage	0.00000	0.00000	0.00000
combined_micro_shift	0.02	Ensemble	fixed_risk_success	0.00000	0.00000	0.00000
combined_micro_shift	0.02	Ensemble	executed_success	0.00000	0.00000	0.00000
combined_micro_shift	0.02	Ensemble	false_safe_rate	0.00000	0.00000	0.00000
combined_micro_shift	0.02	Bayes survival	coverage	0.00000	0.00000	0.00000
combined_micro_shift	0.02	Bayes survival	fixed_risk_success	0.00000	0.00000	0.00000
combined_micro_shift	0.02	Bayes survival	executed_success	0.00000	0.00000	0.00000
combined_micro_shift	0.02	Bayes survival	false_safe_rate	0.00000	0.00000	0.00000
combined_micro_shift	0.02	CVaR guard	coverage	0.00000	0.00000	0.00000
combined_micro_shift	0.02	CVaR guard	fixed_risk_success	0.00000	0.00000	0.00000
combined_micro_shift	0.02	CVaR guard	executed_success	0.00000	0.00000	0.00000
combined_micro_shift	0.02	CVaR guard	false_safe_rate	0.00000	0.00000	0.00000
combined_micro_shift	0.02	Oracle	coverage	0.00000	0.00000	0.00000
combined_micro_shift	0.02	Oracle	fixed_risk_success	0.00000	0.00000	0.00000
combined_micro_shift	0.02	Oracle	executed_success	0.00000	0.00000	0.00000
combined_micro_shift	0.02	Oracle	false_safe_rate	0.00000	0.00000	0.00000
combined_micro_shift	0.05	Conformal	coverage	0.00000	0.00000	0.00000
combined_micro_shift	0.05	Conformal	fixed_risk_success	0.00000	0.00000	0.00000
combined_micro_shift	0.05	Conformal	executed_success	0.00000	0.00000	0.00000
combined_micro_shift	0.05	Conformal	false_safe_rate	0.00000	0.00000	0.00000
combined_micro_shift	0.05	Ensemble	coverage	0.00000	0.00000	0.00000
combined_micro_shift	0.05	Ensemble	fixed_risk_success	0.00000	0.00000	0.00000
combined_micro_shift	0.05	Ensemble	executed_success	0.00000	0.00000	0.00000
combined_micro_shift	0.05	Ensemble	false_safe_rate	0.00000	0.00000	0.00000
combined_micro_shift	0.05	Bayes survival	coverage	0.00000	0.00000	0.00000
combined_micro_shift	0.05	Bayes survival	fixed_risk_success	0.00000	0.00000	0.00000
combined_micro_shift	0.05	Bayes survival	executed_success	0.00000	0.00000	0.00000
combined_micro_shift	0.05	Bayes survival	false_safe_rate	0.00000	0.00000	0.00000
combined_micro_shift	0.05	CVaR guard	coverage	0.00000	0.00000	0.00000
combined_micro_shift	0.05	CVaR guard	fixed_risk_success	0.00000	0.00000	0.00000
combined_micro_shift	0.05	CVaR guard	executed_success	0.00000	0.00000	0.00000
combined_micro_shift	0.05	CVaR guard	false_safe_rate	0.00000	0.00000	0.00000
combined_micro_shift	0.05	Oracle	coverage	-0.02700	0.00392	-0.03092
combined_micro_shift	0.05	Oracle	fixed_risk_success	-0.02500	0.00387	-0.02887
combined_micro_shift	0.05	Oracle	executed_success	-0.92976	0.07233	-1.00209
combined_micro_shift	0.05	Oracle	false_safe_rate	0.00000	0.00000	0.00000
combined_micro_shift	0.10	Conformal	coverage	0.00000	0.00000	0.00000
combined_micro_shift	0.10	Conformal	fixed_risk_success	0.00000	0.00000	0.00000
combined_micro_shift	0.10	Conformal	executed_success	0.00000	0.00000	0.00000
combined_micro_shift	0.10	Conformal	false_safe_rate	0.00000	0.00000	0.00000
combined_micro_shift	0.10	Ensemble	coverage	0.00000	0.00000	0.00000
combined_micro_shift	0.10	Ensemble	fixed_risk_success	0.00000	0.00000	0.00000
combined_micro_shift	0.10	Ensemble	executed_success	0.00000	0.00000	0.00000
combined_micro_shift	0.10	Ensemble	false_safe_rate	0.00000	0.00000	0.00000
combined_micro_shift	0.10	Bayes survival	coverage	0.00000	0.00000	0.00000
combined_micro_shift	0.10	Bayes survival	fixed_risk_success	0.00000	0.00000	0.00000
combined_micro_shift	0.10	Bayes survival	executed_success	0.00000	0.00000	0.00000
combined_micro_shift	0.10	Bayes survival	false_safe_rate	0.00000	0.00000	0.00000
combined_micro_shift	0.10	CVaR guard	coverage	0.00000	0.00000	0.00000
combined_micro_shift	0.10	CVaR guard	fixed_risk_success	0.00000	0.00000	0.00000
combined_micro_shift	0.10	CVaR guard	executed_success	0.00000	0.00000	0.00000
combined_micro_shift	0.10	CVaR guard	false_safe_rate	0.00000	0.00000	0.00000
combined_micro_shift	0.10	Oracle	coverage	-0.08300	0.00531	-0.08831
combined_micro_shift	0.10	Oracle	fixed_risk_success	-0.07100	0.00457	-0.07557
combined_micro_shift	0.10	Oracle	executed_success	-0.85726	0.03878	-0.89604
combined_micro_shift	0.10	Oracle	false_safe_rate	0.00000	0.00000	0.00000
combined_micro_shift	0.20	Conformal	coverage	0.02500	0.00438	0.02062
combined_micro_shift	0.20	Conformal	fixed_risk_success	0.01950	0.00574	0.01376

Table 13: Fixed-risk pairwise differences for Half-life v5. (continued)

Split	Budget	Reference	Metric	Mean diff	CI95	Lower95
combined_micro.shift	0.20	Conformal	executed_success	-0.02888	0.07068	-0.09957
combined_micro.shift	0.20	Conformal	false_safe_rate	0.00000	0.00000	0.00000
combined_micro.shift	0.20	Ensemble	coverage	-0.01000	0.00327	-0.01327
combined_micro.shift	0.20	Ensemble	fixed_risk_success	-0.00800	0.00803	-0.01603
combined_micro.shift	0.20	Ensemble	executed_success	0.00364	0.08953	-0.08589
combined_micro.shift	0.20	Ensemble	false_safe_rate	0.00000	0.00000	0.00000
combined_micro.shift	0.20	Bayes survival	coverage	0.01500	0.00358	0.01142
combined_micro.shift	0.20	Bayes survival	fixed_risk_success	0.01200	0.00688	0.00512
combined_micro.shift	0.20	Bayes survival	executed_success	-0.01004	0.09473	-0.10477
combined_micro.shift	0.20	Bayes survival	false_safe_rate	0.00000	0.00000	0.00000
combined_micro.shift	0.20	CVaR guard	coverage	0.07750	0.00420	0.07330
combined_micro.shift	0.20	CVaR guard	fixed_risk_success	0.06500	0.00584	0.05916
combined_micro.shift	0.20	CVaR guard	executed_success	0.74152	0.21069	0.53083
combined_micro.shift	0.20	CVaR guard	false_safe_rate	0.00000	0.00000	0.00000
combined_micro.shift	0.20	Oracle	coverage	-0.30450	0.02688	-0.33138
combined_micro.shift	0.20	Oracle	fixed_risk_success	-0.26050	0.02931	-0.28981
combined_micro.shift	0.20	Oracle	executed_success	-0.00958	0.08021	-0.08979
combined_micro.shift	0.20	Oracle	false_safe_rate	-0.00251	0.00331	-0.00581
contact_mode.chatter	0.02	Conformal	coverage	0.00000	0.00000	0.00000
contact_mode.chatter	0.02	Conformal	fixed_risk_success	0.00000	0.00000	0.00000
contact_mode.chatter	0.02	Conformal	executed_success	0.00000	0.00000	0.00000
contact_mode.chatter	0.02	Conformal	false_safe_rate	0.00000	0.00000	0.00000
contact_mode.chatter	0.02	Ensemble	coverage	0.00000	0.00000	0.00000
contact_mode.chatter	0.02	Ensemble	fixed_risk_success	0.00000	0.00000	0.00000
contact_mode.chatter	0.02	Ensemble	executed_success	0.00000	0.00000	0.00000
contact_mode.chatter	0.02	Ensemble	false_safe_rate	0.00000	0.00000	0.00000
contact_mode.chatter	0.02	Bayes survival	coverage	0.00000	0.00000	0.00000
contact_mode.chatter	0.02	Bayes survival	fixed_risk_success	0.00000	0.00000	0.00000
contact_mode.chatter	0.02	Bayes survival	executed_success	0.00000	0.00000	0.00000
contact_mode.chatter	0.02	Bayes survival	false_safe_rate	0.00000	0.00000	0.00000
contact_mode.chatter	0.02	CVaR guard	coverage	0.00000	0.00000	0.00000
contact_mode.chatter	0.02	CVaR guard	fixed_risk_success	0.00000	0.00000	0.00000
contact_mode.chatter	0.02	CVaR guard	executed_success	0.00000	0.00000	0.00000
contact_mode.chatter	0.02	CVaR guard	false_safe_rate	0.00000	0.00000	0.00000
contact_mode.chatter	0.02	Oracle	coverage	-0.00050	0.00098	-0.00148
contact_mode.chatter	0.02	Oracle	fixed_risk_success	-0.00050	0.00098	-0.00148
contact_mode.chatter	0.02	Oracle	executed_success	-0.10000	0.19600	-0.29600
contact_mode.chatter	0.02	Oracle	false_safe_rate	0.00000	0.00000	0.00000
contact_mode.chatter	0.05	Conformal	coverage	0.00000	0.00000	0.00000
contact_mode.chatter	0.05	Conformal	fixed_risk_success	0.00000	0.00000	0.00000
contact_mode.chatter	0.05	Conformal	executed_success	0.00000	0.00000	0.00000
contact_mode.chatter	0.05	Conformal	false_safe_rate	0.00000	0.00000	0.00000
contact_mode.chatter	0.05	Ensemble	coverage	0.00000	0.00000	0.00000
contact_mode.chatter	0.05	Ensemble	fixed_risk_success	0.00000	0.00000	0.00000
contact_mode.chatter	0.05	Ensemble	executed_success	0.00000	0.00000	0.00000
contact_mode.chatter	0.05	Ensemble	false_safe_rate	0.00000	0.00000	0.00000
contact_mode.chatter	0.05	Bayes survival	coverage	0.00000	0.00000	0.00000
contact_mode.chatter	0.05	Bayes survival	fixed_risk_success	0.00000	0.00000	0.00000
contact_mode.chatter	0.05	Bayes survival	executed_success	0.00000	0.00000	0.00000
contact_mode.chatter	0.05	Bayes survival	false_safe_rate	0.00000	0.00000	0.00000
contact_mode.chatter	0.05	CVaR guard	coverage	0.00000	0.00000	0.00000
contact_mode.chatter	0.05	CVaR guard	fixed_risk_success	0.00000	0.00000	0.00000
contact_mode.chatter	0.05	CVaR guard	executed_success	0.00000	0.00000	0.00000
contact_mode.chatter	0.05	CVaR guard	false_safe_rate	0.00000	0.00000	0.00000
contact_mode.chatter	0.05	Oracle	coverage	-0.03450	0.00494	-0.03944
contact_mode.chatter	0.05	Oracle	fixed_risk_success	-0.03150	0.00414	-0.03564
contact_mode.chatter	0.05	Oracle	executed_success	-0.91794	0.05589	-0.97383
contact_mode.chatter	0.05	Oracle	false_safe_rate	0.00000	0.00000	0.00000
contact_mode.chatter	0.10	Conformal	coverage	0.00000	0.00000	0.00000
contact_mode.chatter	0.10	Conformal	fixed_risk_success	0.00000	0.00000	0.00000
contact_mode.chatter	0.10	Conformal	executed_success	0.00000	0.00000	0.00000
contact_mode.chatter	0.10	Conformal	false_safe_rate	0.00000	0.00000	0.00000
contact_mode.chatter	0.10	Ensemble	coverage	0.00000	0.00000	0.00000
contact_mode.chatter	0.10	Ensemble	fixed_risk_success	0.00000	0.00000	0.00000
contact_mode.chatter	0.10	Ensemble	executed_success	0.00000	0.00000	0.00000
contact_mode.chatter	0.10	Ensemble	false_safe_rate	0.00000	0.00000	0.00000
contact_mode.chatter	0.10	Bayes survival	coverage	0.00000	0.00000	0.00000
contact_mode.chatter	0.10	Bayes survival	fixed_risk_success	0.00000	0.00000	0.00000
contact_mode.chatter	0.10	Bayes survival	executed_success	0.00000	0.00000	0.00000
contact_mode.chatter	0.10	Bayes survival	false_safe_rate	0.00000	0.00000	0.00000
contact_mode.chatter	0.10	CVaR guard	coverage	0.00000	0.00000	0.00000
contact_mode.chatter	0.10	CVaR guard	fixed_risk_success	0.00000	0.00000	0.00000
contact_mode.chatter	0.10	CVaR guard	executed_success	0.00000	0.00000	0.00000
contact_mode.chatter	0.10	CVaR guard	false_safe_rate	0.00000	0.00000	0.00000
contact_mode.chatter	0.10	Oracle	coverage	-0.10600	0.00616	-0.11216
contact_mode.chatter	0.10	Oracle	fixed_risk_success	-0.09500	0.00716	-0.10216
contact_mode.chatter	0.10	Oracle	executed_success	-0.89576	0.04032	-0.93608

Table 13: Fixed-risk pairwise differences for Half-life v5. (continued)

Split	Budget	Reference	Metric	Mean diff	CI95	Lower95
contact_mode_chatter	0.10	Oracle	false_safe_rate	0.00000	0.00000	0.00000
contact_mode_chatter	0.20	Conformal	coverage	0.02750	0.00468	0.02282
contact_mode_chatter	0.20	Conformal	fixed_risk_success	0.02200	0.00673	0.01527
contact_mode_chatter	0.20	Conformal	executed_success	-0.01684	0.08332	-0.10016
contact_mode_chatter	0.20	Conformal	false_safe_rate	0.00000	0.00000	0.00000
contact_mode_chatter	0.20	Ensemble	coverage	-0.01000	0.00506	-0.01506
contact_mode_chatter	0.20	Ensemble	fixed_risk_success	-0.01050	0.00996	-0.02046
contact_mode_chatter	0.20	Ensemble	executed_success	-0.01783	0.07206	-0.08989
contact_mode_chatter	0.20	Ensemble	false_safe_rate	0.00000	0.00000	0.00000
contact_mode_chatter	0.20	Bayes survival	coverage	0.01950	0.00449	0.01501
contact_mode_chatter	0.20	Bayes survival	fixed_risk_success	0.01350	0.00585	0.00765
contact_mode_chatter	0.20	Bayes survival	executed_success	-0.03832	0.07181	-0.11013
contact_mode_chatter	0.20	Bayes survival	false_safe_rate	0.00000	0.00000	0.00000
contact_mode_chatter	0.20	CVaR guard	coverage	0.09850	0.00638	0.09212
contact_mode_chatter	0.20	CVaR guard	fixed_risk_success	0.08300	0.00719	0.07581
contact_mode_chatter	0.20	CVaR guard	executed_success	0.64439	0.26488	0.37952
contact_mode_chatter	0.20	CVaR guard	false_safe_rate	0.00000	0.00000	0.00000
contact_mode_chatter	0.20	Oracle	coverage	-0.44800	0.05605	-0.50405
contact_mode_chatter	0.20	Oracle	fixed_risk_success	-0.37700	0.04451	-0.42151
contact_mode_chatter	0.20	Oracle	executed_success	-0.00035	0.04303	-0.04338
contact_mode_chatter	0.20	Oracle	false_safe_rate	0.00000	0.00000	0.00000

H FULL PRIOR-WORK PRESSURE TABLE

The table below is a pressure table from the local shared pool, not a claim that every item is a direct baseline. Its purpose is to make the surrounding novelty risk visible and to avoid pretending the half-life framing lives in an empty literature.

Table 14: Hostile prior-work pressure from the local deep-read pool.

Citation	Title	Year	Venue	Hostile score
(Anderson & Warhurst, 2012)	Lost in translation? Skills policy and the shift to skill ecosystems	2012	Strathprints: The University of Strathclyde institutional repository (University of Strathclyde)	21.292
(Kida & Ogi, 2025)	Evaluating how psychological senses and physical motions are affected by avatar shapes in a non-immersive environment	2025	Artificial Life and Robotics	20.600
(Levac et al., 2019)	Learning and transfer of complex motor skills in virtual reality: a perspective review	2019	Journal of NeuroEngineering and Rehabilitation	21.005
(Yandell et al., 2017)	Physical interface dynamics alter how robotic exosuits augment human movement: implications for optimizing wearable assistive devices	2017	Journal of NeuroEngineering and Rehabilitation	19.965
(Andersen et al., 2017)	Integration of a Skill-based Collaborative Mobile Robot in a Smart Cyber-physical Environment	2017	Procedia Manufacturing	19.314
(Urban, 2023)	How Hands-On Assessments Can Boost Retention, Satisfaction, Skill Development, and Career Outcomes in Online Courses	2023	AI, Computer Science and Robotics Technology	19.223
(Abu-Dakka et al., 2015)	Adaptation of manipulation skills in physical contact with the environment to reference force profiles	2015	Autonomous Robots	19.053
(Condino et al., 2011)	How to build patientspecific synthetic abdominal anatomies. An innovative approach from physical toward hybrid surgical simulators	2011	The International Journal of Medical Robotics and Computer Assisted Surgery	18.507
(Usher et al., 2025)	Bridging Virtual and Physical	2025	Journal of Learning Analytics	18.347
(Sepasgozar, 2021)	Differentiating Digital Twin from Digital Shadow: Elucidating a Paradigm Shift to Expedite a Smart, Sustainable Built Environment	2021	Buildings	18.119
(Xerou & Angeli, 2022)	HOW CAN ROBOTICS AND PROGRAMMING AFFECT CHILDREN'S COMPUTATIONAL THINKING SKILLS THROUGH REAL LIFE SCENARIOS?	2022	EDULEARN Proceedings EDULEARN22 Proceedings	18.000
(Yang et al., 2017)	Interface Design of a Physical Human-Robot Interaction System for Human Impedance Adaptive Skill Transfer	2017	IEEE Transactions on Automation Science and Engineering	17.799
(Greenhalgh et al., 2022)	Adapt or die: how the pandemic made the shift from EBM to EBM+ more urgent	2022	BMJ Evidence-Based Medicine	18.600
(Stezek et al., 2021)	A Digital Twin Approach for the Improvement of an Autonomous Mobile Robots (AMR's) Operating Environment-A Case Study	2021	Sensors	17.500

Table 14: Hostile prior-work pressure from the local deep-read pool. (continued)

Citation	Title	Year	Venue	Hostile score
(Bergougui, 2025)	Institutional adaptability, skill-bias technological shifts, and energy efficiency in global decarbonization pathways: Exploring the role of artificial intelligence patents	2025	Technology in Society	17.498
(Li et al., 2022)	ASTEROIDS: Exploring Swarms of Mini-Telepresence Robots for Physical Skill Demonstration	2022	CHI Conference on Human Factors in Computing Systems	17.359
(Persson et al., 2021)	Caregivers' use of robots and their effect on work environment - a scoping review	2021	Journal of Technology in Human Services	18.352
(Shmatko & Volkova, 2020)	Bridging the Skill Gap in Robotics: Global and National Environment	2020	SAGE Open	17.286
(Chen et al., 2023)	HTDet: A Hybrid Transformer-Based Approach for Underwater Small Object Detection	2023	Remote Sensing	17.244
(Averta & Hogan, 2020)	Enhancing Robot-Environment Physical Interaction via Optimal Impedance Profiles	2020		17.117
(Wang et al., 2020)	A Framework of Hybrid Force/Motion Skills Learning for Robots	2020	IEEE Transactions on Cognitive and Developmental Systems	18.044
(Shao et al., 2019)	Learning to Scaffold the Development of Robotic Manipulation Skills	2019	arXiv	18.000
(Liu et al., 2025)	Integrating skill knowledge and multimodal information to enhance 3C assembly skills	2025	Industrial Robot: the international journal of robotics research and application	17.997
(Zeng et al., 2020)	Simultaneously Encoding Movement and sEMG-Based Stiffness for Robotic Skill Learning	2020	IEEE Transactions on Industrial Informatics	16.992
(Moz et al., 2020)	Developing an Interactive Environment through the Teaching of Mathematics with Small Robots	2020	Sensors	16.914
(Chuang, 2020)	An empirical study of displaceable job skills in the age of robots	2020	European journal of training and development	16.869
(Coddington, 2021)	Incompatible with life: Embodied borders, migrant fertility, and the UK's hostile environment'	2021	Environment and Planning C: Politics and Space	17.854
(Ceriscioli & Mohan, 2026)	Planning under Distribution Shifts with Causal POMDPs	2026	arXiv	22.850

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